

Latent Crossroads: Recovering Typographic Robustness in Pretrained CLIP via CLS–Patch Linear Regression

zer0int¹ and GPT-5.2²

¹None. Independent researcher.

²An AI LLM by OpenAI

Correspondence: zer0int@email.gmx

Abstract

CLIP’s strong ability to “read” text in images makes it unusually vulnerable to *typographic attacks*, where a small written word can dominate image–text similarity despite being semantically irrelevant to the dominant visual object. We ask whether *typographic robustness is latent in patch-derived information* in pretrained OpenAI CLIP ViT-L/14, and how this relates to high-norm “register-like” tokens and circuits that store global information in local vision patches. We first characterize CLS–Patch–Register decoupling and investigate attention weights’ tendency to misrepresent effective information flow by leveraging value-path contribution diagnostics. Guided by our findings, we build simple inference-time patch readouts, including a linear *regression probe* to predict CLS from patch features, finding surprisingly robust outcomes for these *constructed* embeddings. We then introduce *Regression-Teacher* fine-tuning with multi-block teachers that enforce patch→CLS alignment during training. Our findings show a depth-localized readout shift: near-perfect final block output alignment with pooled non-register patches alongside strong suppression of alignment with the register subspace. *Regression-Teacher* fine-tunes exhibit substantial typographic attack robustness alongside broad improvements on retrieval benchmarks. We close by dissecting pretrained and fine-tuned model geometry and attention contribution, highlighting a plausible late-block control surface for robustness. **Reproducibility:** Code and model weights are available at <https://github.com/zer0int/CLIP-fine-tune>.

1 Introduction

Large-scale contrastive vision–language pretraining yields an operationally important side effect: models such as CLIP can extract semantic information from text present in images without an explicit OCR objective [11]. As the contrastive loss rewards Vision-Text Encoder-Encoder alignment, glyph patterns that map cleanly onto the text embedding space become unusually salient; OpenAI refer to the resulting failure mode as a *typographic attack vulnerability*, where image–text similarity can be steered toward an unrelated concept using only “pen and paper” [4].

Previous research points to the causal relationship: CLIP’s *image* encoder can entangle the visual appearance of written words with the visual concepts those words denote. Materzyńska et al. [8] show that the model can match images of words to natural images of the corresponding concepts. Recent work has formalized this threat model and its evaluation: The SCAM benchmark, consisting of images of everyday objects with 1. either empty sticky notes on them (NoSCAM), or 2. handwritten text on the note (SCAM) vs. 3. digital text (SynthSCAM), provides a real-world typographic robustness testbed [14].

In parallel, vision-transformer research has uncovered a different internal phenomenon: High-norm outlier tokens that emerge in low- or redundant-information background patch tokens in the latter half of sufficiently deep and well-trained ViTs. Darcet et al. [3] characterize these artifacts and frame them as internal “scratchpads”, introducing *register tokens*—additional non-image tokens prepended to the ViT sequence that absorb the role of these outlier patches, yielding smoother dense features and attention maps [3].

Two adjacent threads sharpened the mechanistic picture: Broadly, Sun et al. [13] study *massive activations*—large, sparse activation spikes that can behave like constant biases and induce “attention sinks”—in LLM; they further show that analogous outlier phenomena appear in transformers beyond language models, including multimodal models such as CLIP. Jiang et al. [6] argue that models do not need to be retrained to ensure registers outside the vision tokens; they identify a sparse set of “register neurons” whose activations concentrate norm onto outlier patch tokens in multiple models including CLIP. They demonstrate a training-free intervention, shifting the high-norm activations into an added *untrained* token prepended to the ViT during inference, effectively creating a register “slot” post hoc [6]. Crucially for our setting, they explicitly connect this mechanism to typographic robustness: By shifting outlier positions onto adversarial text regions, they reduce typographic attack success in CLIP-like models [6].

Further, Lappe and Giese [7] report that the register tokens exhibit extremely high average pairwise cosine similarity (0.9989), as the vector is dominated by high-norm directions that inflate similarity. They further emphasize that this image-invariance does not imply irrelevance: Pre-layer normalization and subsequently keys and values as linear projections of the hidden state allow the model to ignore those extreme coordinates [7]. More broadly, if registers act as privileged global carriers, then patch attention maps can look semantically plausible yet fail to faithfully explain the global representation: The authors show that adding registers in DINOv2 decouples local patch features from global outputs, and demonstrate that this mechanism seems to apply more broadly in deep and overparameterized ViTs. They thus argue that naive patch-based CLS attribution may be misleading even when attention maps appear clean and plausible [7].

Circuit-level analyses of typographic attacks in CLIP provide direct evidence for attention’s role: Hufe et al. [5] identify specialized heads in the latter half of CLIP that causally extract and transmit typographic information to CLS; they propose a training-free defense by selectively ablating this typographic circuit, turning CLIP “dyslexic” [5].

Taken together, these results suggest that typographic vulnerability reflects how CLIP-like models allocate global computation via sparse, high-norm pathways and attention circuits that can over-weight certain (adversarial) information relative to objective visual content.

We ask: If global image embeddings can partially detach from patch-derived evidence in large ViTs, can we recover useful information using alternative, patch-derived image embeddings? First, we apply Lappe and Giese [7] methods to CLIP ViT-L/14; we then investigate the usefulness of methods including linear regression probe mappings between CLS and patch-derived representations; we then apply the findings to inform fine-tuning of pretrained OpenAI CLIP ViT-L/14 using linear CLS-Patch regression teachers; finally, we analyze model geometry and attention contribution in detail.

2 Inference Experiments: Methods

We study whether typographic robustness is *latent* in patch-derived information in pretrained CLIP ViT-L/14 by (i) reproducing CLS–patch decoupling behavior in CLIP and (ii) constructing simple inference-time representations and a linear regression REGPROBE to test patch-to-CLS recoverability.

2.1 Setup and notation

We index ViT blocks by $l \in \{0, \dots, 23\}$. At block l , the CLIP vision transformer yields token states

$$X^{(l)} \in \mathbb{R}^{(P+1) \times d}, \quad (1)$$

where token 0 is the CLS token and tokens $1 \dots P$ correspond to image patches. The standard CLIP image embedding is

$$\mathbf{e}_{\text{CLS}} = \text{norm}\left(\Pi \text{LN}_{\text{post}}\left(X_0^{(23)}\right)\right) \in \mathbb{R}^D, \quad (2)$$

where LN_{post} is CLIP’s final layer normalization and Π is the learned projection to the multimodal embedding space.

2.2 Reproducing CLS–patch decoupling behavior in CLIP

Lappe & Giese [7] show that in DINOv2, global representations can decouple from local patch information in the presence of additional “register” tokens added to the ViT. CLIP ViT-L/14 has no explicit register tokens; we therefore operationalize *register-like* tokens as high-norm patch tokens in late blocks. **All results in this paper use a fixed patch L2 norm threshold $\tau_{\text{reg}} = 70$** (chosen empirically for blocks in CLIP ViT-L/14) Details: [section 10.1](#).

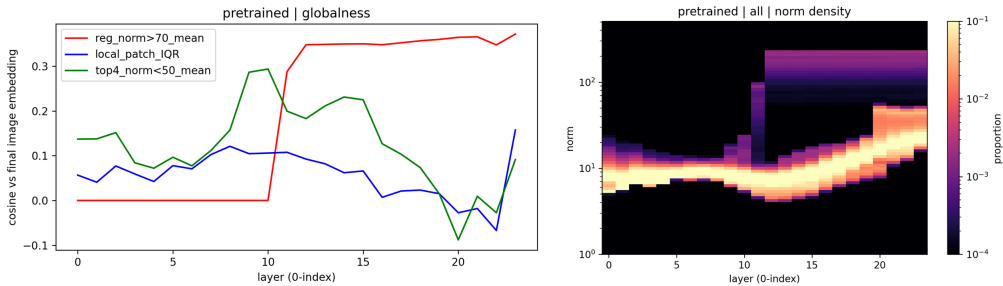


Figure 1: **Pretrained CLIP (ViT-L/14).** *Left:* “Globalness” of patch tokens across depth, measured as cosine similarity to the final image embedding, $\cos(x_p^{(b)}, z_{\text{img}})$, shown for norm-based token groups; high-norm register-like patches (red) emerge around blocks $b \approx 11$ –12. *Right:* token ℓ_2 -norm density over blocks, $\|x_p^{(b)}\|_2$. Dataset: SCAM (full) [14].

Register-like mask (per image). From an *unaltered* forward pass, we compute patch norms at the final block,

$$n_i = \left\| X_i^{(23)} \right\|_2, \quad i \in \{1, \dots, P\}, \quad (3)$$

and define a register-like mask

$$m_{\text{reg}}^{(23)}(i) = \mathbb{I}[n_i > \tau_{\text{reg}}]. \quad (4)$$

Masked last-block cls attention counterfactuals. To probe whether $\mathbf{e}_{\text{CLS}}$ depends primarily on non-register patches versus register-like patches, we construct counterfactual embeddings by re-running attention in a selected block while *masking only the CLS query row* so that CLS may attend only to:

- **patch-only:** keys in $\{i : m_{\text{reg}}^{(23)}(i) = 0\}$,
- **reg-only:** keys in $\{i : m_{\text{reg}}^{(23)}(i) = 1\}$.

Concretely, let $A^{(l)} \in \mathbb{R}^{H \times (P+1) \times (P+1)}$ be the attention probability tensor at block l . For the CLS query (row 0), we set disallowed key logits to $-\infty$ before softmax, producing masked probabilities $\hat{A}^{(l)}$, while leaving all other rows unchanged. We then read out the resulting CLS embedding via LN_{post} and Π as usual (the result remains a CLS embedding, not a pooled patch embedding).

Blockwise “reg leakage” probe. To localize where register-like dependence emerges across depth, we apply the same CLS-row masking at a sequence of blocks $l \in \mathcal{B}$ and forward the modified token states through subsequent blocks to obtain the final embedding. For each l , we compute cosine similarities between FULL, PATCH-only, and reg-only embeddings, and compare representational geometry via centered kernel alignment (CKA). This corresponds to our “REG leakage into FULL over blocks” reproduction experiment of Lappe and Giese [7].

2.3 Attention faithfulness diagnostics: token-mix vs value-mix

Attention weights alone do not necessarily fully explain the attention computation. We therefore report two complementary diagnostics at B_{23} : a *token-mix* proxy (based on attention-weighted token states) and a *value-mix* diagnostic (based on the true attention output from value vectors). The exact formulas and contribution decompositions are provided in [section 10.5](#).

2.4 SCAM evaluation protocol and forced-choice scoring

We evaluate typographic robustness on SCAM [14], which contains real-world images with handwritten adversarial words on post-it notes (SCAM), matched clean images (NoSCAM), and synthetic typography variants (SynthSCAM). For each image, we perform a binary evaluation: The model must prefer the correct object-label prompt over the attack-word prompt. We follow standard CLIP scoring using cosine similarity between an image embedding $\mathbf{e}$ and two text embeddings, and report accuracy and margin statistics per representation.

2.5 Patch-/register-derived representations in CLIP embedding space

Beyond the standard CLS embedding, we construct alternative image embeddings from late-block token states, then map them into the CLIP embedding space using $\text{norm}(\Pi \text{LN}_{\text{post}}(\cdot))$.

Patch pool (Patch-L23). Let $\mathcal{P}_{\neg\text{reg}} = \{i : m_{\text{reg}}^{(23)}(i) = 0\}$ be the non-register patch indices. We compute a pooled hidden vector

$$\mathbf{p}_{23}^{\text{raw}} = \frac{1}{|\mathcal{P}_{\neg\text{reg}}|} \sum_{i \in \mathcal{P}_{\neg\text{reg}}} X_i^{(23)}, \quad (5)$$

and embed

$$\mathbf{e}_{\text{PATCH}} = \text{norm}(\Pi \text{LN}_{\text{post}}(\mathbf{p}_{23}^{\text{raw}})). \quad (6)$$

Register-like pool (Reg-L23) and optional PC subtraction. Let $\mathcal{P}_{\text{reg}} = \{i : m_{\text{reg}}^{(23)}(i) = 1\}$. We pool register-like tokens analogously to obtain $\mathbf{r}_{23}^{\text{raw}}$ and embed $\mathbf{e}_{\text{reg}} = \text{norm}(\Pi \text{LN}_{\text{post}}(\mathbf{r}_{23}^{\text{raw}}))$. As an additional explorative experiment, we apply principal-component subtraction in hidden space before LN_{post} and Π ; the precise transform is deferred to [section 10.6](#).

Patch-delta (Patch Δ). To emphasize late-block updates, we compute a difference of patch pools across adjacent late blocks (final and penultimate). Let $\mathbf{p}_l^{\text{raw}}$ denote the pooled non-register patch hidden vector computed from $X^{(l)}$ as above (with the same masking rule). We define

$$\mathbf{e}_{\Delta} = \text{norm}(\text{norm}(\Pi \text{LN}_{\text{post}}(\mathbf{p}_{23}^{\text{raw}})) - \text{norm}(\Pi \text{LN}_{\text{post}}(\mathbf{p}_{22}^{\text{raw}}))). \quad (7)$$

Least-squares CLS-from-patch regression (CLS-PatchReg). Next, we test whether a simple linear map from pooled patch features to CLS can reconstruct a robust embedding at inference time. We refer to this inference-only map as REGPROBE. Let $\{\mathbf{p}_j\}_{j=1}^N$ be pooled patch hidden vectors (e.g., $\mathbf{p}_{23}^{\text{raw}}$) and $\{\mathbf{c}_j\}_{j=1}^N$ be the corresponding CLS hidden vectors $X_0^{(23)}$, both *before* LN_{post} and Π . Let $\boldsymbol{\mu}_p = \frac{1}{N} \sum_j \mathbf{p}_j$ and $\boldsymbol{\mu}_c = \frac{1}{N} \sum_j \mathbf{c}_j$. We fit an inference-time linear map W by least squares,

$$W = \arg \min_W \sum_{j=1}^N \left\| (\mathbf{p}_j - \boldsymbol{\mu}_p) W^{\top} - (\mathbf{c}_j - \boldsymbol{\mu}_c) \right\|_2^2 \quad (8)$$

and at test time predict

$$\hat{\mathbf{c}} = (\mathbf{p} - \boldsymbol{\mu}_p) W^{\top} + \boldsymbol{\mu}_c, \quad \mathbf{e}_{\text{PATCH}}^{\text{REG}} = \text{norm}(\Pi \text{LN}_{\text{post}}(\hat{\mathbf{c}})). \quad (9)$$

Exact solve details are provided in [section 10.9](#).

Probe data sources: NoSCAM vs ImageNet sampling. To distinguish structural geometry from dataset-specific fit, we fit W using either (i) NoSCAM images (n=1162) [\[14\]](#), or (ii) a small ImageNet-1k (ILSVRC2012) sample drawn uniformly across classes (3 images per class from train/val; n=6000). We then evaluate the resulting inference-time embeddings on SCAM variants and on **MVT**, a benchmark derived from ImageNet/ObjectNet images ranked for difficulty via human minimum viewing time [\[9\]](#). We use a 5k-label subset and evaluate zero-shot accuracy using template prompts (“a photo of a { }”); see our evaluation code [\[15\]](#).

3 Inference Experiments: Results

This section summarizes the empirical conclusions from our inference-time experiments in CLIP ViT-L/14, organized into two parts: (i) reproducing CLS-patch decoupling-style behavior in CLIP using emergent high-norm patch tokens as register-like proxies, and (ii) evaluating whether typographic robustness can be recovered from patch-derived representations via simple inference-time constructions, including a least-squares REGPROBE.

3.1 Reproducing CLS–patch decoupling behavior in CLIP

CLIP exhibits register-like behavior *within* patch tokens. CLIP ViT-L/14 has no dedicated register tokens; instead, for each image we observe a small subset of patch tokens whose norms exceed a fixed threshold and which we treat as *register-like* (a canonical per-image set; cf. Fig. 10.1). On Flickr8k, the number of such tokens above the fixed threshold $\tau_{\text{reg}} = 70$ is stable and low (mean 2.44, range $[1, 6]$), while the remaining patches form a large non-register pool (mean 254 of 256 patches) with markedly lower norms (patch mean 23.85, register-like mean 179.53).

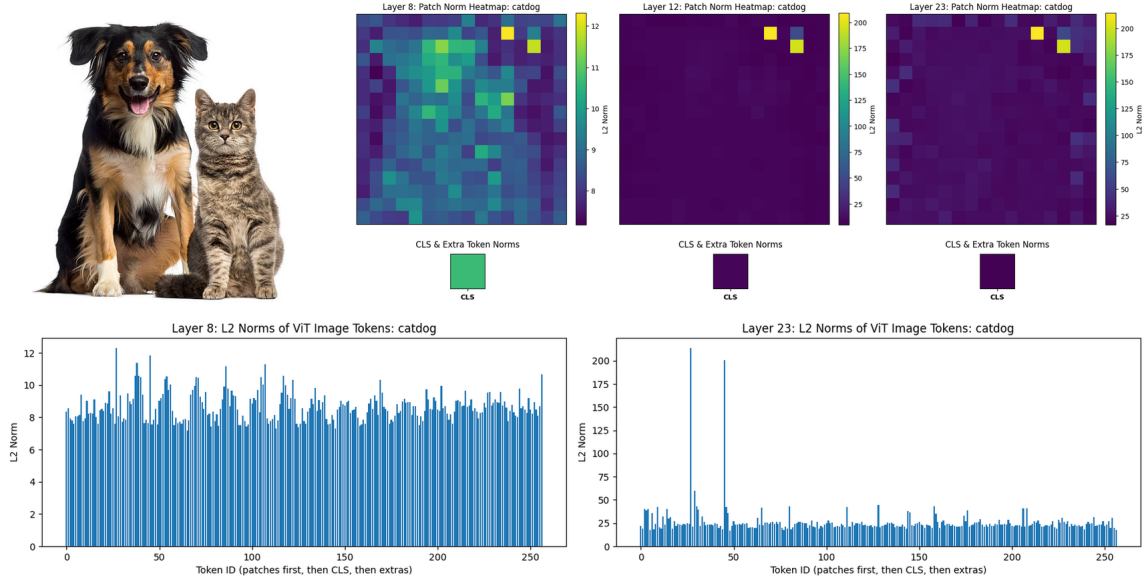


Figure 2: **Example (SCAM).** *Top:* input image and the corresponding high-norm ‘register’ patches in the token grid at blocks $b \in \{8, 12, 23\}$ (CLS shown below). *Bottom:* patch-token ℓ_2 norms, $\|x_p^{(b)}\|_2$, for blocks $b = 8$ and $b = 23$.

Global embedding alignment with non-register patches at the final block. Using the CLS-row masked-attention counterfactual at the final block B_{23} , we find that the standard CLIP embedding (FULL) is almost identical to the PATCH-ONLY counterfactual but substantially less aligned with the REG-ONLY counterfactual:

$$\begin{aligned} \text{CKA}(\text{FULL}, \text{PATCH-ONLY}) &= 0.997868, & \text{CKA}(\text{FULL}, \text{REG-ONLY}) &= 0.888133, \\ \cos(\text{FULL}, \text{PATCH-ONLY}) &= 0.996676, & \cos(\text{FULL}, \text{REG-ONLY}) &= 0.844343. \end{aligned}$$

Thus, at the terminal block, masking CLS attention to non-register patches preserves the final embedding, whereas restricting CLS to register-like patches yields a substantially different representation. Note that high cosine alignment of individual tokens with the final embedding does not imply they dominate the embedding’s construction. Full summary (Flickr8k) in table 13.

Why attention weights alone are misleading: the value path matters. The faithfulness probes separate an “attention-as-weights” story from the actual attention computation. On Flickr8k, the **TOKEN** diagnostic (comparing CLS to an attention-weighted mixture of tokens in LN1 space) is negative on average (-0.3943), whereas the **VALUE** diagnostic (reconstructing the true attention output via value vectors and the output projection) is high (0.9155). Further, value-based decomposition indicates that non-register patches contribute far more strongly to the CLS attention output than register-like patches:

$$\text{VALUE faithfulness (NO-REGS)} = 0.8644, \quad \text{VALUE faithfulness (REGS-ONLY)} = 0.4670.$$

We defer the exact diagnostic formulas and contribution decomposition to [section 10.5](#).

Register-like specialization emerges late. A blockwise masking experiment reveals a depth-dependent transition. When we apply the same PATCH-ONLY vs. REG-ONLY masking at intermediate blocks and then read out the final embedding, REG-ONLY remains very close to FULL in mid blocks but diverges substantially by the final block:

$$\begin{aligned} \text{b12: } \cos(\text{FULL}, \text{REG-ONLY}) &= 0.9931, & \text{CKA}(\text{FULL}, \text{REG-ONLY}) &= 0.9904, \\ \text{b23: } \cos(\text{FULL}, \text{REG-ONLY}) &= 0.8443, & \text{CKA}(\text{FULL}, \text{REG-ONLY}) &= 0.8881. \end{aligned}$$

This indicates a depth-localized transition: the register-only readout remains aligned with FULL at mid-depth but diverges at the terminal block. A full per-block leakage profile (cosine/CKA over blocks) is reported in [table 16](#).

Task-level corroboration: non-register patches carry object identity. A one-shot ImageNet-1k top-5 linear probe (1 train image/class; eval on 1 val image/class), repeated over three random seeds, yields mean accuracies:

$$\text{FULL} = 0.445, \quad \text{PATCH-ONLY} = 0.448, \quad \text{REG-ONLY} = 0.169.$$

The slight improvement of PATCH-ONLY over FULL (within noise expected from a lightweight one-shot setup) combined with the collapse of REG-ONLY suggests that the non-register patch pool carries most of the linearly accessible object-class signal in this one-shot probe, whereas the register-only subset is insufficient on its own. See [table 14](#) for the mean $\pm$ std over seeds.

Sensitivity to typographic attacks: SCAM vs. NoSCAM. Finally, a targeted SCAM probe shows that the alignment between FULL and REG decreases under typographic attacks relative to clean NoSCAM, while remaining clearly nonzero:

$$\begin{aligned} \cos(\text{FULL}, \text{REG}) (\text{NoSCAM}) &= 0.7934 \\ \cos(\text{FULL}, \text{REG}) (\text{SCAM}) &= 0.7406, \quad \cos(\text{FULL}, \text{REG}) (\text{SynthSCAM}) = 0.7445. \end{aligned}$$

This split-wise comparison shows a modest decrease in $\cos(\text{FULL}, \text{REG})$ under typographic attacks relative to NoSCAM, while remaining clearly nonzero. Full SCAM probe statistics (including auxiliary REG contribution summaries) are provided in [table 15](#).

Representation similarity does not imply zero-shot equivalence. The decoupling counterfactuals above compare internal representations under controlled masking at a fixed depth. In contrast, the inference-time patch-derived embeddings used below change the *readout* from token states into the multimodal space (pooling, optional projections, and regression), and therefore change the induced similarity geometry used for zero-shot classification. High cosine/CKA alignment with FULL can thus co-exist with worse clean accuracy: the patch-derived readout may preserve class-relevant object signal while discarding calibration and global invariances encoded by the canonical CLS pathway.

3.2 Recovering typographic robustness from patch-derived representations

We evaluate inference-time image embeddings on SCAM (NoSCAM/SCAM/SynthSCAM), where typographic attacks exploit CLIP’s emergent ability to “read” text in images. Results are reported as zero-shot accuracy between the correct object-label prompt and the attack-word prompt (wrong). We additionally report the ‘very difficult dataset’ zero-shot accuracy on MVT [\[9\]](#) to expose the robustness–generalization trade-off.

Standard CLS vs. patch-derived readouts. The standard CLIP embedding (CLS) is near-ceiling accuracy on clean NoSCAM but collapses under typographic attacks. A simple patch pooling readout (PATCH-L23) substantially improves robustness on SCAM/SynthSCAM but reduces clean NoSCAM accuracy, consistent with the decoupling evidence in §3.1. As a strong baseline, PATCH Δ recovers much of the robustness while largely preserving clean accuracy, suggesting that a substantial fraction of typographic brittleness is injected by late-block updates to the pooled patch representation and can be attenuated by emphasizing the incremental late-layer refinement rather than the full final-layer patch state. Full results are provided in table 17.

Readout	Acc (NoSCAM/SCAM/SynthSCAM)	MVT Acc
CLS	NoSCAM: 0.9905, SCAM: 0.4182, SynthSCAM: 0.3219	0.8830
PATCH-L23	NoSCAM: 0.7349, SCAM: 0.6024, SynthSCAM: 0.6317	0.3414
PATCH Δ	NoSCAM: 0.9690, SCAM: 0.8778, SynthSCAM: 0.9217	0.6881

Table 1: Inference results. CLS: standard CLIP image embedding; PATCH-L23: Final block patch pooling (excluding high-norm registers); PATCH Δ : L23-L22 patch-difference readout.

Probe readouts: cls-from-patch regression. Our strongest typographic attack robustness recovery uses a least-squares (optionally ridge-regularized) REGPROBE trained on NoSCAM, mapping pooled non-register patch features to CLS, then projecting into CLIP’s embedding space:

Probe readout	Acc (NoSCAM/SCAM/SynthSCAM)	MVT Acc
CLS-PATCHREG-S	NoSCAM: 0.9871, SCAM: 0.9286, SynthSCAM: 0.9449	0.5695
CLS-PATCHREG-I	NoSCAM: 0.9466, SCAM: 0.8761, SynthSCAM: 0.8890	0.7166

Table 2: We denote the probe by its fit distribution: CLS-PATCHREG-S when fit on NoSCAM (n=1162), and CLS-PATCHREG-I when fit on ImageNet-1k (3 samples/class, n=6000).

Robustness-generalization trade-off controlled by probe fit. On MVT, simple robustness-oriented readouts reduce accuracy relative to CLS (PATCH-L23, PATCH Δ). For the regression probe, the trade-off depends strongly on the probe-fit distribution: CLS-PATCHREG-S collapses on MVT (0.5695), whereas CLS-PATCHREG-I preserves substantially more of the global structure required for MVT (0.7166). Thus, regression-derived embeddings hint towards promising generalization across domains, although shown to be sensitive to probe-fit in our small-scale probe training sets.

Interpretation in the context of prior work. These inference results connect three strands of prior evidence: (i) typographic text can dominate CLIP similarity despite being semantically unrelated; (ii) typographic information is transmitted by specific attention-head circuits to CLS; and (iii) high-norm pathways can interact with typographic robustness in CLIP-like models. Our results add an inference-time perspective: patch-derived readouts improve typographic robustness but trade off performance on globally demanding recognition, suggesting that some invariances expressed by the standard CLS pathway are attenuated by these projections/poolings. Simple linear reconstruction from patches can yield a CLS-like embedding that is accurate on clean data and substantially more robust to typography, while trading off some of this higher-level, scene-level invariance.

Hypothesis. We hypothesize that late-block updates to CLS contain a structured component that supports *scene-level invariances and compositional cues* (e.g., inferring “cat” from partial evidence, such as a lump under a duvet with only a visible tail) but also mediates “*reading*”-driven concept activation from embedded text (e.g., glyph shapes $\rightarrow$ positional order $\rightarrow$ “reading” a word left-to-right to form $\rightarrow$ activation of the corresponding concept (such as *feline* concepts for the word “cat” in an image)). We hypothesize that a late-block component of the CLS update simultaneously supports scene-level invariances and mediates text-driven concept activation, and that this component is preferentially expressed along directions associated with the register-like patch subset. Manipulating access to these directions may therefore modulate both (i) typographic robustness and (ii) performance on globally demanding recognition.

Transition to training with regression teachers. Inference-time projections expose an opportunity and a limitation. On the one hand, typographic robustness can be recovered to a surprising extent from patch-derived information. On the other hand, MVT reveals that aggressive projections can discard global structure needed for difficult viewpoint recognition, and that the probe-fit distribution meaningfully shapes this trade-off. These findings motivate the next stage of our work: rather than relying solely on inference-time transformations, we investigate whether fine-tuning CLIP with regression teachers REGTEACHER constraints can internalize the robustness benefit while reducing the associated loss in globally demanding recognition.

4 Regression-Teacher Fine-Tuning

4.1 From inference to a train-time hypothesis

Our inference-only experiments expose a consistent pattern: (i) the native CLIP global embedding (CLS) is highly effective on clean data but collapses under typographic perturbations, (ii) object identity remains substantially recoverable from the broader patch-token manifold, and (iii) a simple least-squares map from patch-derived features to the CLS embedding can restore typographic robustness even when fit on clean or out-of-domain images. We therefore convert the inference-time regression into a *train-time constraint*: during fine-tuning, we encourage CLIP’s internal geometry to keep the CLS embedding *linearly predictable* from patch-derived features, reducing –but not eliminating– reliance on a hijackable carrier route while maintaining CLIP’s contrastive alignment.

4.2 Starting point and overall objective

We start from pretrained CLIP ViT-L/14 OpenAI [10]. Training proceeds in two phases:

Phase 0 (base fine-tune). To fine-tune the model, we first perform standard CLIP contrastive training (InfoNCE loss) on paired image–text data using Geometric Parametrization (GmP) [1]. This stage produces an aligned initialization on COCO-SPRIGT (SPRIGT-T2I [12]) (n=36000); we train both encoders (full model, all weights require gradient) to convergence (20 Epochs) with additional injection of our custom typographic attack dataset adversarial triplet (available from our codebase; only used during Phase 0) every 25 batches. The selected model checkpoint at Epoch 14 was chosen by best-overall performance vs. pretrained, and resembles a ‘maximum possible’ typographic attack robustness we can archive by conventional (no teacher) fine-tuning. This model will be our starting point for teacher-training; we’ll subsequently keep the text encoder frozen and only train the ViT using REGTEACHER using the same dataset the text encoder is now already aligned to.

Readout	Acc (NoSCAM/SCAM/SynthSCAM)	MVT Acc
CLS	NoSCAM: 0.9880, SCAM: 0.6928, SynthSCAM: 0.6790	0.8794

Table 3: Inference results for Phase 0 model, ckpt 14, using CLS standard CLIP image embedding. Config reference to reproduce model: `normal_finetune_stage_zero.json` [15]

Phase 1 (RegTeacher fine-tune). We continue from the PHASE 0 checkpoint and optimize

$$\mathcal{L} = \mathcal{L}_{\text{CLIP}} + \sum_{\ell \in \mathcal{T}} \lambda_{\ell} \mathcal{L}_{\text{teacher}}^{(\ell)}, \quad (10)$$

where $\mathcal{T}$ is a set of teacher layers (late ViT blocks, e.g. $\{21, 22, 23\}$) and λ_{ℓ} are per-layer weights. In our implementation, the total loss is accumulated as a base contrastive term (InfoNCE loss) plus a weighted sum of regression-consistency terms across layers. Config reference: `reproduce_regr_norm_model_teachers.json` and `reproduce_regr_brut_model_teachers.json` [15]

4.3 Base contrastive objective

Let $\mathbf{i}_b \in \mathbb{R}^D$ and $\mathbf{t}_b \in \mathbb{R}^D$ be the normalized image and text embeddings for a minibatch of size B . We form logits

$$\mathbf{z}_{b,b'} = \frac{\langle \mathbf{i}_b, \mathbf{t}_{b'} \rangle}{\tau}, \quad (11)$$

with temperature τ , and apply a symmetric cross-entropy objective over image→text and text→image directions with label smoothing. This is the standard CLIP-style contrastive training objective with additional label smoothing (smoothing factor 0.1).

4.4 Regression teachers

Each REGTEACHER at layer ℓ is a linear map that predicts a CLS-space target from patch-derived features extracted at the target layer, fit on COCO-SPRIGT 'val' (n=4500) + ImageNet-1k images (3 per class, n=6000).

Token extraction and CLIP-space embeddings. For each image, we run the visual transformer to block ℓ and obtain token states $X^{(\ell)} \in \mathbb{R}^{(1+P) \times d}$ (CLS plus P patch tokens). We map intermediate tokens into the final CLIP embedding space using the model’s post-layer normalization and projection (via LN_{post} and Π). We denote:

- $\mathbf{c}_{\ell} \in \mathbb{R}^D$: the CLS embedding at layer ℓ ,
- $\mathbf{p}_{\ell} \in \mathbb{R}^D$: a pooled patch embedding at layer ℓ .

Register-aware patch pooling. CLIP does not prepend explicit register tokens; instead, we operationalize *register-like* tokens as a per-image subset of patch tokens with unusually large norms. Using the register threshold $\tau_{\text{reg}} = 70$, we define a mask over patches at layer ℓ :

$$m_j^{(\ell)} = \mathbb{I}[\|x_j^{(\ell)}\|_2 \geq \tau_{\text{reg}}], \quad j \in \{1, \dots, P\}. \quad (12)$$

The default patch pool $\mathbf{p}_{\ell}$ averages *non-register* patches ($m_j^{(\ell)} = 0$), with a per-sample fallback if the mask is degenerate (implementation detail in our codebase).

Teacher targets and target mixing. Rather than always targeting the target layer- ℓ CLS embedding, we allow the teacher target to blend the local CLS with an auxiliary signal. In our implementation, the target is a normalized convex combination of the local CLS and either (i) a centered register-mean (derived from register-like tokens) embedding at the same layer (REG-MIX) or (ii) the final image embedding (FINAL-MIX). We write the target as $\mathbf{c}_\ell^*$.

Least-squares fitting. We fit teachers by centering training pairs and solving a linear least-squares problem. In the simplest case, the teacher uses patch features only:

$$W_\ell = \arg \min_W \sum_n \left\| (\mathbf{p}_\ell^{(n)} - \mu_{p,\ell}) W^\top - (\mathbf{c}_\ell^{*(n)} - \mu_{c,\ell}) \right\|_2^2 \quad (13)$$

Optionally, a reg-aware teacher concatenates patch features with a centered register embedding $\Delta \mathbf{r}_\ell$ before regression, i.e. $[\mathbf{p}_\ell - \mu_{p,\ell}, \Delta \mathbf{r}_\ell]$.

The resulting teacher stores $(W_\ell, \mu_{p,\ell}, \mu_{c,\ell})$ (and, when applicable, register statistics) in a cache for reuse.

4.5 Teacher consistency loss

We compute the teacher prediction

$$\hat{\mathbf{c}}_\ell = (\mathbf{p}_\ell - \mu_{p,\ell}) W_\ell^\top + \mu_{c,\ell}, \quad (14)$$

(or the analogous reg-aware form), and penalize mismatch between $\hat{\mathbf{c}}_\ell$ and the target $\mathbf{c}_\ell^*$ using an MSE over *normalized* embeddings, scaled by embedding dimension D :

$$\mathcal{L}_{\text{teacher}}^{(\ell)} = D \cdot \left\| \frac{\hat{\mathbf{c}}_\ell}{\|\hat{\mathbf{c}}_\ell\|_2} - \frac{\mathbf{c}_\ell^*}{\|\mathbf{c}_\ell^*\|_2} \right\|_2^2. \quad (15)$$

4.6 Teacher refresh and staleness control

Teachers are constructed from a stream of embeddings up to a maximum sample budget with an internal train/validation split. Teachers may be rebuilt periodically or conditionally (e.g. every epoch, or when validation cosine drops below thresholds). This ensures the regression constraint remains aligned with the evolving model during fine-tuning. We set teachers to be re-trained every Epoch for our experiments.

4.7 Trainable parameters (Phase 1)

For teacher-training, we freeze the text encoder and train only the vision tower, keeping the text anchor fixed while modifying how images map into the shared embedding space. This design isolates the intervention to image-side geometry and explicitly targets the CLS-patch relationship under the same text embedding reference.

Summary. PHASE 1 REGTEACHER fine-tuning enforces a distributed set of linear recoverability constraints across late transformer blocks while preserving CLIP’s contrastive alignment objective. Conceptually, the model is encouraged to maintain a global embedding that is also explainable from the patch manifold while simultaneously being nudged to rely less on the register subspace. Additional details can be found in appendix [section 10.3](#) and in our codebase [15].

5 Evaluating the Regression-Teacher fine-tuned CLIP model

Geometry and Representation. We analyze how REGRESSION-REGTEACHER fine-tuning reorganizes CLIP token geometry, and how these changes relate to typographic robustness and standard zero-shot / retrieval performance. We compare three CLIP models: pretrained baseline (**pretrained**) and two regression-teacher fine-tuned variants (**regr-norm**, **regr-brut**):

Regression-RegTeacher fine-tuned variants.

- **regr-norm (35 epochs):** three auxiliary teachers at blocks $\{21, 22, 23\}$. At blocks $l \in \{21, 22\}$, the local CLS^(l) is trained toward a reg-derived target at the same depth using a blended objective with reg-weight $\lambda_{\text{reg}}^{(l)} = 0.3$ (and complementary patch-supported weight $1 - \lambda_{\text{reg}}^{(l)} = 0.7$). At the terminal block $l = 23$, the teacher targets the *final* embedding using a full CLS-to-Patch objective with weight $\lambda^{(23)} = 1.0$ and *no* reg component ($\lambda_{\text{reg}}^{(23)} = 0$).
- **regr-brut (30 epochs):** two auxiliary teachers at blocks $\{22, 23\}$. Both teachers target the final CLS via CLS-to-Patch objectives (i.e., CLS [23]→Patch[23] and CLS [23]→Patch[22]) with unit weight and *no* reg *blending* ($\lambda_{\text{reg}}^{(22)} = \lambda_{\text{reg}}^{(23)} = 0$).

All other training details (exact optimizer/schedule/augmentations/teacher loss weight, etc.) are provided with the released codebase [15].

5.1 Decoupling: Teacher training localizes register–readout decoupling in final block

Methods: Same as for inference tests; please refer to the previously established procedures Section 2.2.

REGRESSION-REGTEACHER fine-tuning further tightens $\mathbf{z}_{\text{full}} - \mathbf{z}_{\text{patch}}^{(23)}$ alignment and sharply reduces $\mathbf{z}_{\text{full}} - \mathbf{z}_{\text{reg}}^{(23)}$ alignment at the terminal block. As a coarse task-level sanity check, reg-only embeddings retain some linearly accessible ImageNet signal in the pretrained model (one-shot top-5: 0.154).

Patch alignment strengthens. Both teacher-trained models preserve and tighten correspondence between the final embedding and pooled non-reg patches:

$$\begin{aligned} \text{pretrained: } & \mathbb{E} \left[\cos \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{patch}}^{(23)} \right) \right] = 0.9967, & \text{CKA} \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{patch}}^{(23)} \right) &= 0.9979, \\ \text{regr-norm: } & \mathbb{E} \left[\cos \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{patch}}^{(23)} \right) \right] = 0.9986, & \text{CKA} \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{patch}}^{(23)} \right) &= 0.9997, \\ \text{regr-brut: } & \mathbb{E} \left[\cos \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{patch}}^{(23)} \right) \right] = 0.9996, & \text{CKA} \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{patch}}^{(23)} \right) &= 0.9999. \end{aligned}$$

In contrast, teacher training suppresses $\mathbf{z}_{\text{full}} - \mathbf{z}_{\text{reg}}^{(23)}$ alignment:

$$\begin{aligned} \text{pretrained: } & \mathbb{E} \left[\cos \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{reg}}^{(23)} \right) \right] = 0.8443, & \text{CKA} \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{reg}}^{(23)} \right) &= 0.8881, \\ \text{regr-norm: } & \mathbb{E} \left[\cos \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{reg}}^{(23)} \right) \right] = 0.2526, & \text{CKA} \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{reg}}^{(23)} \right) &= 0.6005, \\ \text{regr-brut: } & \mathbb{E} \left[\cos \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{reg}}^{(23)} \right) \right] = 0.1961, & \text{CKA} \left(\mathbf{z}_{\text{full}}, \mathbf{z}_{\text{reg}}^{(23)} \right) &= 0.7374. \end{aligned}$$

Table 4: One-shot ImageNet top-5 (same protocol as §3.1). reg-only performance collapses after teacher fine-tuning; FULL and PATCH-only performance increase in these models:

Metric	pretrained	regr-norm	regr-brut
Top-5 (FULL)	0.445	0.602	0.538
Top-5 (PATCH-only)	0.448	0.603	0.538
Top-5 (REG-only)	0.169	0.009	0.015

Persistence of high-norm tokens. Crucially, this factorization is not explained by the disappearance of high-norm tokens. The reg-count remains nonzero for all images (zero fraction 0), with mean counts 2.44 (**pretrained**, range $[1, 6]$), 2.056 (**regr-norm**), and 1.747 (**regr-brut**). Token magnitudes remain sharply bimodal: the mean non-reg patch norm is 23.85 (**pretrained**), while reg-like token norms remain large (mean 179.53 for **pretrained**, and means ≈ 172 and ≈ 170 for **regr-norm** and **regr-brut**, respectively).

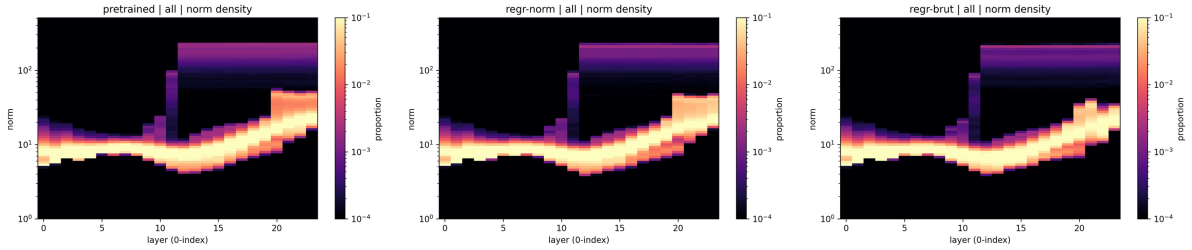


Figure 3: Token ℓ_2 -norm density across depth, $\|x_p^{(b)}\|_2$, for pretrained, **regr-norm**, **regr-brut**.

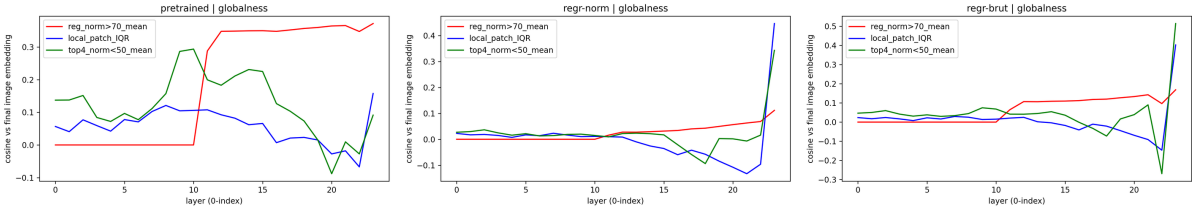


Figure 4: Patch “globalness” across depth for pretrained, **regr-norm**, and **regr-brut**, measured by $\cos(x_p^{(b)}, z_{\text{img}})$. Fine-tuning reduces alignment of patches with the final embedding, with the strongest suppression for register-like (high-norm) patches (red).

This supports a *geometric rerouting* interpretation: the model still forms a small high-norm subset, but its contribution to the final embedding is strongly attenuated.

Register counts across depth. Consistent with Fig. 3, the mean number of register-like tokens remains nonzero across representative mid/late blocks (Table 5).

Block	pretrained	regr-norm	regr-brut
12	3.620	3.130	2.695
13	3.620	3.130	2.695
20	3.625	3.085	2.695
21	3.640	3.115	2.695
22	3.645	3.115	2.710
23	3.695	3.135	2.745

Table 5: Mean number of register-like tokens per image (threshold $\tau_{\text{reg}} = 70$), MVT.

5.2 Depth-wise leakage localizes the transition to the terminal block

The blockwise leakage diagnostic (Section 10.8) reveals a depth-localized effect in teacher-trained models. For both **regr-norm** and **regr-brut**, REGpathways remain highly similar to FULL through most of the trunk (up to block 22), followed by an abrupt collapse at the final block 23.

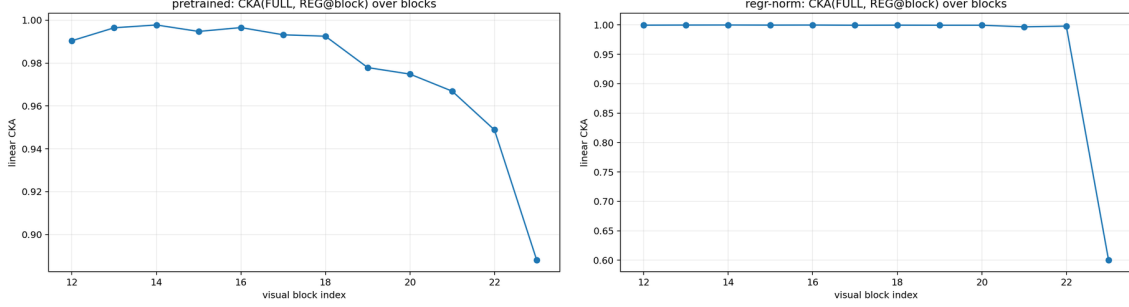


Figure 5: REG-leakage over blocks, linear CKA. Left, pretrained; right: regr-norm.

The **pretrained** model already shows a depth-dependent transition:

$$\text{pretrained b12: } \cos(\text{FULL}, \text{REG}) = 0.9931, \text{ CKA}(\text{FULL}, \text{REG}) = 0.9904,$$

compared to final:

$$\text{pretrained b23: } \cos(\text{FULL}, \text{REG}) = 0.8443, \text{ CKA}(\text{FULL}, \text{REG}) = 0.8881.$$

For the teacher-trained variants:

$$\text{regr-norm b12: } \cos(\text{FULL}, \text{REG}) = 0.9985, \text{ CKA}(\text{FULL}, \text{REG}) = 0.9996,$$

and similarly, in the penultimate block:

$$\text{regr-norm b22: } \cos(\text{FULL}, \text{REG}) = 0.9937, \text{ CKA}(\text{FULL}, \text{REG}) = 0.9980,$$

However, the diagnostic becomes sharply localized at the final block:

$$\text{regr-norm b23: } \cos(\text{FULL}, \text{REG}) = 0.2526, \text{ CKA}(\text{FULL}, \text{REG}) = 0.6005.$$

A parallel pattern holds for **regr-brut**, where **regr-brut** b22: $\cos(\text{FULL}, \text{REG}) \approx 0.995$ drops to **regr-brut** b23: $\cos(\text{FULL}, \text{REG}) = 0.196$. Meanwhile, $\cos(\text{FULL}, \text{PATCH})$ remains near 1.0 at every tested block in both teacher-trained models. For the full per-block results, refer to Table 16.

These results suggest a terminal-block intervention effect: internal reg-like signals persist, but their contribution to the terminal embedding is strongly reduced relative to pretrained, while non-reg patch readouts remain stable. One possible explanation is a learned terminal-block gating/projection that suppresses components not predictable from patch-derived features, consistent with the auxiliary teacher objective.

5.3 Attention faithfulness metrics shift consistently with rerouting

We report token- versus value-faithfulness diagnostics for the terminal-block attention computation (definitions in Section 10.5). In **pretrained**, token faithfulness is strongly negative on average (-0.394), whereas in **regr-norm** it becomes positive ($+0.150$), and in **regr-brut** it is near zero (-0.015). Value faithfulness remains high in all models, indicating that the value-path reconstruction matches the true attention output. The token-faithfulness shift (negative $\rightarrow$ near-zero/positive) is consistent with attention weights becoming a more reliable proxy for which tokens span the final embedding direction in the LN1-space diagnostic.

These metrics are diagnostic rather than definitive; however, the change in token faithfulness is consistent with a regime where the subset of tokens emphasized by attention more reliably spans the final embedding direction – at least in the scope of our LN1-space token diagnostic. For the full results table, we defer to Table 13.

5.4 Implications for typographic robustness and semantic performance

5.4.1 Robustness gains co-occur with suppression of reg-like shortcut directions

On SCAM-style typographic benchmarks, REGRESSION-REGTEACHER fine-tuning yields large gains relative to the pretrained baseline, and significant gains compared to the Phase 0 baseline GmP model (GmP acc: SCAM(0.6928), SynthSCAM(0.6790)):

Table 6: Typographic attack robustness, comparison across models. In the pretrained model, SynthSCAM is harder than SCAM; the fine-tuned models reduce this gap.

Dataset (n)	Method	pretrained	regr-norm	regr-brut
NoSCAM (1162)	CLS	0.9905	0.9897	0.9897
SCAM (1162)	CLS	0.4182	0.8038	0.8830
SynthSCAM (1162)	CLS	0.3219	0.8021	0.8804

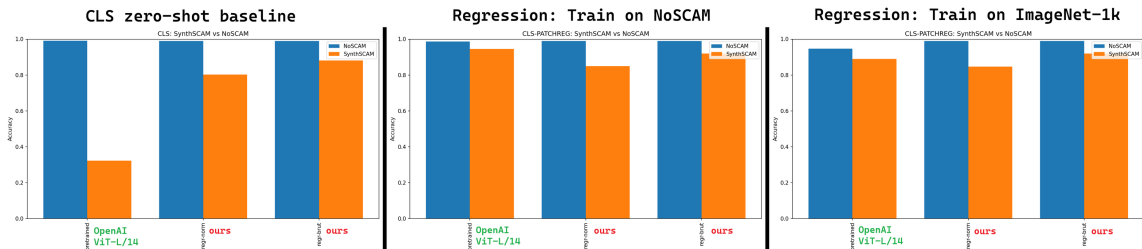


Figure 6: Zero-shot accuracies (left-to-right methods): CLS, CLS-PATCHREG-S (probe fit on NoSCAM), and CLS-PATCHREG-I (probe fit on ImageNet). Colors indicate evaluation split: NoSCAM (blue) and SynthSCAM (orange). Panels: pretrained (left), **regr-norm** (center), **regr-brut** (right). Notably, pretrained benefits most from the regression-probe embedding on SynthSCAM.

A natural geometric mechanism is consistent with these observations: typography can still induce high-norm, reg-like activations, but REGRESSION-REGTEACHER training enforces a terminal readout that is substantially less aligned with the pooled reg-like direction. This interpretation remains correlational: we establish a tight empirical coupling between (i) terminal-layer suppression of $\mathbf{z}_{\text{full}} - \mathbf{z}_{\text{reg}}^{(23)}$ alignment and (ii) improved typographic robustness, but do not claim that geometry alone proves causation.

5.4.2 Broader improvements in retrieval and zero-shot performance

Beyond typographic attack robustness, both teacher-trained models improve standard CLIP metrics, particularly cross-modal retrieval. On COCO2014 retrieval (R@5), **regr-norm** improves substantially over **pretrained** for both image-to-text and text-to-image, with **regr-brut** also improving (typically slightly less than **regr-norm**). Similar gains are observed on XM3600 cross-modal retrieval. On SugarCrepe, both REGRESSION-REGTEACHER models improve accuracy and margin statistics across perturbation types (e.g. **add_obj**, **replace_obj**, **swap_att**), indicating enhanced sensitivity to compositional caption perturbations. Zero-shot classification improves strongly on ImageNet-1k (acc@5) and on VOC-2007 multilabel zero-shot accuracy (mAP).

Table 7: **Excerpts from benchmarks (our code).** For full results, see Table 18 ff.

Metric	pretrained	regr-norm	regr-brut
COCO2014 Retrieval I2T (R@5)	0.788	0.871	0.850
COCO2014 Retrieval T2I (R@5)	0.590	0.785	0.782
SugarCrepe (add_obj) (acc)	0.7842	0.9627	0.9515
SugarCrepe (add_att) (acc)	0.7168	0.9205	0.8743

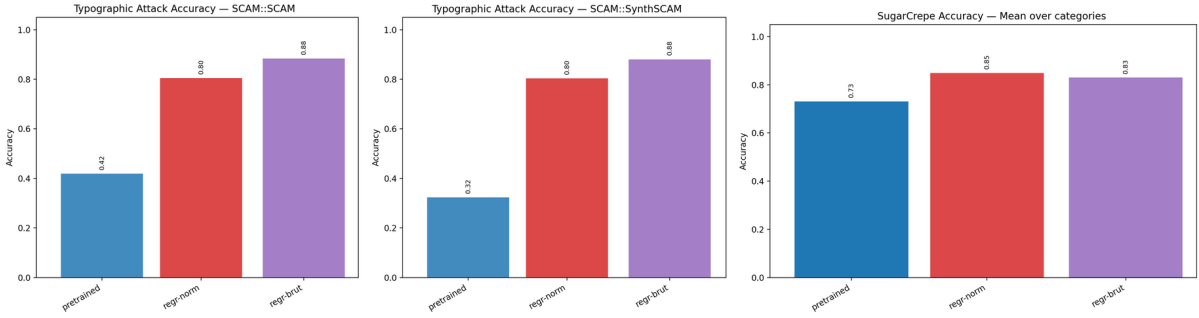


Figure 7: Accuracy across benchmarks (left-to-right): SCAM, SynthSCAM, and SugarCrepe (mean over categories). Colors: pretrained (blue), **regr-norm** (red), **regr-brut** (lavender).

Table 8: Benchmark evaluation: LAION CLIP Benchmark [2]: zero-shot and retrieval.

Benchmark	Metric	pretrained	regr-norm	regr-brut
ImageNet-1k (train)	Zero-shot acc@1	0.3270	0.4566	0.4100
ImageNet-1k (train)	Zero-shot acc@5	0.5300	0.6817	0.6513
ImageNet-1k (train)	Zero-shot mean per-class recall	0.3261	0.4547	0.4078
MSCOCO Captions	image retrieval R@5	0.2196	0.3510	0.3308
MSCOCO Captions	text retrieval R@5	0.3032	0.5042	0.4758
XM3600	image retrieval R@5	0.3059	0.4254	0.4138
XM3600	text retrieval R@5	0.2429	0.4091	0.3874

5.4.3 Trade-offs: Constraints can reduce generalization and linear separability

Our objective explicitly targets alignment under perturbation; this can reduce linear-probe separability on ImageNet while improving zero-shot and compositional robustness. Despite improved retrieval and zero-shot performance, the more aggressive teacher configuration (**regr-brut**) exhibits stronger degradations on certain measures:

Table 9: Teacher-dependent small (regr-norm) / large (regr-brut) performance impact on MVT and LP. For full results, see [Table 18](#) ff.

Benchmark	Metric	pretrained	regr-norm	regr-brut
ImageNet-1k (Linear Probe)	Top-1 (%)	72.35	70.94	65.09
ImageNet-1k (Linear Probe)	Top-5 (%)	93.42	93.29	89.60
MVT (200382)	ZS-acc	0.8830	0.8730	0.8573

These outcomes are consistent with a constraint-strength trade-off: Imposing stronger patch→CLS recoverability pressure can reduce representational degrees of freedom available for integrating unusual local evidence into a stable global representation, which may be important for hard viewpoint recognition. We will explore the phenomenon in detail (e.g. anisotropy) in the next section [Section 6.0.1](#).

5.5 Conclusion: Regression-Teacher Fine-Tuning

REGRESSION-REGTEACHER fine-tuning induces a distinct, measurable reorganization of CLIP ViT token geometry. The most distinctive effect of REGRESSION-REGTEACHER fine-tuning is the terminal localization of reg-only leakage: reg-like signals persist throughout the trunk yet decouple sharply at the final block, coinciding with large robustness gains.

6 Excursion: Attention and Register-Subspace Geometry

Building on the results in Sec. 5, we collect complementary probes that sharpen *where* and *how* late-depth rerouting arises.

This section collects complementary probes that sharpen *where* and *how* this behavior arises. Rather than a single pipeline, we use three views that cross-check one another: (a) causal subspace edits inside the vision tower, (b) patch-space geometry relative to a fixed reference “register basis,” and (c) attention-routing diagnostics that explicitly separate *attention weights* from *effective contribution*. The latter follows the caution of Lappe & Giese that attention maps alone need not faithfully explain global representations when the value pathway (and output projection) can be reparameterized independently [7].

6.0.1 Model geometry analysis

We report cone concentration (cone_R, κ), anisotropy (sphericity, effective rank), and cross-modal subspace mismatch (principal angles); all metrics are defined in section B. CLIP exhibits strongly anisotropic embedding geometry rather than a uniformly hyperspherical one: both modalities concentrate in pronounced cones (large cone_R / κ) with low sphericity and reduced effective rank relative to the ambient dimension, and the dominant image/text variance subspaces are substantially misaligned (mean principal angles $\sim 60^\circ$ at $k=64$).

Metric	pretrained	regr-norm	regr-brut
I: cone_R	0.748186	0.451401	0.379880
I: sphericity	0.0979	0.0339	0.0190
I: eff_rank (entropy)	178.45	120.70	79.04
T: cone_R	0.590802	0.448189	0.283353
T: sphericity	0.0659	0.0629	0.0728
T: eff_rank (entropy)	146.79	154.00	178.68
IT: principal_angles mean (deg)	60.08	42.09	51.91

This yields a geometric account of the modality gap via mean-direction mismatch, covariance-shape mismatch, and a similarity-structure mismatch in which within-modality similarities (II/TT) exceed paired IT similarities (table 21).

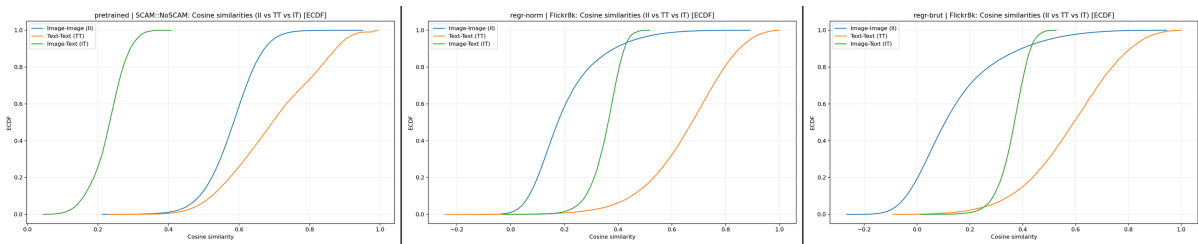


Figure 8: Flickr8k empirical CDFs of cosine similarities $\cos(\cdot, \cdot)$ for image-image (II, blue), text-text (TT, orange), and paired image-text (IT, green). Left-to-right: pretrained, **regr-norm**, **regr-brut**. Fine-tuning shifts the paired IT distribution toward higher similarity.

The regression fine-tunes move the joint system into a less modality-separated regime—paired IT similarity increases and distributional gap metrics shrink—yet they realize this via distinct geometric trade-offs: regr-norm most strongly reduces cross-modal subspace angles (more shared dominant subspace), whereas regr-brut more aggressively compresses image-side effective dimensionality while increasing text-side spread.

To make these distributional shifts visually interpretable, [figure 9](#) shows a PCA projection of pooled text and image embeddings. While PCA is only a low-dimensional slice of a high-dimensional structure, it helps distinguish whether changes reflect a near-rigid cross-modal re-centering versus a reshaping into multi-cluster structure. In pretrained, text embeddings form an elongated, curved manifold with comparatively compact image embeddings. In contrast, both fine-tunes exhibit more symmetric, multi-cluster geometry in *both* modalities, suggesting a change in coarse-scale structure (e.g., increased multimodality under this projection), rather than a pure mean re-centering.

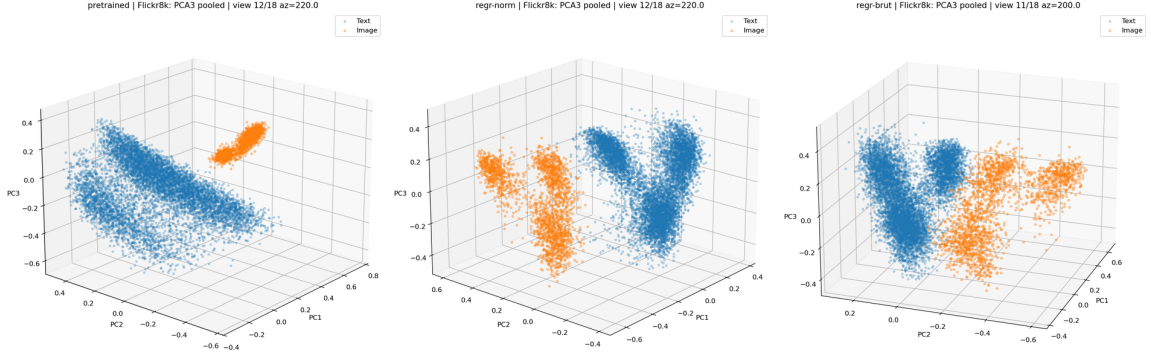


Figure 9: Flickr8k PCA of pooled embeddings in a shared PCA basis. Text embeddings (blue) and image embeddings (orange); left-to-right: pretrained, **regr-norm**, **regr-brut**.

On SCAM, switching NoSCAM→SynthSCAM produces only modest global second-order geometric shifts—text-side statistics are nearly invariant and image-side changes remain small even in pretrained—despite catastrophic behavioral failure in pretrained under typographic attack (32% vs. ~99% on NoSCAM). This dissociation is consistent with a conditional, high-leverage mechanism (i.e., behavior can hinge on a small number of directions not reflected in bulk second-order shifts), rather than implying a wholesale distribution shift ([table 22](#)).

6.1 Causal recoverability via subspace surgery

We test where the final image embedding draws its support by intervening at transformer block b and editing only the *patch-token* states. Let X_{patch} denote the patch-token matrix at block b , let U_{reg} be a fixed reference basis (constructed once from the pretrained model). We replace patch states with either (a) their projection onto U_{reg} (“reg-only”) or (b) the orthogonal complement (“nonreg-only”):

$$X_{\text{patch}} \leftarrow \Pi_{U_{\text{reg}}}(X_{\text{patch}}) \quad \text{or} \quad X_{\text{patch}} \leftarrow X_{\text{patch}} - \Pi_{U_{\text{reg}}}(X_{\text{patch}}),$$

leaving the class token unchanged. We then measure cosine similarity between the post-intervention embedding and the unmodified baseline embedding,

$$\cos\left(f_{\theta}^{(b, \text{mode})}(I), f_{\theta}(I)\right),$$

so preservation (or destruction) under the edit provides a depth-local, causal probe of whether downstream embedding computation depends on the removed subspace components at that depth.

Findings I. Recoverability is strongly depth-dependent. In early blocks ($b \leq 8$), NONREG-ONLY is nearly lossless (all models ≥ 0.99), whereas REG-ONLY preserves the embedding only partially (pretrained ~ 0.50 – 0.63 ; fine-tunes ~ 0.18 – 0.31), indicating that early embedding-relevant signal is carried predominantly by patch components not aligned to the reference reg basis. From mid-depth onward ($b \approx 9$ – 21), NONREG-ONLY degrades while REG-ONLY increases, consistent with progressively stronger reliance on reference-reg-aligned directions. In the final blocks ($b \geq 22$), both edits become highly preserving, indicating that this particular U_{reg} vs. $U_{\text{reg}}^\perp$ decomposition is not strongly diagnostic at late depth (e.g., due to redundancy or mixing).

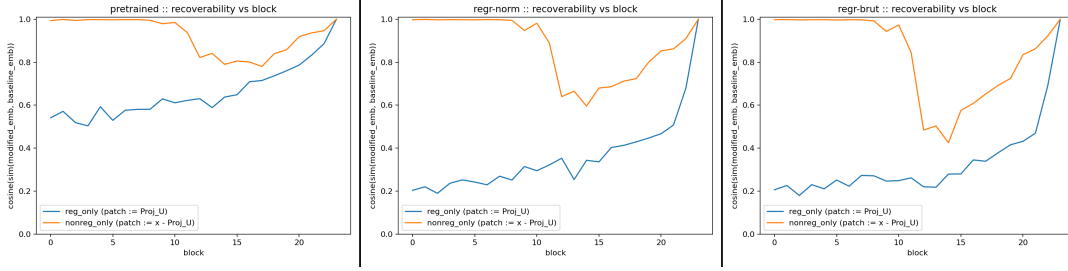


Figure 10: Recoverability under patch-token subspace surgery, projection $\Pi_{U_{\text{reg}}}(X_{\text{patch}})$. Left-to-right: pretrained, **regr-norm**, **regr-brut**.

6.2 Register-projection geometry in patch space

To characterize register-like directions in patch-token space, we measure per block: (i) the projected energy fraction onto U_{reg} for each patch token $x_{i,p}^{(b)}$, averaged over (i,p) ; and (ii) principal-angle geometry between the patch PCA subspace and U_{reg} (computed from PCA bases, $k=5$).

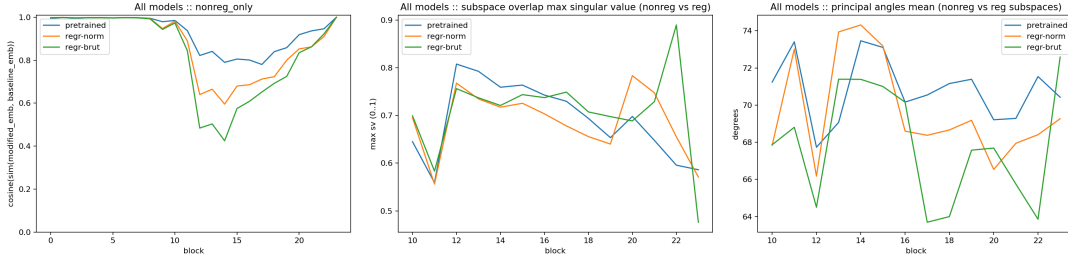


Figure 11: *Left*: NONREG-ONLY recoverability (mean cosine similarity to the baseline embedding). *Center*: best-case subspace overlap $\sigma_{\max} = \cos(\theta_{\min})$ between nonreg and reg subspaces. *Right*: mean principal angle $\frac{1}{k} \sum_{i=1}^k \theta_i$ between the nonreg and reg subspaces (here $k=5$).

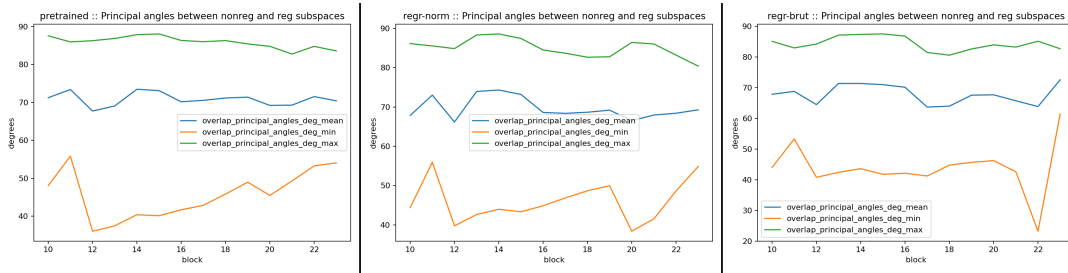


Figure 12: Per-model principal-angle statistics between nonreg and reg subspaces (min/mean/max over the $k=5$ angles). **regr-brut** exhibits a minimum-angle dip to $\sim 25^\circ$ at block 22, consistent with the $\sigma_{\max}$ spike in Fig. 11.

Findings II. Across models, the projected energy fraction onto the pretrained reference basis is modest in early blocks ($\text{fracMean} \approx 0.10\text{--}0.14$), dips around $b \approx 7\text{--}9$, and rises substantially at mid-late depth ($b \approx 11\text{--}16$), with the strongest increase in **regr-brut** (peak $\text{fracMean} \sim 0.25$). In contrast, overlap summaries based on subspace angles are comparatively stable: mean angles remain large ($\sim 64\text{--}74^\circ$) with only localized best-case alignment events, notably the regr-brut minimum-angle dip at block 22. Together, these patterns suggest that fine-tuning can increase the *amount* of patch energy aligned with the pretrained register basis (without implying global reg/nonreg collapse), while allowing occasional late-block coupling between the two subspaces along a small number of directions.

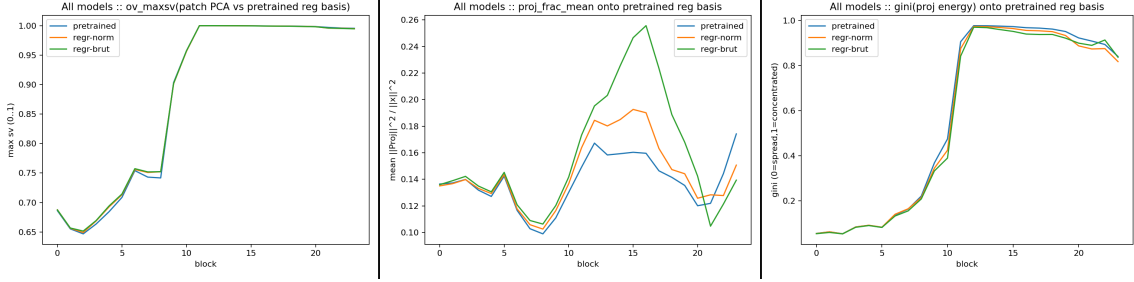


Figure 13: Patch-space alignment to the *pretrained* register basis. *Left*: best-case overlap $\sigma_{\max}$ between nonreg and reg subspaces. *Center*: projected energy fraction onto the pretrained register basis, $\|\Pi_{U(0)} x\|_2^2 / \|x\|_2^2$. *Right*: Gini coefficient of projection energy across patches.

6.3 Cross-image consistency and late-depth coupling events

In (**regr-brut**), cross-image detection yields a localized increase in best-case overlap between the detected reg and nonreg subspaces ($\text{ovMaxSV} = 0.890$ at block 22; ≈ 0.596 pretrained; 0.655 regr-norm), while mean principal angles remain high. Thus, the evidence supports an episodic late-block coupling rather than a global reg/nonreg subspace collapse.

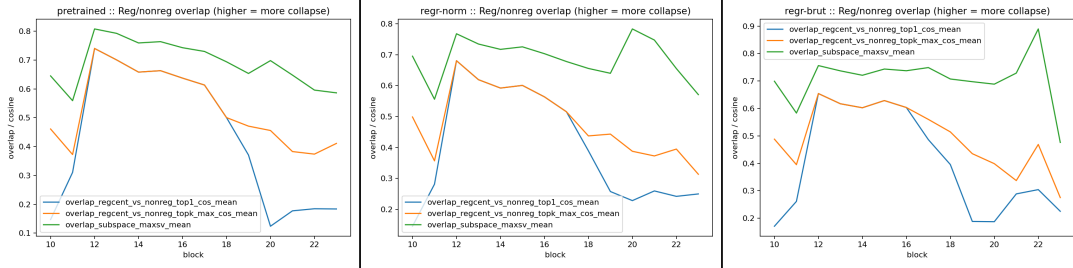


Figure 14: Cross-image overlap diagnostics (reg vs. nonreg): reg-centroid vs. nonreg similarity (top-1 / top- k) and subspace overlap $\sigma_{\max}$. Left-to-right: pretrained, **regr-norm**, **regr-brut**.

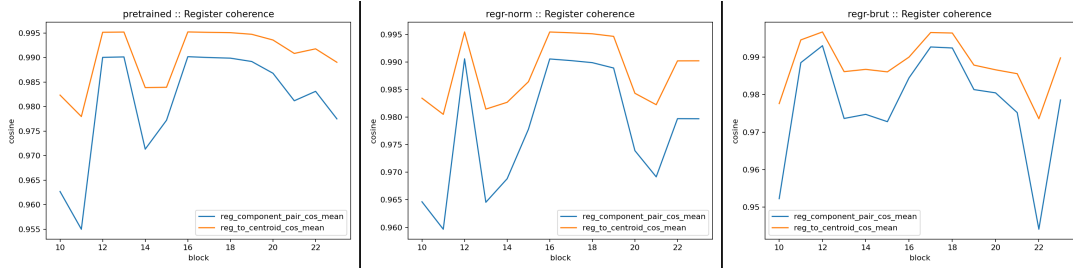


Figure 15: Cross-image register coherence. Orange: register-to-centroid cosine similarity. Blue: within-component pairwise cosine similarity. Left-to-right: pretrained, **regr-norm**, **regr-brut**.

6.4 Attention routing diagnostics: entropy, norms, and effective contribution

Simple per-head ℓ_2 norm trends provide a coarse but useful “anomaly detector”: they reveal which heads undergo depth-localized changes under fine-tuning, and therefore which heads merit detailed routing and contribution analysis. Figure 20 summarizes per-head mean ℓ_2 norms across blocks for the pretrained model, and deviations Δ for both fine-tunes. Notably, a single head (Head 10; blue) exhibits an isolated mid-depth spike in the pretrained model (block 12) and a pronounced, selective suppression in both fine-tunes (stronger for **regr-brut**), while most heads remain near baseline until late blocks. We treat this as a signal: a head-specific routing mechanism is being modified, which we dissect in the following subsections using token-resolved attention (attention weights) and effective contribution.

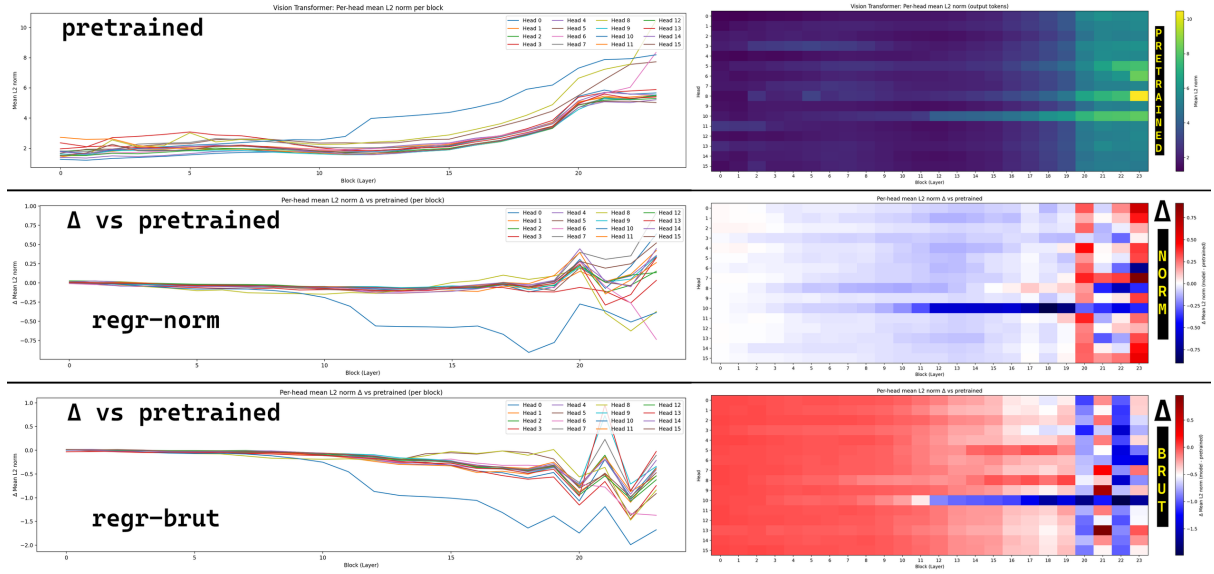


Figure 16: Per-head ℓ_2 -norm overview reveals a head-specific fine-tuning signature. *Top*: pretrained per-head mean ℓ_2 norm across blocks. *Middle/Bottom*: change relative to pretrained, $\Delta(\cdot)$, for **regr-norm** and **regr-brut**. Head 10 (blue) shows (i) an isolated pretrained spike around block ≈ 12 and (ii) selective norm suppression under both fine-tunes (stronger for **regr-brut**).

6.5 Register-focused attention vs. attenuated value-path transmission

The terminal decoupling above does not require eliminating register-attraction in *attention weights* A ; instead, it is compatible with suppressing the *value-path transmission* of register content into the head outputs. In addition, the substantially larger Attn/skip ratio in the fine-tunes suggests altered reliance on explicit attention pathways relative to skip-dominated propagation, without implying that attention weights alone identify the causally relevant carriers.

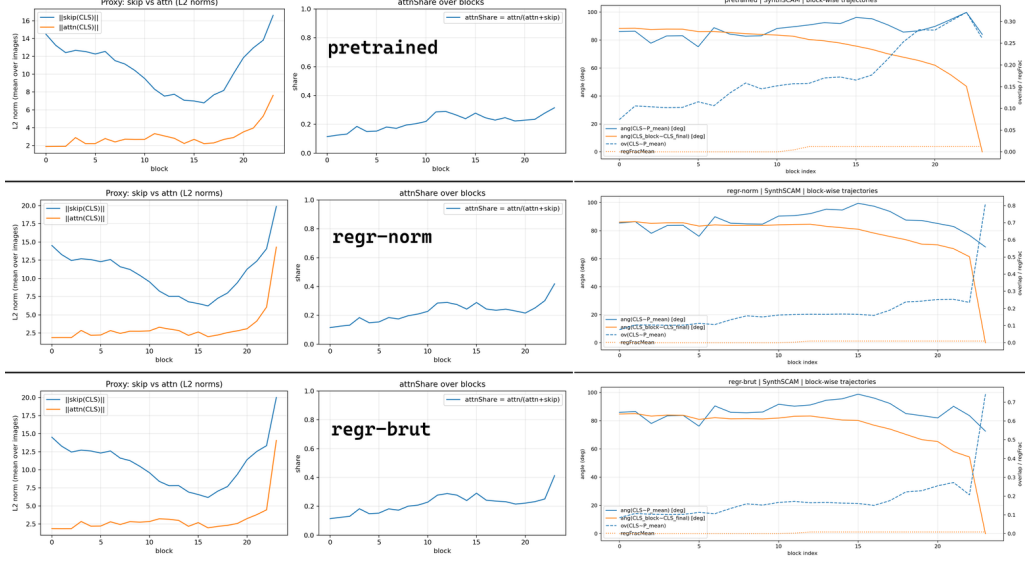


Figure 17: **CLS readout geometry across depth.** *Left:* ℓ_2 norm of the residual/skip contribution to CLS (blue) versus the attention write to CLS (orange). *Center:* attention contribution ratio $\|w_{\text{attn}}\|_2 / (\|w_{\text{attn}}\|_2 + \|w_{\text{skip}}\|_2)$. *Right:* angles (degrees) $\angle(\text{CLS}, \mu_P)$ (blue, solid) and $\angle(\text{CLS}^{(b)}, \text{CLS}^{(23)})$ (orange, solid), overlap $\cos(\text{CLS}, \mu_P)$ (blue, dashed), and mean register fraction. Fine-tunes: Attention share of the CLS update > 0.4 vs. ~ 0.35 pretrained.

We therefore interpret the qualitative signature as *persistent register-focused attention with attenuated value-path transmission*: certain heads may continue to assign stable attention mass to register-like indices (a $QK^\top$ -level attractor), while the corresponding attention-weighted values and output projections contribute little along the terminal embedding direction (a V /out-proj level suppression).

Figure 18 provides an visual example of the phenomenon: raw attention weights can remain sharply structured even when the corresponding attention-weighted value pathway yields weak or redistributed magnitude, illustrating the need to separate A from AV -level influence.

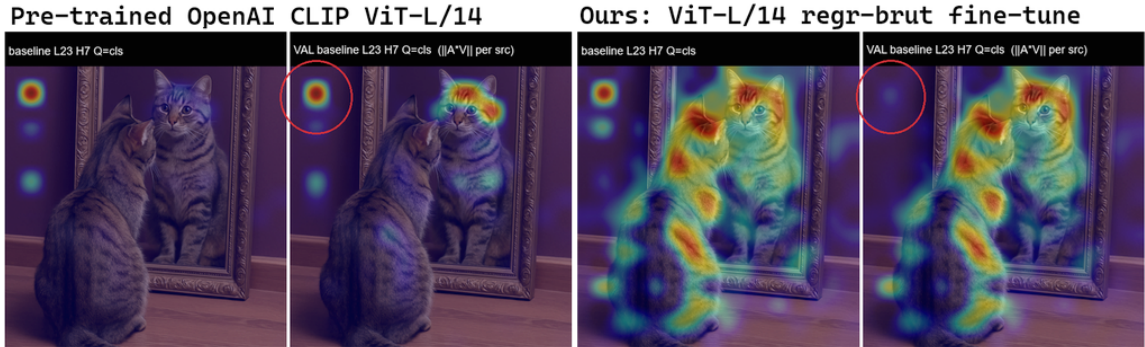


Figure 18: Raw attention diagnostics (block 23, head 7; query token $Q = \text{CLS}$). *Left:* attention weights A . *Right:* $\|A * V\|$. (per key position).

Complementarily, gradient-based rollout summarizes how attention-mediated pathways aggregate across heads and layers. Figure 19 shows the divergent qualitative attention heatmap patterns in pretrained vs. regr-brut.



Figure 19: Gradient-based attention rollout in the final ViT block for text queries “a coffee”, “a cup”, and “a mouth”. Left: pretrained, showing peak attention on background register-like regions. Right: **regr-brut**, with a more on-target attention distribution.

6.6 Attention–contribution decoupling

MLP→K routing and attention–contribution decoupling. Consider a CLIP-like ViT with blocks $\ell = 0, \dots, L - 1$, embedding width d , and H attention heads (head dimension D , $d = HD$). In block ℓ , the MLP projects back to the residual stream via $W_c^{(\ell)} \in \mathbb{R}^{d \times m}$ (`mlp.c_proj.weight`); in the next block $\ell + 1$, the attention key projection is $W_k^{(\ell+1)} \in \mathbb{R}^{d \times d}$ (`attn.k_proj.weight`). Writing the head slice as $W_{k,h}^{(\ell+1)} \in \mathbb{R}^{D \times d}$, we score how strongly MLP feature $f \in \{1, \dots, m\}$ routes into head h ’s keys via

$$S_{h,f}^{(\ell)} = \|W_{k,h}^{(\ell+1)} W_c^{(\ell)}[:, f]\|_2, \quad S^{(\ell)} \in \mathbb{R}^{H \times m}. \quad (16)$$

We summarize routing using per-head quantiles of $S_{h,\cdot}^{(\ell)}$ (details in [section A](#)).

For attention–contribution decoupling, let a_h denote the standard CLS→patch attention distribution for head h at layer ℓ (renormalized over patches). We define an effective-contribution proxy by value-norm weighting:

$$c_{h,p}^{\text{raw}} = a_{h,p} \|V^{(\ell,h)}[p]\|_2, \quad c_h = \frac{c_h^{\text{raw}}}{\sum_p c_{h,p}^{\text{raw}}}.$$

We report the decoupling score $\text{KL}(a_h \| c_h)$ and the entropies $H(a_h), H(c_h)$, together with register-mass summaries based on high-norm patch tokens (register proxy). Full definitions and implementation details are given in [section A](#).

Attention entropy. Let $A^{(\ell,h)}(q, \cdot)$ denote the attention distribution over keys at layer ℓ , head h , for query token q . We report attention entropy

$$H\left(A^{(\ell,h)}(q, \cdot)\right) = - \sum_{i=1}^T A^{(\ell,h)}(q, i) \log A^{(\ell,h)}(q, i),$$

where $T \approx 257$ for ViT-L/14. Entropy near $\log T \approx 5.55$ indicates diffuse mixing; low entropy indicates more concentrated (“pointer-like”) routing.

To test whether attention *weights* reflect information flow, we compute an *effective contribution* distribution $\tilde{c}$ per head from attention-weighted values, and summarize it via (i) contribution entropy $H(\tilde{c})$, (ii) top- k /tail mass statistics, and (iii) mismatch between attention and contribution (e.g. $D_{\text{KL}}(A \| \tilde{c})$). This follows the decoupling premise that high attention mass need not imply high write impact if value/output pathways suppress or rotate content. [7] Finally, we track ℓ_2 norms of projected $Q/K/V$ vectors ($\|Q\|, \|K\|, \|V\|$) for (i) all tokens and (ii) CLS-query only.

6.7 Findings: Attention–contribution

We evaluate these diagnostics on $N=236$ typographic adversarial images from our dataset [15] and compare **pretrained**, **regr-norm**, and **regr-brut**. Tables [tables 30, 32, 34, 36, 38, 40](#) and [42](#) report, per block: (i) $\text{KL}(\text{attn} \parallel \text{contrib})$, (ii) $\Delta\text{reg} = \text{reg}_{\text{contrib}} - \text{reg}_{\text{attn}}$, (iii) $\Delta H = H_{\text{contrib}} - H_{\text{attn}}$ (summarized via absolute means), and (iv) $\text{corr}(x, y)$ between attention weights and effective contribution. As register indices are defined per image from the unaltered terminal-layer norms, we propagate these indices to earlier blocks to track the same patch locations across depth.

Early and mid depth: stable outlier identities (Blocks 0 and 10). At block 0, the same heads dominate mismatch across models: H11 maximizes KL, while H10 maximizes both $|\Delta\text{reg}|$ and $|\Delta H|$ (Tab. [table 30](#)). Fine-tuning slightly lowers the *average* mismatch at this depth (mean KL $0.0453 \rightarrow 0.0391/0.0407$), while leaving the outlier identities unchanged (Tab. [table 30](#)).

At block 10, a different shared outlier emerges: H01 is maximal in KL, $|\Delta\text{reg}|$, and $|\Delta H|$ for all three models, with relatively small changes in mean KL (≈ 0.02) but a larger increase in mean $|\Delta H|$ for the fine-tunes (Tab. [table 32](#)). Together, blocks 0 and 10 indicate that fine-tuning can modulate mismatch magnitude without necessarily changing which heads are most decoupled.

A mid-depth “routing bottleneck” (Block 12) with modest mitigation. Block 12 shows a distinct regime with higher average mismatch than early/mid layers (mean KL ≈ 0.10) and large entropy shifts ($\max |\Delta H| \approx 1.26\text{--}1.31$), with the same maximal-KL head (H14) across models (Tab. [table 34](#)). Unlike block 10, the fine-tunes slightly reduce mean KL and mean $|\Delta\text{reg}|$ at this block ($0.0994 \rightarrow 0.0954 \rightarrow 0.0924$; $0.00607 \rightarrow 0.00565 \rightarrow 0.00529$), suggesting partial mitigation while preserving the dominant head identity (Tab. [table 34](#)). Of note, this is the block in which Head 10 out_proj norm spikes, as seen in Figure [20](#).

Late depth: global mismatch inflation (Blocks 18–20). Mismatch becomes substantially larger in late depth. Block 18 has the largest mean KL in the reported set ($0.508 \rightarrow 0.532 \rightarrow 0.567$), accompanied by large mean $|\Delta H|$ and mean $|\Delta\text{reg}|$ (Tab. [table 36](#)). Block 20 remains in a high-mismatch regime (mean KL $0.328\text{--}0.424$) with large entropy shifts in the fine-tunes (Tab. [table 38](#)). In this late-depth region, fine-tuning does not uniformly reduce decoupling; instead, it can coincide with increased average mismatch (Tabs. [tables 36](#) and [38](#)).

(a) Block 10.				(b) Block 18.			
	pretrained	regr-norm	regr-brut		pretrained	regr-norm	regr-brut
max KL (H, val.)	H01, 0.065	H01, 0.066	H01, 0.071	max KL (H, val.)	H07, 0.683	H09, 0.739	H09, 0.759
max $ \Delta\text{reg} $ (H, val.)	H01, 0.010	H01, 0.009	H01, 0.009	max $ \Delta\text{reg} $ (H, val.)	H15, 0.063	H15, 0.076	H15, 0.088
max $ \Delta H $ (H, val.)	H01, 0.146	H01, 0.186	H01, 0.235	max $ \Delta H $ (H, val.)	H07, 1.751	H07, 2.039	H07, 2.248
mean KL	0.020	0.020	0.021	mean KL	0.508	0.532	0.567
mean $ \Delta\text{reg} $	0.003	0.003	0.003	mean $ \Delta\text{reg} $	0.035	0.038	0.044
mean $ \Delta H $	0.049	0.054	0.060	mean $ \Delta H $	1.081	1.343	1.568

Table 10: Attention routing diagnostics overview Block 10, Block 18.

Penultimate block (22): divergence in register involvement. At block 22, pretrained and regr-norm show near-zero *average* register-mass discrepancy (mean $|\Delta\text{reg}| = 3 \times 10^{-5}$ and 2.2×10^{-4}), whereas regr-brut is substantially larger (0.0135; Tab. [table 40](#)). The per-head table clarifies that regr-brut’s register effect is *heterogeneous*: Δreg spans both signs across heads (11/16 positive, 5/16 negative; Tab. [table 41](#)), while pretrained is mostly negative/near-zero and regr-norm is mostly positive but very small in magnitude (Tab. [table 41](#)). Entropy shifts at block 22 are predominantly positive across models (15/16 heads in pretrained; 16/16 in both fine-tunes; Tab. [table 41](#)), consistent with effective contribution being more diffuse than raw

attention at this depth, even as its *magnitude* increases in the fine-tunes (mean $|\Delta H| \approx 0.9$ vs. 0.135 pretrained; Tab. [table 40](#)).

Final block (23): attention–contribution consistency and sign inversion. At the terminal block 23, fine-tuning reduces average attention–contribution divergence (mean KL 0.273 $\rightarrow$ 0.109 $\rightarrow$ 0.087), while the maximal-KL head remains H14 in all three models (Tab. [table 42](#)). Unlike the summary (which reports absolute means), the per-head table supports a *sign-structure* change in ΔH : ΔH is predominantly positive in pretrained (15/16 heads) but predominantly negative in regr-norm (11/16) and regr-brut (12/16; Tab. [table 43](#)), indicating a shift from contribution being typically *more diffuse* than attention to being typically *more peaked* for most heads. Register-mass shifts at block 23 also become highly structured: Δreg is overwhelmingly negative across heads in both fine-tunes (15/16 negative), with a single exception (H13) that is positive and also attains the maximal $|\Delta \text{reg}|$ in both fine-tunes (Tab. [tables 42](#) and [43](#)); in pretrained, H13 is slightly negative (Tab. [table 43](#)).

(a) Block 22 excerpt.									(b) Block 23 excerpt.										
H	pretrained			regr-norm			regr-brut			H	pretrained			regr-norm			regr-brut		
	d_ent	KL	d_reg	d_ent	KL	d_reg	d_ent	KL	d_reg		d_ent	KL	d_reg	d_ent	KL	d_reg	d_ent	KL	d_reg
0	0.090	0.253	-0.000	1.204	0.309	0.000	1.363	0.354	0.009	14	0.220	0.533	0.000	0.069	0.356	-0.001	0.178	0.278	-0.003
3	0.118	0.143	-0.000	0.982	0.235	0.000	0.732	0.206	0.040	13	-0.035	0.057	-0.001	0.057	0.053	0.012	0.134	0.074	0.018
11	0.033	0.240	-0.000	1.157	0.311	0.000	1.172	0.360	0.009	11	0.241	0.269	0.000	-0.276	0.070	-0.000	-0.182	0.089	-0.001

Table 11: Attention routing diagnostics overview Block 22, Block 23.

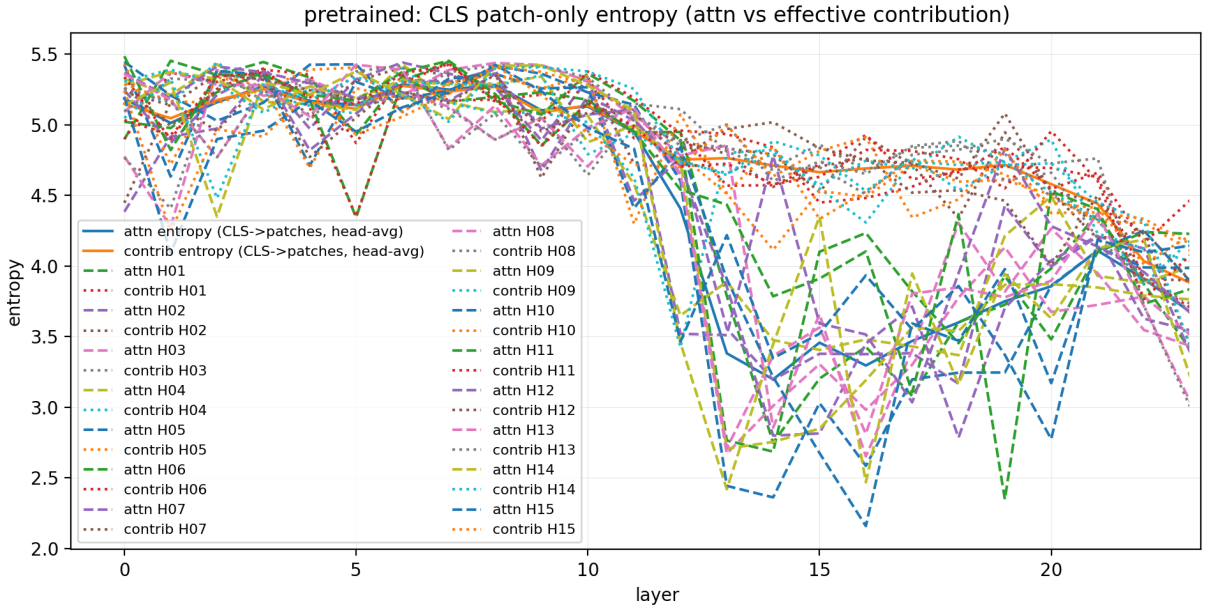


Figure 20: Pretrained CLIP ViT-L/14: CLS patch-only entropy comparison between attention weights and effective contribution (value-path transmission).

6.8 Registers and head-wise register routing

Using the same $N=236$ typographic attack adversarial images as above, we analyze how different CLIP ViT-L/14 variants route information from (*emergent registers*) into the final image embedding. We run identical forward-hook instrumentation on three models: (i) **pretrained** (ViT-L/14), (ii) **regr-norm**, and (iii) **regr-brut**. We evaluate a dataset of attacked images over a selected set of transformer blocks $\mathcal{B}$ ($\mathcal{B} = \{10, \dots, 23\}$). Each block has $H = 16$ attention heads. We define registers by their previously defined L2 norm $\tau_{\text{reg}}=70$ and quantify (i) how strongly each head *attends to* these tokens, (ii) how strongly each head *writes from* them into width space, and (iii) whether that write *survives* to the final CLIP image embedding.

Method: Registers and head-wise register routing.

Register mask. Let the post-block token stream be $X^{(b)} \in \mathbb{R}^{T \times B \times C}$ with $T = 1 + P$ (CLS + P patches). For sample i and patch p , define a patch norm

$$n_{i,p}^{(b)} = \|X_{\text{patch}}^{(b)}[i, p, :]\|_2.$$

We mark registers by thresholding and restricting to a sparse set:

$$m_{i,p}^{(b)} = \mathbf{1}[n_{i,p}^{(b)} > \tau] \quad \text{then keep at most } K \text{ patches (and ensure } K_{\min} \geq 1).$$

We lift this patch mask to token indices $s \in \{0, \dots, P\}$ by setting $r_{i,0}^{(b)} = 0$ (CLS is never a register) and $r_{i,1+p}^{(b)} = m_{i,p}^{(b)}$.

REG attention mass and REG write norm. For head h at block b , let $\alpha_{i,\text{cls} \rightarrow s}^{(b,h)}$ be the attention probability from the CLS query to source token s . The *REG attention mass* is

$$A^{(b,h)} = \mathbb{E}_i \left[\sum_{s=0}^P \alpha_{i,\text{cls} \rightarrow s}^{(b,h)} r_{i,s}^{(b)} \right],$$

i.e., how much probability mass that head assigns to register-marked sources.

Let $v_{i,s}^{(b,h)} \in \mathbb{R}^d$ be the value vector at source s (head dimension d). The *register-restricted* head output for CLS is

$$o_{i,\text{reg}}^{(b,h)} = \sum_{s=0}^P \alpha_{i,\text{cls} \rightarrow s}^{(b,h)} r_{i,s}^{(b)} v_{i,s}^{(b,h)} \in \mathbb{R}^d.$$

With head-slice output projection $W_{O,h}^{(b)} \in \mathbb{R}^{C \times d}$, the corresponding *width-space write* is

$$w_i^{(b,h)} = W_{O,h}^{(b)} o_{i,\text{reg}}^{(b,h)} \in \mathbb{R}^C, \quad W^{(b,h)} = \mathbb{E}_i [\|w_i^{(b,h)}\|_2].$$

Intuition: $A^{(b,h)}$ measures *where attention looks* (probability mass), while $W^{(b,h)}$ measures *how much signal is actually written* from register sources.

Survival to the final embedding. To test whether a head’s register write at block b propagates to the final CLIP image embedding, we inject $w_i^{(b,h)}$ into the CLS token at that block and run the remainder of the network $F_{>b}(\cdot)$:

$$\tilde{x}_{i,\text{cls}}^{(b,h)} = x_{i,\text{cls}}^{(b)} + w_i^{(b,h)}, \quad e_i = F_{>b}(X_i^{(b)}), \quad \tilde{e}_i^{(h)} = F_{>b}(\tilde{X}_i^{(b,h)}).$$

The *survival score* is the mean embedding displacement:

$$S^{(b,h)} = \mathbb{E}_i [\|\tilde{e}_i^{(h)} - e_i\|_2].$$

For full details, we defer to App. A.9.

6.9 Findings

Across all three models, late blocks exhibit a highly non-uniform distribution of (i) attention mass allocated to high-norm REG-like tokens and (ii) the norm of the value-path REG write. Rather than reflecting diffuse, weak effects spread across heads, the REG channel is repeatedly dominated by a small subset of heads, and fine-tuning primarily manifests as a partial reassignment and consolidation of which heads carry this channel.

Table 12: REG-routing changes under TypoAttack (n=236): top heads and shifts.

Finding	Block/Metric	pretrained	regr-norm	regr-brut
Persistent REG-writer	b18: top $\ w_{\text{REG}}\ $	h12=0.3546	h12=0.3395	h12=0.4027
REG-attn dominance	b18: top m_{REG}	h7=0.6346	h7=0.7255	h7=0.7642
Late routing hub (write)	b22: top $\ w_{\text{REG}}\ $	h7=0.1411	h10=0.4807	h10=0.2479
Late routing hub (attn)	b22: top m_{REG}	h7=0.2051	h11=0.4594	h0=0.3883
Fine-tune “gating” head	b21: $\Delta\ w_{\text{REG}}\ $ vs pre	–	h4:+0.5132	h4:+0.6979
Pruned late spray	b23: Δm_{REG} vs pre	–	(shift to h10/h0)	h1:-0.2491, h8:-0.2489
Survival-stable routing	b22: top write-survival	h7=0.1023	h10=0.2430	h0=0.1401

A persistent mid-late REG-writer emerges across models. At Block 18, the same head (Head 12) is the dominant contributor to the REG write norm in the pretrained model as well as both fine-tuned variants. This stability indicates that the REG-writing pathway is not an incidental artifact of a specific checkpoint, but reflects a reproducible mechanism that remains available after fine-tuning. Notably, fine-tuning does not monotonically reduce attention–contribution decoupling across depth: mismatch can increase in late-mid blocks (e.g., 18–20) while decreasing sharply at the terminal block (23), consistent with a depth-localized reconfiguration of the final readout.

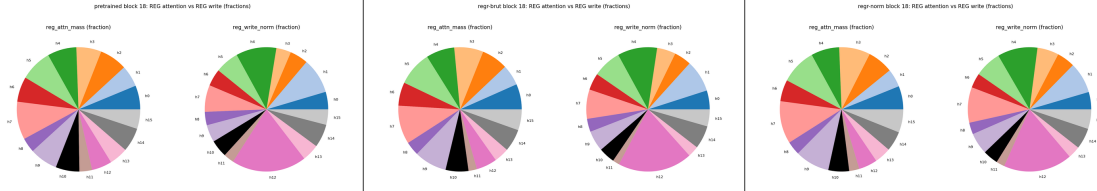


Figure 21: Block 18: per-head fractions of register attention mass and register write norm. Left-to-right: pretrained, **regr-norm**, **regr-brut**. Head 12 has the largest write-norm fraction in all three models.

A gating head dominates downstream-relevant REG writes in fine-tuned models. Inter-model deltas further reveal a striking increase for Head 4 at Block 21 in both fine-tuned models, including the largest observed increases in REG write norm as well as a strong increase in survival-weighted REG writing. Because survival-weighted scores emphasize writes that remain influential in subsequent blocks, this pattern indicates not only larger writes, but writes that persist and remain functionally relevant downstream. This is consistent with the emergence of a late-block control point (a gating or routing head) through which the fine-tuned models alter the propagation of REG-aligned information.

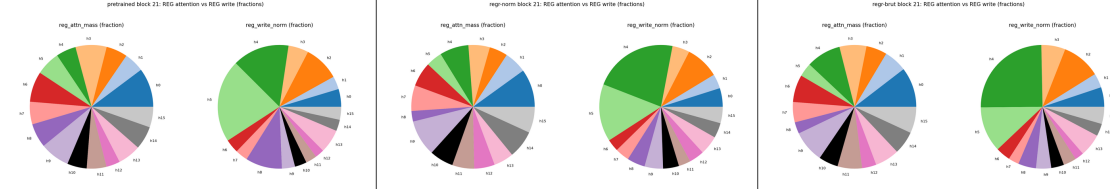


Figure 22: Block 21: per-head fractions of register attention mass and register write norm. Left-to-right: pretrained, **reg-norm**, **reg-brut**. Under fine-tuning, Head 4 exhibits an inflated write-norm fraction relative to pretrained (where Head 5 dominates).

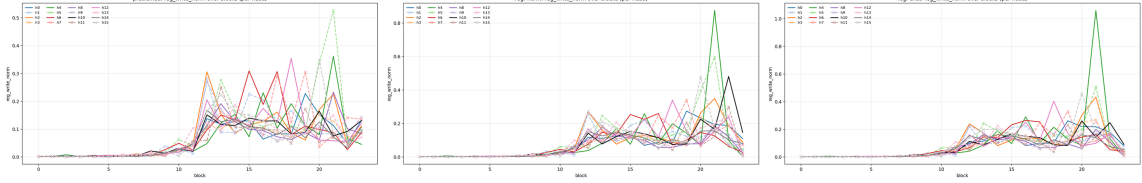


Figure 23: Register write-norm fraction across blocks by head (left-to-right: pretrained, **reg-norm**, **reg-brut**). Pretrained shows a Head 5 spike at block 21 (fraction ≈ 0.5), whereas both fine-tunes exhibit a pronounced Head 4 peak (fraction ≈ 0.8 for **reg-norm** and ≈ 1.0 for **reg-brut**).

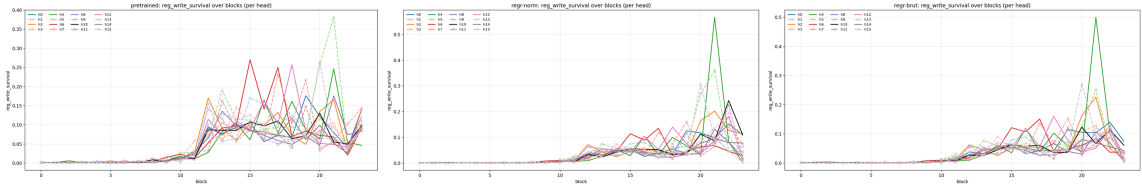


Figure 24: Register write-survival analogue of Fig. 23, shown per head across blocks (left-to-right: pretrained, **reg-norm**, **reg-brut**).

Fine-tuning induces late-stage consolidation. The most pronounced qualitative change occurs at Block 22, where **reg-norm** displays a sharp concentration of both REG write norm and REG attention mass into a small set of heads (centered on Heads 10–13). In contrast, the pretrained model exhibits comparatively modest REG writes at Block 22 and a more weakly concentrated attention profile. This shift suggests that **reg-norm** learns to implement a structured penultimate-block routing for REG-related information, i.e., a controlled conduit through which REG-aligned signal is routed. The existence of a concentrated hub is consistent with an interpretation in which robustness is achieved not by globally suppressing REG-like tokens, but by restricting and shaping the pathways that contribute to the final direction.

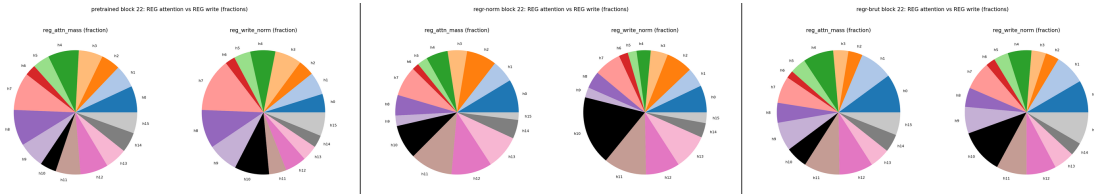


Figure 25: Block 22: per-head fractions of register attention mass and register write norm. Left-to-right: pretrained, **reg-norm**, **reg-brut**. Late “hub” heads (H10–H13) are highlighted (H10: black; H11: brown; H12: pink; H13: rose).

Fine-tuning alters the character of the final block. At Block 23, the pretrained model allocates substantial REG attention mass across several heads, whereas both fine-tuned models reallocate this mass toward a smaller and different set of heads, with the **regr-brut** model in particular exhibiting large negative deltas for pretrained-dominant attention heads. Such a redistribution is compatible with a robustness-oriented reconfiguration: reducing the number of high-gain late-stage pathways that could be exploited by spurious typographic features to dominate image–text similarity.

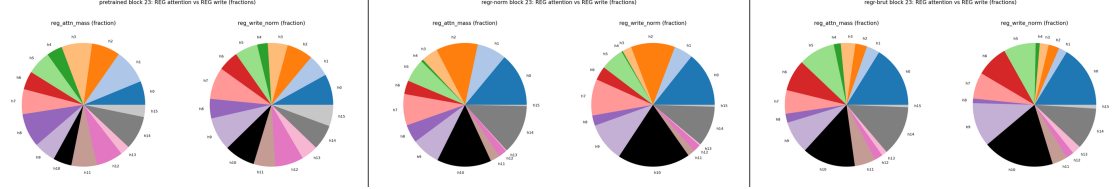


Figure 26: Block 23: per-head fractions of register attention mass and register write norm. Left-to-right: pretrained, **regr-norm**, **regr-brut**. Fine-tuning increases the relative mass and write fractions of Head 10 (black) compared to pretrained.

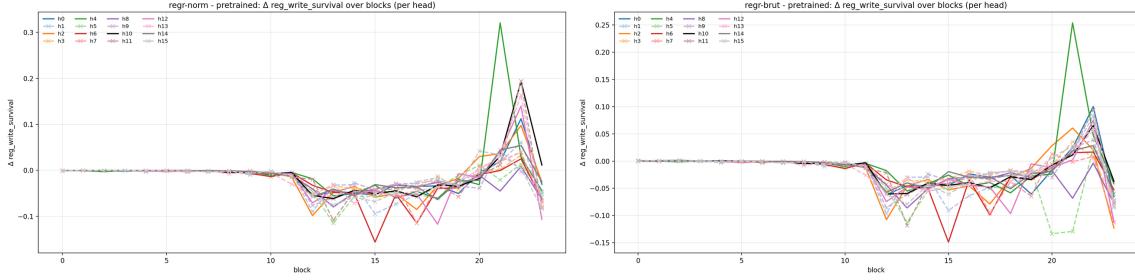


Figure 27: Change in register write survival relative to pretrained, $\Delta(\cdot)$, for **regr-norm** (left) and **regr-brut** (right). Head 4 shows the dominant increase in both models; **regr-norm** additionally exhibits an increase at block 22, Head 10. Both fine-tunes decrease Head 6, while **regr-brut** further decreases Head 5 around blocks 20–21.

Block	Head	pretrained		regr-norm		regr-brut	
		attn	write	attn	write	attn	write
23	0	0.1880	0.1314	0.1586	0.1097	0.1233	0.0794
	1	0.2754	0.0960	0.0874	0.0379	0.0263	0.0108
	2	0.2263	0.1120	0.1241	0.0889	0.0235	0.0138
	3	0.2422	0.0850	0.0495	0.0185	0.0292	0.0114
	4	0.1250	0.0454	0.0066	0.0027	0.0139	0.0053
	5	0.1890	0.0977	0.0692	0.0470	0.0732	0.0411
	6	0.1449	0.0876	0.0403	0.0279	0.0628	0.0410
	7	0.1979	0.1364	0.0930	0.0730	0.0470	0.0335
	8	0.2669	0.0835	0.0540	0.0206	0.0179	0.0059
	9	0.1733	0.1428	0.0849	0.0786	0.0650	0.0563
	10	0.1470	0.1310	0.1620	0.1462	0.1065	0.0911
	11	0.2025	0.0927	0.0283	0.0144	0.0416	0.0193
	12	0.2125	0.1219	0.0247	0.0151	0.0167	0.0100
	13	0.0754	0.0714	0.0028	0.0029	0.0108	0.0112
	14	0.2662	0.1073	0.1459	0.0808	0.1001	0.0533
	15	0.0965	0.0913	0.0037	0.0039	0.0050	0.0048

We defer to the full results in Tables [tables 28](#) and [29](#).

Implications for typographic robustness. Taken together, these findings support a view in which typographic robustness is associated with *structural re-routing* of REG-aligned pathways rather than a uniform attenuation of REG-related activity. The pretrained model exhibits multiple late-stage heads that can carry REG attention and REG writes, consistent with a more permissive routing regime. In contrast, **regr-norm** concentrates REG traffic into a late-block hub, while **regr-brut** combines strong late gating with suppression of specific pretrained-dominant final-block heads. This pattern motivates the mechanistic analyses that follow: if fine-tuning improves robustness by constraining and reshaping the effective REG channel, then the decisive evidence should appear in head-level pathway attribution, qualitative attention patterns, and targeted interventions that test whether these heads act as controllable routing points.

Conclusion: Attention and Register-Subspace Geometry Across pretrained and regression-teacher fine-tunes, global embedding geometry remains anisotropic and cross-modal alignment improves under fine-tuning without implying large attack-induced distribution shifts. Subspace surgery and patch-space projection diagnostics show a depth-dependent increase in reliance on directions aligned with a fixed pretrained register basis, with fine-tuning often increasing aligned energy while largely preserving broad reg/nonreg separation (large mean angles) and exhibiting only localized best-case coupling events late in depth. Attention diagnostics reinforce that attention maps alone don’t identify effective carriers: late depth can show inflated attention–contribution mismatch while the terminal block exhibits qualitative sign-structure changes and reduced divergence under fine-tuning. Head-wise routing concentrates the REG channel into a small set of dominant heads; some heads remain stable carriers across models while fine-tuning reassigns and consolidates late-stage transmission, consistent with robustness emerging from rerouting/constraining late pathways rather than eliminating register-like activity.

7 Discussion

Across experiments, we find that CLIP exhibits persistent *register-like tokens*: high-norm patch outliers that act as a measurable proxy for global, image-level information. Regression-teacher fine-tuning does not remove these tokens (their counts remain nonzero), but it *reduces their causal influence* on the terminal image embedding. The evidence localizes this change to the final transformer block(s), consistent with late-stage control of the *readout* from the residual stream into the embedding space.

Patch-supported backbone with a high-leverage shortcut axis. Pretrained CLIP retains substantial terminal alignment between the final embedding and pooled reg-like directions, and reg-only constructions can retain nontrivial semantic content. This combination seems to underly a failure mode in which most of the representation is patch-supported in a distributional sense – yet a small, high-leverage register-correlated direction can dominate similarity and rotate the final embedding towards spurious / adversarial directions. This dissociation supports a conditional, high-leverage mechanism: attacks can act by amplifying a small set of directions or token pathways rather than by inducing a wholesale distribution shift in embedding space. Consistent with this, regression fine-tuning reduces modality-gap metrics and increases paired IT similarity, but these global geometry changes alone do not explain the depth-localized collapse of FULL-reg alignment at the terminal block.

Depth-localized decoupling under regression-teacher training. Both regression-teacher variants preserve high-norm token statistics (nonzero reg counts and large reg norms) but sharply reduce the terminal-layer alignment between the final embedding and pooled reg-like directions. The leakage diagnostic localizes this transition: reg-only pathways remain highly similar to FULLthrough the trunk and even the penultimate block, followed by an abrupt collapse towards the final block. This depth-localized behavior is not explained by broad representational drift alone; instead, it is consistent with a late-stage attention readout that suppresses, rotates, or projects away components that are not supported by patch-derived information at readout.

regr-norm vs. regr-brut: two robustness regimes. Although both fine-tunes realize terminal decoupling from pooled reg directions, they exhibit distinct secondary signatures. **regr-norm** primarily shows structured late-block routing control: reg integration becomes concentrated into a small set of heads at late depth (routing hub), consistent with restricting high-gain pathways while preserving a heterogeneous patch-token basis. **regr-brut** shows stronger late-depth restructuring, consistent with increased late-depth redundancy and episodic coupling between reg/nonreg subspaces along a small number of directions, potentially reducing the degrees of freedom available for spurious information to dominate similarity, but at the expense of broader representational trade-offs.

Attention weights are not sufficient; value-path transmission is the control surface. Across models, attention can remain sharply structured toward high-norm tokens even when those tokens cease to influence the terminal embedding. Our attention–contribution diagnostics show that late blocks can exhibit mismatch inflation, and the sign-structure of entropy shifts changes at the terminal block under fine-tuning. These observations are compatible with the decoupling premise that $QK^\top$ attention patterns need not reflect causal influence when value mixtures and output projections reparameterize what is actually written to the residual stream [7].

7.1 Working theories suggested by the evidence

(1) Late-stage gating/projection onto a patch-supported subspace. The most direct implication of the depth-wise results is a late-stage operator that constrains the terminal embedding to lie near a subspace predictable from patch-derived features. This is suggested by the abrupt collapse of FULL-reg alignment at block 23 while reg statistics persist and reg-only remains highly similar to FULLthrough block 22 in the fine-tuned models.

(2) Value-path reparameterization that decouples register keys from decision-relevant writes. The persistence of register-focused attention alongside reduced terminal reg influence is explained if attention mass on register-like keys persists in $QK^\top$ geometry while V vectors and/or W_O rotate or suppress the attended mixtures along decision-relevant directions. This would yield structured attention maps with attenuated AV -level transmission to the terminal embedding, consistent with attention-contribution decoupling [7] and with mechanistic defenses that target late write pathways [5].

7.2 Conclusion.

CLIP remains patch-supported in a variance-explained sense yet is vulnerable to typographic attacks that exploit a narrow, register-correlated direction with high leverage on similarity. Regression-teacher training improves robustness primarily by imposing late readout control—attenuating or rerouting register-sourced value-path influence—rather than by eliminating the register-like subsystem itself.

8 Limitations, Implications and Further Research Directions

Model fine-tuning. We first trained the full CLIP model and subsequently froze the text encoder, with teacher fine-tuning on ViT only; our single 24GB GPU setup forced us to use very small batches once Regression-Teachers were attached, making end-to-end optimization markedly more brittle. In this regime, updating the text tower alongside the vision tower risked a degenerate shortcut solution in the learned geometry (the text encoder adapting to explain away supervision rather than preserving a stable language-vision alignment). Freezing the text encoder after initial joint fit therefore served as a practical stabilization / regularization step.

Compute and coverage. A comprehensive characterization of the robustness-generalization trade-off would ideally require a controlled sweep over Regression-Teacher variants (e.g., teacher family and loss weighting), with multiple converged checkpoints per setting. This would enable a phase-diagram of robustness versus transfer and retrieval metrics – and an explicit mapping of the Pareto frontier.

Analysis. We provide mechanistic correlates observed especially in late blocks (e.g. geometry vs. attention analysis). These findings are consistent with the *hypothesis* that pushing typographic robustness beyond $\sim 80\%$ increasingly trades off out-of-distribution transfer, but do not by themselves provide a complete frontier.

Training scale. Training was conducted under constrained resources (single 24GB GPU), which necessitated small batch sizes. While Geometric Parametrization mitigated instability and enabled reliable optimization, larger batches and broader training corpora could plausibly yield further gains and/or shift the robustness-generalization balance. A particularly informative follow-up would be to train probes and/or teachers with substantially increased data scale (ideally approaching pre-training scale, or a meaningful fraction thereof) to test whether the identified trends persist.

Scope of intervention analysis. The mechanistic analyses in this paper identify late-block “control” attention heads associated with typographic sensitivity and general routing / integration of information. We treat our findings and hypotheses as a testable prediction rather than a fully validated causal account, and leave a more systematic intervention sweep to future work.

Implications for defenses. Our results suggest a principled design space for defenses: interventions that constrain *terminal readout geometry* (or late write pathways into CLS) may be sufficient to reduce typographic vulnerability while preserving trunk representations and broad zero-shot capabilities. This aligns with mechanistic head-level defenses [5] while also highlighting an alternative lever: late-stage gating/projection induced by training objectives. On the other hand, because *where global information is represented* is trainable rather than fixed, such defenses can reshape the *controllability* and *failure structure* of the embedding space: stronger constraints may replace a brittle shortcut with a more *low-dimensional, stable* error frontier that persists under benign transforms (crop/resize/recompression) and can be easier to exploit in retrieval and ranking than a diffuse “hard tail.”

Further research directions. Further experiments could sharpen causal claims and surface failure modes: (i) localizing the relevant late writes via controlled value-path interventions; (ii) testing the terminal gating hypothesis directly by estimating the effective Jacobian from token states to the final embedding and probing for a low-rank or subspace-selective projection emerging at the late / final blocks; (iii) characterizing robustness–locality trade-offs by measuring effective dimensionality/anisotropy across depth and by auditing whether errors (e.g. retrieval) become more *concentrated and repeatable* (stable under small transforms); and (iv) disambiguating whether regression-trained models exhibiting late token proliferation reflect increased global-carrier capacity or reduced representational degrees of freedom by tracking patch-level mutual information with labels and cross-image component structure across blocks.

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9 Appendix

10 Appendix: Method Details

10.1 Canonical Register Set (per-image)

Canonical per-image register set. Let a ViT layer/block index be $b \in \{0, \dots, L\}$, and let the post-block token stream be

$$X^{(b)} \in \mathbb{R}^{T \times C}, \quad T = 1 + P,$$

with $x_{\text{cls}}^{(b)} := X_0^{(b)} \in \mathbb{R}^C$ and patch tokens $X_{\text{patch}}^{(b)} := (X_1^{(b)}, \dots, X_P^{(b)}) \in \mathbb{R}^{P \times C}$. For each image i , we define a *canonical* (image-specific) register-like index set from an *unaltered* forward pass at a reference block $\bar{b}$ (by default, the terminal block $\bar{b} = L$):

$$R_i := \left\{ p \in \{1, \dots, P\} : \|x_{i,p}^{(\bar{b})}\|_2 \geq \tau_{\text{reg}} \right\}, \quad p_i := |R_i|.$$

Thus, both the register count p_i and the register positions R_i may vary *across images*, but are treated as fixed *within* an image once inferred from the unperturbed pass.

Blockwise set stability (empirical). For completeness, we can also define the block-local threshold set

$$R_i^{(b)} := \left\{ p \in \{1, \dots, P\} : \|x_{i,p}^{(b)}\|_2 \geq \tau_{\text{reg}} \right\}.$$

In our data, register-like tokens *emerge* around a characteristic block $b_\star$ and then stabilize: $R_i^{(b)} = R_i$ for all images and all $b \geq b_\star$. Accordingly, all interventions and auxiliary objectives in this work are parameterized by the canonical per-image set R_i (rather than recomputing $R_i^{(b)}$ at every block), ensuring that “register” versus “non-register” denotes a consistent partition of patches *within each image*.

10.2 Raw feature norms (unnormalized encoder outputs)

For completeness, the code also tracks norms of the *unnormalized* encoder outputs (before L2 normalization). Let $\tilde{f}_i$ and $\tilde{g}_j$ denote raw `encode_image` and `encode_text` outputs. We summarize

$$\|\tilde{f}_i\|_2, \quad \|\tilde{g}_j\|_2$$

by mean/std and selected percentiles (5/50/95). These statistics capture changes in embedding scale that are otherwise removed by normalization and may affect downstream calibration (e.g., temperature, thresholds).

10.3 Teacher target mixing

We allow the teacher target $\mathbf{c}_\ell^\star$ to mix the layer- ℓ CLS embedding with an auxiliary global signal. Let $\mathbf{c}_\ell$ be the layer- ℓ CLS embedding and let $\mathbf{g}_\ell$ denote an auxiliary embedding (either a centered register-mean at layer ℓ or the final image embedding). For mixing coefficient $\alpha \in [0, 1]$, we form

$$\mathbf{c}_\ell^\star = \text{norm}((1 - \alpha)\mathbf{c}_\ell + \alpha\mathbf{g}_\ell). \quad (17)$$

For additional details please refer to our codebase [15].

10.4 REG-aware teacher features (concatenation and optional whitening)

A reg-aware teacher augments patch features with a register-derived feature $\Delta \mathbf{r}_\ell$. One instantiation concatenates centered features:

$$\mathbf{f}_\ell = [\mathbf{p}_\ell - \mu_{p,\ell}, \Delta \mathbf{r}_\ell],$$

and predicts $\mathbf{c}_\ell^\star$ via least squares or ridge regression on $\mathbf{f}_\ell$. When enabled, diagonal whitening rescales the register-derived coordinates to reduce dominance from high-variance dimensions; exact scaling follows the implementation.

10.5 Attention faithfulness diagnostics (token-mix and value-mix)

We report two diagnostics at B_{23} .

(A) Token-mix diagnostic (LN1 space). Let $X_{\text{LN1}}^{(23)}$ denote the token states in the pre-attention layer-normalized space (LN1) for block B_{23} . Let $A^{(23,h)} \in \mathbb{R}^{(1+P) \times (1+P)}$ be the attention matrix (after softmax) for head $h \in \{1, \dots, H\}$ at block 23. We define the mean CLS-to-patch attention distribution (renormalized over patches) by

$$\alpha_i \propto \frac{1}{H} \sum_{h=1}^H A_{0,i}^{(23,h)}, \quad i \in \{1, \dots, P\} = \{1, \dots, P\}, \quad \sum_{i=1}^P \alpha_i = 1.$$

We form an attention-weighted token mixture

$$\mathbf{z}_{\text{tok}} = \sum_{i=1}^P \alpha_i X_{\text{LN1},i}^{(23)},$$

and report $\cos(X_{\text{LN1},0}^{(23)}, \mathbf{z}_{\text{tok}})$ as “TOKEN faithfulness.”

(B) Value-mix diagnostic (true attention output). Let $V_s^{(23,h)} \in \mathbb{R}^{d_h}$ denote the value vector at source token $s \in \{0, \dots, P\}$ for head h in block 23. The per-head CLS attention output (pre-residual) is

$$o_{\text{cls}}^{(23,h)} = \sum_{s=0}^P A_{0,s}^{(23,h)} V_s^{(23,h)} \in \mathbb{R}^{d_h}.$$

Let W_O, b_O be the attention output projection parameters. The true CLS attention output in width-space is

$$\mathbf{z}_{\text{val}} = W_O \text{concat}_{h=1}^H (o_{\text{cls}}^{(23,h)}) + b_O.$$

We compare $\mathbf{z}_{\text{val}}$ to the script-traced CLS attention output, and decompose contributions from self ($s = 0$), register-like patches ($s \in \mathcal{P}_{\text{reg}}$), and non-register patches ($s \in \mathcal{P}_{\text{-reg}}$) by restricting the summation over s to the corresponding index sets [15].

10.6 Principal-component subtraction for register-like pooling

When enabled, we suppress dominant directions in the register-like hidden space before LN_{post} and Π . Given a register mean $\boldsymbol{\mu}_{\text{reg}}$ and top- k orthonormal principal directions $U_k \in \mathbb{R}^{k \times d}$ learned from register-like token samples (with principal directions stacked as rows), we transform each register token as

$$\tilde{X}_i^{(23)} = X_i^{(23)} - \left((X_i^{(23)} - \boldsymbol{\mu}_{\text{reg}}) U_k^\top \right) U_k,$$

pool $\mathbf{r}_{23}^{\text{raw}} = \frac{1}{|\mathcal{P}_{\text{reg}}|} \sum_{i \in \mathcal{P}_{\text{reg}}} \tilde{X}_i^{(23)}$, and embed $\mathbf{e}_{\text{reg}} = \text{norm}(\Pi \text{LN}_{\text{post}}(\mathbf{r}_{23}^{\text{raw}}))$.

10.7 Patch-subspace energy diagnostics

Using a set of clean images, we collect centered patch pools $\mathbf{p}_{23}^{\text{raw}} - \boldsymbol{\mu}_p$ and compute a PCA basis $B \in \mathbb{R}^{k \times d}$ (default $k = 128$) with basis directions stacked as rows. We project a centered CLS hidden vector $\mathbf{c}^{\text{raw}} = X_0^{(23)}$ onto the patch subspace and map to CLIP space:

$$\hat{\mathbf{c}}_{\text{sub}} = \boldsymbol{\mu}_p + \left((\mathbf{c}^{\text{raw}} - \boldsymbol{\mu}_p) B^\top \right) B, \quad \mathbf{e}_{\text{PATCH}}^{\text{SUB}} = \text{norm}(\Pi \text{LN}_{\text{post}}(\hat{\mathbf{c}}_{\text{sub}})).$$

For CLS-PATCHSUB, we report the fraction of centered CLS energy captured by the patch PCA subspace. Let $\mathbf{u} = \mathbf{c}^{\text{raw}} - \boldsymbol{\mu}_p$ and $\mathbf{u}_{\parallel} = ((\mathbf{u})B^{\top})B$. We compute

$$\text{EnergyFrac} = \frac{\|\mathbf{u}_{\parallel}\|_2^2}{\|\mathbf{u}\|_2^2}.$$

10.8 Leakage diagnostic: cls-row attention masking

We define a blockwise counterfactual that isolates how information from a selected patch subset can propagate into the final image embedding. At each block ℓ , let $A^{(\ell,h)} \in \mathbb{R}^{(1+P) \times (1+P)}$ denote the attention matrix (after softmax) for head $h \in \{1, \dots, H\}$, indexed with token 0 as CLS and tokens 1: P as patches. Let $S \subseteq \{1, \dots, P\}$ be a subset of patch indices (either $\mathcal{P}$ or $\mathcal{R}$).

For a chosen start block b , we apply a CLS-row restriction at *every* block $\ell \geq b$ by zeroing attention mass from CLS to all patch tokens not in S :

$$\tilde{A}_{0,j}^{(\ell,h)} = \begin{cases} A_{0,j}^{(\ell,h)}, & j \in S \cup \{0\}, \\ 0, & \text{otherwise,} \end{cases} \quad \text{then renormalize so } \sum_{j=0}^P \tilde{A}_{0,j}^{(\ell,h)} = 1.$$

All other rows of $A^{(\ell,h)}$ are left unchanged, and all other computations (values V , output projection, MLP, residual paths) are unmodified. We then run the transformer forward with $\tilde{A}^{(\ell,h)}$ for all $\ell \geq b$ and read out the final image embedding as usual.

We denote the resulting embedding by FULL when $S = \{1, \dots, P\}$, by PATCH-ONLY when $S = \mathcal{P}$, and by REG-ONLY when $S = \mathcal{R}$. We report cosine similarity and CKA between these counterfactual embeddings and the unmodified baseline.

10.9 Regression-probe solve details (least squares and ridge)

Let $P \in \mathbb{R}^{N \times d}$ and $C \in \mathbb{R}^{N \times d}$ be the centered design matrices whose rows are $(\mathbf{p}_j - \boldsymbol{\mu}_p)$ and $(\mathbf{c}_j - \boldsymbol{\mu}_c)$, respectively. We fit an *unregularized* linear least-squares map by minimizing

$$W = \arg \min_W \|PW^{\top} - C\|_F^2, \quad (18)$$

and compute the solution using a numerically stable least-squares solver (e.g., QR/SVD; `torch.linalg.lstsq`). The resulting map is used at test time via

$$\hat{\mathbf{c}} = (\mathbf{p} - \boldsymbol{\mu}_p)W^{\top} + \boldsymbol{\mu}_c. \quad (19)$$

A Instrumenting MLP→K routing, register-feature discovery, and attention diagnostics

A.1 Setup and notation

We consider a CLIP-like vision transformer with L residual blocks. Each block operates on a token sequence of length $T = 1 + P$, consisting of one CLS token and P patch tokens. For an input image i , let $X_i^{(\ell)} \in \mathbb{R}^{T \times d}$ denote the token stream at the *input* of the attention module in block ℓ (equivalently, the tensor passed into `block.attn` after LN_1 in standard pre-norm designs). We write the CLS token as index 0 and patch tokens as indices $1..P$.

The attention in block ℓ uses H heads with head dimension D and $d = HD$. With projections $W_q^{(\ell)}, W_k^{(\ell)}, W_v^{(\ell)} \in \mathbb{R}^{d \times d}$, the per-head slices are $W_{q,h}^{(\ell)}, W_{k,h}^{(\ell)}, W_{v,h}^{(\ell)} \in \mathbb{R}^{D \times d}$ such that

$$W_q^{(\ell)} = \begin{bmatrix} W_{q,1}^{(\ell)} \\ \vdots \\ W_{q,H}^{(\ell)} \end{bmatrix}, \quad W_k^{(\ell)} = \begin{bmatrix} W_{k,1}^{(\ell)} \\ \vdots \\ W_{k,H}^{(\ell)} \end{bmatrix}, \quad W_v^{(\ell)} = \begin{bmatrix} W_{v,1}^{(\ell)} \\ \vdots \\ W_{v,H}^{(\ell)} \end{bmatrix}.$$

The MLP in block ℓ expands to width m and projects back to d . We denote the MLP output projection (a.k.a. `c_proj`) as $W_c^{(\ell)} \in \mathbb{R}^{d \times m}$, and the MLP input projection (a.k.a. `c_fc`) as $W_f^{(\ell)} \in \mathbb{R}^{m \times d}$. For an activation function ϕ (QuickGELU in the implementation), the MLP mapping for token t is

$$\text{MLP}^{(\ell)}(x_t) = W_c^{(\ell)} \phi(W_f^{(\ell)} x_t).$$

A.2 $\text{MLP}^{(\ell)} \rightarrow \text{K}^{(\ell+1)}$ routing scores

The script defines a purely weight-based routing score capturing how each expanded MLP feature in block ℓ would influence the *next* block’s key projection. Let $W_c^{(\ell)} \in \mathbb{R}^{d \times m}$ be the output projection of block ℓ ’s MLP, and let $W_k^{(\ell+1)} \in \mathbb{R}^{d \times d}$ be the key projection of block $\ell+1$ ’s attention. For head h , take the head slice $W_{k,h}^{(\ell+1)} \in \mathbb{R}^{D \times d}$ (contiguous in the row dimension under the standard stacked-head parameterization). For feature index $f \in \{1, \dots, m\}$, define the routing vector

$$u_{h,f}^{(\ell)} = W_{k,h}^{(\ell+1)} W_c^{(\ell)}[:, f] \in \mathbb{R}^D,$$

and the routing magnitude score

$$S_{h,f}^{(\ell)} = \|u_{h,f}^{(\ell)}\|_2 = \|W_{k,h}^{(\ell+1)} W_c^{(\ell)}[:, f]\|_2. \quad (20)$$

Equivalently, stacking features yields

$$S^{(\ell)} = \left[\|W_{k,h}^{(\ell+1)} W_c^{(\ell)}\|_{2,\text{col}} \right]_{h=1}^H \in \mathbb{R}^{H \times m},$$

where $\|\cdot\|_{2,\text{col}}$ denotes the vector of column-wise ℓ_2 norms.

Optional LN_1 - γ approximation. The code optionally rescales key projection columns by the next block’s LN_1 gain $\gamma^{(\ell+1)} \in \mathbb{R}^d$ via

$$\widetilde{W}_k^{(\ell+1)} = W_k^{(\ell+1)} \text{diag}(\gamma^{(\ell+1)}),$$

and then uses $\widetilde{W}_{k,h}^{(\ell+1)}$ in (20). This ignores mean/variance terms and is treated as a heuristic.

Per-head thresholding and feature roles. Fix a quantile $\tau \in (0, 1)$ (e.g. $\tau = 0.98$). For each head h , set a head-specific threshold

$$\theta_h^{(\ell)} = \text{Quantile}_\tau(\{S_{h,f}^{(\ell)}\}_{f=1}^m), \quad \mathcal{F}_h^{(\ell)} = \{f : S_{h,f}^{(\ell)} \geq \theta_h^{(\ell)}\}. \quad (21)$$

A feature f is called a *collaborator* if it is high for more than one head, and a *specialist* if it is high for exactly one head:

$$\text{collaborator: } |\{h : f \in \mathcal{F}_h^{(\ell)}\}| > 1, \quad \text{specialist: } |\{h : f \in \mathcal{F}_h^{(\ell)}\}| = 1.$$

The script also reports per-head top- K lists $\text{TopK}(S_{h,\cdot}^{(\ell)})$.

Cross-model comparisons. Let $S^{(\ell)}(a)$ and $S^{(\ell)}(b)$ be routing-score matrices for two models (e.g. pretrained vs fine-tuned). For each head h , define:

$$\rho_h^{(\ell)} = \text{Spearman}(S_{h,\cdot}^{(\ell)}(a), S_{h,\cdot}^{(\ell)}(b)), \quad J_h^{(\ell)}(K) = \frac{|\text{TopK}(S_{h,\cdot}^{(\ell)}(a)) \cap \text{TopK}(S_{h,\cdot}^{(\ell)}(b))|}{|\text{TopK}(S_{h,\cdot}^{(\ell)}(a)) \cup \text{TopK}(S_{h,\cdot}^{(\ell)}(b))|}.$$

A.3 Register-feature discovery in MLP space

The script identifies “register-like” MLP features for a given layer ℓ by combining (i) a *token-level* register proxy based on high patch-token norms in the block output space, (ii) MLP feature magnitudes on selected tokens, and (iii) per-feature scaling by column norms of $W_c^{(\ell)}$.

Token proxy for registers. Given a batch of images, the block output tensor at layer ℓ is captured (via a forward hook) and reshaped to $[B, T, d]$. For each sample i and patch token $p \in \{1, \dots, P\}$, define the patch norm

$$n_{i,p}^{(\ell)} = \|X_i^{(\ell)}[p, :]\|_2.$$

Given a threshold $\eta > 0$ (e.g. $\eta = 70$), define a mask of “register proxy” tokens

$$r_{i,p}^{(\ell)} = \mathbf{1}[n_{i,p}^{(\ell)} > \eta].$$

If no patch tokens satisfy the threshold for a sample, the code falls back to selecting the top- K patch tokens by $n_{i,p}^{(\ell)}$.

MLP feature magnitudes on selected tokens. The pre-activation at the MLP expansion (output of $W_f^{(\ell)}$) is captured as $Z_i^{(\ell)} \in \mathbb{R}^{T \times m}$, and the activation is

$$H_i^{(\ell)} = \phi(Z_i^{(\ell)}) \in \mathbb{R}^{T \times m},$$

with ϕ being the model’s activation if available and QuickGELU as a fallback: $\phi(z) = z \sigma(1.702z)$.

Restrict to patch tokens and apply the register mask. Let $\mathcal{I}^{(\ell)} = \{(i, p) : p \in \{1, \dots, P\}, r_{i,p}^{(\ell)} = 1\}$ be the selected token set. For each feature f , define a batchwise maximal (absolute) activation over selected tokens:

$$a_f^{(\ell)} = \max_{(i,p) \in \mathcal{I}^{(\ell)}} |H_i^{(\ell)}[p, f]|.$$

Column-norm weighting and impact score. Let $c_f^{(\ell)} = \|W_c^{(\ell)}[:, f]\|_2$ be the column norm of the MLP output projection. The script defines a per-feature impact score

$$\text{impact}_f^{(\ell)} = a_f^{(\ell)} c_f^{(\ell)}. \quad (22)$$

Across batches, it maintains a running maximum (so (22) is effectively computed as a maximum over all processed images/tokens).

Selecting register features. Let $\text{impact}^{(\ell)} \in \mathbb{R}^m$ be the final impact vector. Register features are selected by one of:

$$\mathcal{R}^{(\ell)} = \begin{cases} \text{TopN}(\text{impact}^{(\ell)}) & \text{if } N > 0, \\ \{f : \text{impact}_f^{(\ell)} \geq \text{Quantile}_q(\text{impact}^{(\ell)})\} & \text{if } q > 0, \\ \{f : \text{impact}_f^{(\ell)} \geq \alpha\} & \text{otherwise (absolute threshold } \alpha\text{)}. \end{cases}$$

If thresholding yields $\emptyset$ and neither TopN nor quantile is specified, it falls back to a small TopN.

A.4 Per-head output norms: heuristic and “true” pre/post out_proj

Heuristic block-output slice. For block outputs $Y^{(\ell)} \in \mathbb{R}^{B \times T \times d}$, reshape $d = HD$ into $Y^{(\ell)} \rightarrow \mathbb{R}^{B \times T \times H \times D}$ and compute per-head mean norms

$$L2_h^{(\ell)} = \frac{1}{BT} \sum_{i=1}^B \sum_{t=1}^T \|Y_i^{(\ell)}[t, h, :]\|_2.$$

True attention output norms (pre/post out_proj). Within attention at layer ℓ , compute per-head

$$Q_{i,h} = X_i^{(\ell)} W_{q,h}^{(\ell)\top} \in \mathbb{R}^{T \times D}, \quad K_{i,h} = X_i^{(\ell)} W_{k,h}^{(\ell)\top} \in \mathbb{R}^{T \times D}, \quad V_{i,h} = X_i^{(\ell)} W_{v,h}^{(\ell)\top} \in \mathbb{R}^{T \times D}.$$

Define attention weights

$$A_{i,h} = \text{softmax}\left(\frac{Q_{i,h} K_{i,h}^\top}{\sqrt{D}}\right) \in \mathbb{R}^{T \times T}.$$

The *pre* out_proj head output is

$$O_{i,h}^{\text{pre}} = A_{i,h} V_{i,h} \in \mathbb{R}^{T \times D}.$$

Let $W_o^{(\ell)} \in \mathbb{R}^{d \times d}$ be the output projection. The script applies per-head *input-column* slices:

$$W_{o,h}^{(\ell)} = W_o^{(\ell)}[:, hD : (h+1)D] \in \mathbb{R}^{d \times D},$$

yielding the *post* out_proj per-head contribution in embedding space:

$$O_{i,h}^{\text{post}} = O_{i,h}^{\text{pre}} W_{o,h}^{(\ell)\top} \in \mathbb{R}^{T \times d}.$$

It reports mean norms both over all query tokens and for CLS-only:

$$\text{pre_all}_h^{(\ell)} = \frac{1}{BT} \sum_{i,t} \|O_{i,h}^{\text{pre}}[t, :]\|_2, \quad \text{pre_cls}_h^{(\ell)} = \frac{1}{B} \sum_i \|O_{i,h}^{\text{pre}}[0, :]\|_2,$$

and analogously $\text{post_all}_h^{(\ell)}$, $\text{post_cls}_h^{(\ell)}$ from $O_{i,h}^{\text{post}}$.

A.5 Attention diagnostics: entropy, register-mass, and Q/K/V norms

For each layer/head, the script also records:

- **Q/K/V norms:** $\|Q_{i,h}[t]\|_2, \|K_{i,h}[t]\|_2, \|V_{i,h}[t]\|_2$, averaged either over all tokens t or for $t = 0$ (CLS).
- **Attention entropy:** for each query token t , entropy over keys $H(A_{i,h}[t, :]) = -\sum_s A_{i,h}[t, s] \log A_{i,h}[t, s]$; averaged over all t or for CLS only.
- **Register-mass (key-side):** define a per-sample mask of register-proxy keys on patch tokens, using $\|X_i^{(\ell)}[p]\|_2 > \eta$ (or a top- K fallback). For CLS query,

$$\text{regmass}_{i,h,\text{cls}}^{(\ell)} = \sum_{p=1}^P A_{i,h}[0, p] \mathbf{1}[p \in \mathcal{K}_i^{(\ell)}],$$

and similarly averaged over all queries t .

A.6 Effective contribution diagnostics (attention vs. contribution)

The code defines a patch-only CLS distribution and an “effective contribution” distribution that weights keys by the value norm.

Patch-only attention for CLS. Let $A_{i,h} \in \mathbb{R}^{T \times T}$ be attention weights. Excluding the CLS key, define the patch-only attention for query token t :

$$\tilde{A}_{i,h}[t, p] = \frac{A_{i,h}[t, p]}{\sum_{p'=1}^P A_{i,h}[t, p'] + \varepsilon}, \quad p \in \{1, \dots, P\}.$$

For CLS, write $a_{i,h} \in \Delta^{P-1}$ as $a_{i,h}[p] = \tilde{A}_{i,h}[0, p]$.

Contribution proxy. Compute value vectors $V_{i,h}[p] \in \mathbb{R}^D$ and their norms $\nu_{i,h}[p] = \|V_{i,h}[p]\|_2$. Define unnormalized contribution weights

$$c_{i,h}^{\text{raw}}[p] = a_{i,h}[p] \nu_{i,h}[p],$$

and the normalized contribution distribution $c_{i,h} \in \Delta^{P-1}$:

$$c_{i,h}[p] = \frac{c_{i,h}^{\text{raw}}[p]}{\sum_{p'=1}^P c_{i,h}^{\text{raw}}[p'] + \varepsilon}.$$

Decoupling metrics. For each head h and layer ℓ , the script reports averages over images of:

$$\begin{aligned} H(a_{i,h}) &= -\sum_{p=1}^P a_{i,h}[p] \log a_{i,h}[p], & H(c_{i,h}) &= -\sum_{p=1}^P c_{i,h}[p] \log c_{i,h}[p], \\ \text{KL}(a_{i,h} \| c_{i,h}) &= \sum_{p=1}^P a_{i,h}[p] \log \frac{a_{i,h}[p] + \varepsilon}{c_{i,h}[p] + \varepsilon}. \end{aligned}$$

Register mass under attention vs. contribution. Using the same register-proxy key set $\mathcal{K}_i^{(\ell)} \subseteq \{1, \dots, P\}$ defined by high $\|X_i^{(\ell)}[p]\|_2$ (or a top- K fallback), define

$$m_{i,h}^A = \sum_{p \in \mathcal{K}_i^{(\ell)}} a_{i,h}[p], \quad m_{i,h}^C = \sum_{p \in \mathcal{K}_i^{(\ell)}} c_{i,h}[p].$$

The script reports per-layer/head means of $m_{i,h}^A$ and $m_{i,h}^C$, and their across-image Pearson correlation $\text{corr}(m_{i,h}^A, m_{i,h}^C)$.

Optional outcome correlations. Given an outcome scalar o_i per image, the script optionally computes correlations $\text{corr}(m_{:,h}^A, o)$ and $\text{corr}(m_{:,h}^C, o)$ per layer/head.

A.7 Controlled ablation variants (hook-defined interventions)

The script runs multiple forward-pass variants by attaching hooks:

- **baseline:** no hooks.
- **reg_nuke_only:** for selected layers $\ell \in \mathcal{L}_{\text{nuke}}$ and feature indices $f \in \mathcal{R}_{\text{nuke}}^{(\ell)}$, zero the MLP expansion output (pre-activation) at those feature coordinates:

$$Z_i^{(\ell)}[t, f] \leftarrow 0 \quad \text{for all tokens } t \text{ and } f \in \mathcal{R}_{\text{nuke}}^{(\ell)}.$$

- **zero_attn_only:** replace the attention module output with zero, so the attention residual contribution is removed.
- **zero_mlp_only:** replace the MLP module output with zero, so the MLP residual contribution is removed.
- **zero_attn_mlp:** apply both `zero_attn_only` and `zero_mlp_only`.
- **reg_nuke_plus_zero_attn_mlp:** combine `reg_nuke_only` with `zero_attn_mlp`.

A.8 Register-mover candidate scan

The script includes a non-invasive scan that hooks the MLP activation output in each block and seeks dimensions that spike on high-norm patch tokens relative to a per-image median patch activation.

For a layer ℓ , let $H_i^{(\ell)} \in \mathbb{R}^{T \times m}$ denote the post-activation MLP expanded activations (for a batch). For each image i , define high-norm patch tokens by an *adaptive* threshold: compute patch-token norms in activation space, take their median med_i , and select tokens with norm $> \kappa \text{med}_i$ for a multiplier κ . For each feature f , compare selected tokens to the median patch activation vector:

$$\Delta_{i,f}^{(\ell)} = \max_{p \in \mathcal{H}_i} \left(H_i^{(\ell)}[p, f] - \text{median}_{p' \in \{1, \dots, P\}} H_i^{(\ell)}[p', f] \right),$$

(or a mean instead of max, depending on a flag). A feature is marked as a candidate if $\Delta_{i,f}^{(\ell)} > \delta$ for any image i ; the script tracks counts and maximal observed Δ across the dataset.

A.9 Registers and head-wise register routing

Let the token stream after block $b \in \mathcal{B}$ be

$$X^{(b)} \in \mathbb{R}^{T \times B \times C}, \quad T = 1 + P$$

with CLS token $x_{\text{cls}}^{(b)} \in \mathbb{R}^{B \times C}$ and patch tokens $X_{\text{patch}}^{(b)} \in \mathbb{R}^{B \times P \times C}$. Define per-patch norms

$$n_{i,p}^{(b)} = \left\| X_{\text{patch}}^{(b)}[i, p, :] \right\|_2.$$

For each sample i , we mark registers via a thresholded, top- K mask

$$m_{i,p}^{(b)} = \mathbf{1} \left[n_{i,p}^{(b)} > \tau \right] \text{ then keep at most } K \text{ true entries, enforcing at least } K_{\min} \geq 1.$$

(If a block has no patches above threshold for a sample, an optional fallback reuses the register indices inferred at a reference block, default $b_{\text{ref}} = 13$.)

Head-wise REG attention mass. The code relies on capturing each block’s attention probabilities and values for the CLS query. Let $\alpha_{i,\text{cls} \rightarrow s}^{(b,h)}$ be the attention probability from CLS to source token $s \in \{0, \dots, P\}$ in head h for sample i . Define the source register indicator $r_{i,s}^{(b)}$ as $r_{i,0}^{(b)} = 0$ (CLS is not a register) and $r_{i,1+p}^{(b)} = m_{i,p}^{(b)}$. Then the *REG attention mass* per head is

$$A^{(b,h)} = \mathbb{E}_i \left[\sum_{s=0}^P \alpha_{i,\text{cls} \rightarrow s}^{(b,h)} r_{i,s}^{(b)} \right].$$

This measures how much probability mass the CLS query assigns to register-marked patch tokens.

Head-wise REG write norm (effective contribution proxy). Let $v_{i,s}^{(b,h)} \in \mathbb{R}^d$ be the value vector at source token s (with head dimension d). The *register-restricted* head output (for the CLS query) is

$$o_{i,\text{reg}}^{(b,h)} = \sum_{s=0}^P \alpha_{i,\text{cls} \rightarrow s}^{(b,h)} r_{i,s}^{(b)} v_{i,s}^{(b,h)} \in \mathbb{R}^d.$$

Let $W_O^{(b)} \in \mathbb{R}^{C \times C}$ be the output projection, partitioned into head slices $W_{O,h}^{(b)} \in \mathbb{R}^{C \times d}$. The code defines the corresponding *write vector* into model width-space as

$$w_i^{(b,h)} = W_{O,h}^{(b)} o_{i,\text{reg}}^{(b,h)} \in \mathbb{R}^C,$$

and summarizes each head by the mean write norm

$$W^{(b,h)} = \mathbb{E}_i [\|w_i^{(b,h)}\|_2].$$

Intuitively, $A^{(b,h)}$ captures *where attention looks* (probability mass), while $W^{(b,h)}$ captures *how much width-space signal is actually written* from register sources.

Survival-aware head influence. To estimate whether a head’s register write at block b causally propagates to the final CLIP image embedding, we run a controlled injection experiment. Let $F_{>b}(\cdot)$ denote the computation of the remaining transformer blocks ($b+1, \dots, L-1$), followed by the standard CLIP image projection (LayerNorm + linear projection) applied to the CLS token. Given the captured post-block stream $X^{(b)}$ and per-sample write vectors $w_i^{(b,h)}$, we form an intervened stream by adding the head write into the CLS token:

$$\tilde{X}_i^{(b,h)} = X_i^{(b)}, \quad \tilde{x}_{i,\text{cls}}^{(b,h)} = x_{i,\text{cls}}^{(b)} + w_i^{(b,h)}.$$

Let the baseline and intervened final embeddings be

$$e_i = F_{>b}(X_i^{(b)}) \in \mathbb{R}^D, \quad \tilde{e}_i^{(h)} = F_{>b}(\tilde{X}_i^{(b,h)}) \in \mathbb{R}^D.$$

The *survival score* per head is the mean embedding displacement magnitude:

$$S^{(b,h)} = \mathbb{E}_i[\|\tilde{e}_i^{(h)} - e_i\|_2].$$

This is computed efficiently by batching all heads as a single expanded batch ($B \times H$) through the remainder of the network.

Per-model outputs (core tensors). For each model tag t and probed block set $\mathcal{B}$, we produce matrices

$$A_t \in \mathbb{R}^{|\mathcal{B}| \times H}, \quad W_t \in \mathbb{R}^{|\mathcal{B}| \times H}, \quad S_t \in \mathbb{R}^{|\mathcal{B}| \times H} \quad (\text{if survival is enabled on those blocks}),$$

where rows correspond to blocks b and columns to heads h .

Normalized mismatch diagnostic. To compare *relative* head allocation rather than absolute scale, the code also computes per-block head fractions

$$\hat{A}_t^{(b,h)} = \frac{A_t^{(b,h)}}{\sum_{h'} A_t^{(b,h')} + \varepsilon}, \quad \hat{W}_t^{(b,h)} = \frac{W_t^{(b,h)}}{\sum_{h'} W_t^{(b,h')} + \varepsilon},$$

and the per-head mismatch

$$\Delta_t^{(b,h)} = \hat{A}_t^{(b,h)} - \hat{W}_t^{(b,h)},$$

which highlights heads whose attention mass to registers is disproportionate to their effective width-space write from registers.

Inter-model delta analysis. Finally, to quantify mechanistic changes induced by fine-tuning, we compute aligned inter-model differences on the intersection of probed blocks. For two model tags t_a, t_b , define

$$\Delta A_{t_a-t_b} = A_{t_a} - A_{t_b}, \quad \Delta W_{t_a-t_b} = W_{t_a} - W_{t_b}, \quad \Delta S_{t_a-t_b} = S_{t_a} - S_{t_b} \quad (\text{if available}),$$

and the delta-of-mismatch

$$\Delta(\Delta)_{t_a-t_b} = (\hat{A}_{t_a} - \hat{W}_{t_a}) - (\hat{A}_{t_b} - \hat{W}_{t_b}).$$

These inter-model deltas isolate which blocks and heads change most strongly in (i) register-directed attention mass, (ii) register-sourced write magnitude, and (iii) causal survival to the final embedding.

B Modality Gap and Embedding Geometry Metrics

This appendix defines the modality-gap and geometry statistics reported in Sec. 6 (“Model geometry analysis”). All metrics below are computed from *L2-normalized* CLIP embeddings unless stated otherwise.

Embeddings and normalization. Let $\{f_i\}_{i=1}^{N_I} \subset \mathbb{R}^D$ be image features and $\{g_j\}_{j=1}^{N_T} \subset \mathbb{R}^D$ be text features, where each is normalized:

$$\|f_i\|_2 = 1, \quad \|g_j\|_2 = 1.$$

We denote modality means

$$\mu_I = \frac{1}{N_I} \sum_{i=1}^{N_I} f_i, \quad \mu_T = \frac{1}{N_T} \sum_{j=1}^{N_T} g_j,$$

and their unit directions

$$\hat{\mu}_I = \frac{\mu_I}{\|\mu_I\|_2 + \varepsilon}, \quad \hat{\mu}_T = \frac{\mu_T}{\|\mu_T\|_2 + \varepsilon},$$

with $\varepsilon = 10^{-12}$.

B.1 Center-based modality gap metrics

Euclidean mean gap.

$$\text{gap}_{\text{center,euclid}} = \|\mu_I - \mu_T\|_2.$$

Cosine mean gap. Define

$$\cos(\mu_I, \mu_T) = \hat{\mu}_I^\top \hat{\mu}_T, \quad \text{gap}_{\text{center,cosine}} = 1 - \cos(\mu_I, \mu_T).$$

Diagonal Mahalanobis mean gap. Let per-dimension variances be

$$(\sigma_I^2)_d = \text{Var}((f_i)_d) + 10^{-6}, \quad (\sigma_T^2)_d = \text{Var}((g_j)_d) + 10^{-6},$$

and define pooled diagonal variance

$$(\sigma_P^2)_d = \frac{1}{2}((\sigma_I^2)_d + (\sigma_T^2)_d).$$

Then

$$\text{gap}_{\text{mahalanobis,diag}} = \left(\sum_{d=1}^D \frac{(\mu_{I,d} - \mu_{T,d})^2}{(\sigma_P^2)_d} \right)^{1/2}.$$

Diagonal Frechet-style gap (FID with diagonal covariance). With the same diagonal variances,

$$\text{gap}_{\text{frechet,diag}} = \|\mu_I - \mu_T\|_2^2 + \sum_{d=1}^D \left[(\sigma_I^2)_d + (\sigma_T^2)_d - 2\sqrt{(\sigma_I^2)_d(\sigma_T^2)_d} \right].$$

(We report this scalar as an additional shape+shift discrepancy proxy.)

B.2 Distributional gap metrics (MMD on the sphere)

RBF Maximum Mean Discrepancy (MMD). Let $X = \{f_i\}_{i=1}^m$ and $Y = \{g_j\}_{j=1}^m$ be equal-size random subsets sampled without replacement from image/text embeddings (we use $m = \min(\text{MMD_SUBSET}, N_I, N_T)$). For a Gaussian kernel $k_\gamma(x, y) = \exp\{-\gamma\|x - y\|_2^2\}$, the (biased) MMD estimator is

$$\text{MMD}^2(X, Y) = \frac{1}{m(m-1)} \sum_{i \neq i'} k_\gamma(f_i, f_{i'}) + \frac{1}{m(m-1)} \sum_{j \neq j'} k_\gamma(g_j, g_{j'}) - \frac{2}{m^2} \sum_{i=1}^m \sum_{j=1}^m k_\gamma(f_i, g_j).$$

We follow the code by setting γ via the median heuristic on cross distances: let $D_{ij} = \|f_i - g_j\|_2^2$ and $\tilde{d} = \text{median}\{D_{ij}\}$, then

$$\gamma = \frac{1}{2(\tilde{d} + \varepsilon)}.$$

We report $\text{gap}_{\text{mmd}, \text{rbf}} := \text{MMD}^2(X, Y)$.

B.3 Similarity distributions: II, TT, and IT

All similarities are cosine similarities because embeddings are normalized.

Matched image–text similarity (IT). Given a mapping $\pi(j) \in \{1, \dots, N_I\}$ associating each text embedding g_j to its corresponding image embedding $f_{\pi(j)}$, define

$$s_j^{\text{IT}} = f_{\pi(j)}^\top g_j.$$

We report mean and std: $\text{sim}_{\text{IT}, \text{mean}} = \frac{1}{N_T} \sum_j s_j^{\text{IT}}$.

Within-image text–text similarity (TT). Let $\mathcal{T}(i) = \{j : \pi(j) = i\}$ be the caption indices for image i . For each i with $|\mathcal{T}(i)| \geq 2$ we form all unordered caption pairs $\{(j, j') : j < j', j, j' \in \mathcal{T}(i)\}$ and compute

$$s_{(j, j')}^{\text{TT}} = g_j^\top g_{j'}.$$

We pool these values across images and summarize by mean/std.

Image–image similarity (II). To avoid quadratic cost, we sample random unordered pairs (i, i') , $i \neq i'$, and compute $s_{(i, i')}^{\text{II}} = f_i^\top f_{i'}$.

Mean-gap summaries (used as descriptive statistics). When finite,

$$\text{gap}_{\text{mean}, \text{II-IT}} = \mathbb{E}[s^{\text{II}}] - \mathbb{E}[s^{\text{IT}}], \quad \text{gap}_{\text{mean}, \text{TT-IT}} = \mathbb{E}[s^{\text{TT}}] - \mathbb{E}[s^{\text{IT}}].$$

Histogram-based Jensen–Shannon distance (JSD) between similarity distributions.

Let $p_{\text{IT}}, p_{\text{TT}}, p_{\text{II}}$ be discrete distributions obtained by histogramming each similarity set into a common set of bins over $[-1, 1]$, normalized to sum to 1 (with a small ε floor, as in the code). The Jensen–Shannon distance between p and q is

$$\text{JSD}(p, q) = \sqrt{\frac{1}{2} \text{KL}(p \parallel m) + \frac{1}{2} \text{KL}(q \parallel m)}, \quad m = \frac{1}{2}(p + q),$$

where KL is the discrete Kullback–Leibler divergence.

Wasserstein-1 and Kolmogorov–Smirnov statistics. For empirical samples S and S' , the code also reports: (i) $W_1(S, S')$ (1D Earth-Mover distance), and (ii) the two-sample KS statistic $\sup_x |\hat{F}(x) - \hat{F}'(x)|$, computed on the cosine samples.

B.4 Alignment and uniformity (Wang–Isola)

Alignment (matched IT distance). Using matched cosines s_j^{IT} , define squared Euclidean distance on the sphere:

$$\|f_{\pi(j)} - g_j\|_2^2 = 2 - 2 f_{\pi(j)}^\top g_j = 2 - 2 s_j^{\text{IT}}.$$

We report

$$\text{align}_{\text{IT}} = \mathbb{E}_j [2 - 2 s_j^{\text{IT}}].$$

Uniformity. For a modality embedding set $\{z_n\}_{n=1}^N$ on the unit sphere, uniformity at temperature t is estimated by sampling random pairs (n, n') , $n \neq n'$, and computing

$$\text{unif}(z; t) = \log \mathbb{E} \left[\exp(-t \|z_n - z_{n'}\|_2^2) \right], \quad \|z_n - z_{n'}\|_2^2 = 2 - 2 z_n^\top z_{n'}.$$

We use $t = 2$ (Wang & Isola) and a fixed number of sampled pairs.

B.5 Covariance geometry: cone-R, kappa, sphericity, and effective rank

Centered covariance eigen-decomposition. Given normalized embeddings $z_n \in \mathbb{R}^D$ in a modality, define

$$\mu = \frac{1}{N} \sum_{n=1}^N z_n, \quad C = \frac{1}{N-1} \sum_{n=1}^N (z_n - \mu)(z_n - \mu)^\top.$$

Let $(\lambda_1, \dots, \lambda_D)$ be the eigenvalues of C (nonnegative, sorted ascending in code), and $U \in \mathbb{R}^{D \times D}$ the eigenvector matrix.

Cone-ness: $\text{cone_R} = \|\mu\|_2$. We report

$$\text{cone_R} = \|\mu\|_2,$$

separately for image and text. Large $\|\mu\|_2$ indicates embeddings concentrate around a dominant mean direction.

vMF-style concentration proxy κ (heuristic). The code defines a rough concentration proxy (motivated by vMF asymptotics on S^{D-1}): for $R = \text{cone_R}$,

$$\kappa \approx \frac{R(D - R^2)}{1 - R^2}, \quad R \in (0, 1).$$

(This is used descriptively; it is not a fitted vMF parameter.)

Effective rank (entropy form). Let $p_i = \lambda_i / \sum_{j=1}^D \lambda_j$. The spectral entropy is

$$H(p) = - \sum_{i=1}^D p_i \log(p_i),$$

and the effective rank is

$$\text{eff_rank}_{\text{entropy}} = \exp(H(p)).$$

Participation ratio.

$$\text{eff_rank}_{\text{PR}} = \frac{\left(\sum_{i=1}^D \lambda_i\right)^2}{\sum_{i=1}^D \lambda_i^2}.$$

Sphericity proxy. We report a scale-invariant sphericity proxy in $[0, 1]$:

$$\text{sphericity} = \frac{\left(\prod_{i=1}^D (\lambda_i + \varepsilon)\right)^{1/D}}{\frac{1}{D} \sum_{i=1}^D \lambda_i + \varepsilon} = \frac{\text{GM}(\lambda)}{\text{AM}(\lambda)}.$$

Degenerate spectra (strong anisotropy) yield values near 0.

Additional anisotropy summaries. The code also uses:

$$\text{eig_cv} = \frac{\text{Std}(\lambda)}{\text{Mean}(\lambda)}, \quad \text{cond}_{95/05} = \frac{Q_{0.95}(\lambda) + \varepsilon}{Q_{0.05}(\lambda) + \varepsilon}, \quad \text{top_eig_frac} = \frac{\max_i \lambda_i}{\sum_j \lambda_j + \varepsilon}.$$

Centroid-direction cosine distribution. Define the centroid direction $\hat{\mu} = \mu/(\|\mu\|_2 + \varepsilon)$. For each embedding z_n we compute

$$c_n = z_n^\top \hat{\mu} = \cos(z_n, \hat{\mu}),$$

and report histograms/ECDFs of $\{c_n\}$ for images and texts separately.

B.6 PCA explained variance ratios and cross-modal principal angles

Explained variance ratio (EVR). Using covariance eigenvalues λ_i in *descending* order $\lambda_{(1)} \geq \dots \geq \lambda_{(D)}$, define

$$\text{EVR}_k = \frac{\lambda_{(k)}}{\sum_{j=1}^D \lambda_{(j)}}, \quad \text{cumEVR}_k = \sum_{j=1}^k \text{EVR}_j.$$

We report $\{\text{cumEVR}_k\}_{k=1}^K$ for $K = \text{PCA_TOPK}$ for image, text, and pooled (image+text) embeddings.

Principal angles between top- k covariance subspaces (image vs. text). Let $U_I \in \mathbb{R}^{D \times k}$ and $U_T \in \mathbb{R}^{D \times k}$ be matrices whose columns are the top- k covariance eigenvectors (largest eigenvalues) for image and text, respectively. Compute singular values of $M = U_I^\top U_T$:

$$\sigma_1 \geq \dots \geq \sigma_k, \quad \sigma_\ell \in [0, 1].$$

The principal angles are

$$\theta_\ell = \arccos(\sigma_\ell), \quad \theta_\ell^\circ = \frac{180}{\pi} \theta_\ell.$$

We summarize by min / median / mean / max over $\{\theta_\ell^\circ\}_{\ell=1}^k$ with $k = \text{PRINCIPAL_ANGLES_K}$.

C Appendix: Tables / Data.

Table 13: Decoupling experiments: Flickr8k: final-layer patch/REG decoupling and contribution diagnostics (threshold $\tau = 70$).

Metric	pretrained	regr-norm	regr-brut
CKA(full, patch-only)	0.997868	0.999664	0.999888
CKA(full, reg-only)	0.888133	0.600534	0.737377
Attn/skip ratio (mean)	0.400625	0.711000	0.675470
Reg count $> \tau$ (mean)	2.444	2.056	1.747
Reg count $> \tau$ (min / max)	1 / 6	1 / 4	1 / 4
Reg count $> \tau$ (zero frac.)	0.0000	0.0000	0.0000
Patch count $\leq \tau$ (mean)	253.556	253.944	254.253
Patch norms $\leq \tau$ (mean)	23.852	24.333	22.567
Reg norms $> \tau$ (mean)	179.532	172.074	170.016
Faith (TOKEN) mean	-0.394301	0.149746	-0.014661
Faith (TOKEN) no-REGs mean	-0.357839	0.138118	-0.019703
Faith (TOKEN) REGs-only mean	-0.008859	0.108917	0.132816
Faith (VALUE) mean	0.915536	0.826883	0.863328
Faith (VALUE) no-REGs mean	0.864358	0.820621	0.858823
Faith (VALUE) REGs-only mean	0.466978	0.238490	0.380568
cos(FULL, PATCH) mean	0.996676	0.998648	0.999597
cos(FULL, REG) mean	0.844343	0.252561	0.196064
cos(PATCH, REG) mean	0.831843	0.247820	0.192458
Margin below $[\tau-10, \tau]$: frac_any	0.0450	0.0476	0.0420
Margin above $(\tau, \tau+30]$: frac_any	0.1468	0.1478	0.1548

Table 14: Decoupling experiments: ImageNet-1k one-shot top-5 accuracy (mean $\pm$ std over 3 seeds) for FULL, PATCH-only, and REG-only embeddings.

Metric	pretrained	regr-norm	regr-brut
Top-5 (FULL)	0.445333 $\pm$ 0.014522	0.602000 $\pm$ 0.012675	0.537667 $\pm$ 0.009104
Top-5 (PATCH-only)	0.448333 $\pm$ 0.013275	0.602667 $\pm$ 0.011086	0.538333 $\pm$ 0.009568
Top-5 (REG-only)	0.169000 $\pm$ 0.003742	0.008667 $\pm$ 0.001247	0.015000 $\pm$ 0.002944

Table 15: Decoupling experiments: SCAM probe results reported per split (NoSCAM, SCAM, SynthSCAM) for each model. Each split has $n = 1162$ images.

Split	Metric	pretrained	regr-norm	regr-brut
NoSCAM	cos(FULL, REG)	0.793406	0.289478	0.303599
	Additional REG norm (mean)	4.166453	4.234234	4.245158
	Additional REG cosine contribution (mean)	0.552401	0.304923	0.328850
SCAM	cos(FULL, REG)	0.740587	0.260496	0.288447
	Additional REG norm (mean)	4.173426	4.236292	4.246686
	Additional REG cosine contribution (mean)	0.527288	0.311570	0.339518
SynthSCAM	cos(FULL, REG)	0.744529	0.260824	0.285420
	Additional REG norm (mean)	4.171109	4.234714	4.245596
	Additional REG cosine contribution (mean)	0.532084	0.312496	0.342577

Table 16: Decoupling experiments: REG leakage into FULL over blocks (patch/REG masking at each block, final embedding readout): mean cosines and CKA.

Block	Metric	pretrained	regr-norm	regr-brut
b12	cos(FULL, PATCH)	0.9995	0.9998	0.9999
	cos(FULL, REG)	0.9931	0.9985	0.9986
	cos(PATCH, REG)	0.9913	0.9981	0.9983
	CKA(FULL, PATCH)	0.9994	1.0000	1.0000
	CKA(FULL, REG)	0.9904	0.9996	0.9997
	Reg count (mean)	2.420	2.050	1.730
b13	cos(FULL, PATCH)	0.9986	0.9995	0.9994
	cos(FULL, REG)	0.9972	0.9991	0.9992
	cos(PATCH, REG)	0.9950	0.9983	0.9982
	CKA(FULL, PATCH)	0.9982	0.9999	0.9999
	CKA(FULL, REG)	0.9964	0.9998	0.9998
	Reg count (mean)	2.420	2.050	1.729
b19	cos(FULL, PATCH)	0.9978	0.9969	0.9968
	cos(FULL, REG)	0.9800	0.9974	0.9967
	cos(PATCH, REG)	0.9735	0.9934	0.9919
	CKA(FULL, PATCH)	0.9977	0.9996	0.9996
	CKA(FULL, REG)	0.9779	0.9994	0.9993
	Reg count (mean)	2.426	2.048	1.729
b20	cos(FULL, PATCH)	0.9977	0.9898	0.9932
	cos(FULL, REG)	0.9757	0.9973	0.9919
	cos(PATCH, REG)	0.9703	0.9859	0.9821
	CKA(FULL, PATCH)	0.9977	0.9987	0.9993
	CKA(FULL, REG)	0.9748	0.9994	0.9970
	Reg count (mean)	2.422	2.044	1.725
b21	cos(FULL, PATCH)	0.9980	0.9932	0.9951
	cos(FULL, REG)	0.9660	0.9912	0.9882
	cos(PATCH, REG)	0.9593	0.9890	0.9829
	CKA(FULL, PATCH)	0.9980	0.9994	0.9994
	CKA(FULL, REG)	0.9668	0.9968	0.9930
	Reg count (mean)	2.429	2.047	1.730
b22	cos(FULL, PATCH)	0.9991	0.9916	0.9974
	cos(FULL, REG)	0.9478	0.9937	0.9946
	cos(PATCH, REG)	0.9452	0.9878	0.9936
	CKA(FULL, PATCH)	0.9991	0.9995	0.9998
	CKA(FULL, REG)	0.9488	0.9980	0.9978
	Reg count (mean)	2.431	2.046	1.736
b23	cos(FULL, PATCH)	0.9967	0.9986	0.9996
	cos(FULL, REG)	0.8443	0.2526	0.1961
	cos(PATCH, REG)	0.8318	0.2478	0.1925
	CKA(FULL, PATCH)	0.9979	0.9997	0.9999
	CKA(FULL, REG)	0.8881	0.6005	0.7374
	Reg count (mean)	2.444	2.056	1.747

Table 17: Inference Experiments: Overview of zero-shot accuracy on Typographic Attacks and MVT. CLS-PATCHREG-I: Probe trained on ImageNet-1k (ILSVRC2012), 3 images per class, n=6000; CLS-PATCHREG-N: Probe trained on NoSCAM, n=1162.

Dataset (n)	Method	pretrained	regr-norm	regr-brut
NoSCAM (1162)	CLS	0.9905	0.9897	0.9897
NoSCAM (1162)	CLS-PATCHSUB	0.9544	0.9845	0.9811
NoSCAM (1162)	CLS-PATCHREG-I	0.9466	0.9888	0.9888
NoSCAM (1162)	CLS-PATCHREG-N	0.9871	0.9897	0.9888
NoSCAM (1162)	REG-L23-NOPC	0.9380	0.9613	0.9570
NoSCAM (1162)	REG-L23-1PC	0.9630	0.9802	0.9802
NoSCAM (1162)	REG-L23-8PC	0.9509	0.9664	0.9604
NoSCAM (1162)	PATCH-L23	0.7349	0.9725	0.9716
NoSCAM (1162)	PATCH Δ	0.9690	0.9905	0.9888
SCAM (1162)	CLS	0.4182	0.8038	0.8830
SCAM (1162)	CLS-PATCHSUB	0.4957	0.8632	0.9002
SCAM (1162)	CLS-PATCHREG-I	0.8761	0.8537	0.9174
SCAM (1162)	CLS-PATCHREG-N	0.9286	0.8537	0.9165
SCAM (1162)	REG-L23-NOPC	0.7410	0.8244	0.7719
SCAM (1162)	REG-L23-1PC	0.7539	0.8726	0.7943
SCAM (1162)	REG-L23-8PC	0.7057	0.8038	0.7143
SCAM (1162)	PATCH-L23	0.6024	0.7470	0.8623
SCAM (1162)	PATCH Δ	0.8778	0.8451	0.8744
SynthSCAM (1162)	CLS	0.3219	0.8021	0.8804
SynthSCAM (1162)	CLS-PATCHSUB	0.4406	0.8580	0.9071
SynthSCAM (1162)	CLS-PATCHREG-I	0.8890	0.8460	0.9200
SynthSCAM (1162)	CLS-PATCHREG-N	0.9449	0.8494	0.9200
SynthSCAM (1162)	REG-L23-NOPC	0.7823	0.8382	0.7771
SynthSCAM (1162)	REG-L23-1PC	0.8055	0.8812	0.8072
SynthSCAM (1162)	REG-L23-8PC	0.7289	0.8167	0.7126
SynthSCAM (1162)	PATCH-L23	0.6317	0.7470	0.8632
SynthSCAM (1162)	PATCH Δ	0.9217	0.8614	0.8769
MVT (200382)	CLS	0.8830	0.8730	0.8573
MVT (200382)	CLS-PATCHSUB	0.4720	0.8246	0.8057
MVT (200382)	CLS-PATCHREG-I	0.7166	0.8703	0.8518
MVT (200382)	CLS-PATCHREG-N	0.5695	0.8675	0.8478
MVT (200382)	REG-L23-NOPC	0.7640	0.7935	0.7680
MVT (200382)	REG-L23-1PC	0.7921	0.8193	0.8032
MVT (200382)	REG-L23-8PC	0.7724	0.8057	0.7812
MVT (200382)	PATCH-L23	0.3414	0.8652	0.8191
MVT (200382)	PATCH Δ	0.6881	0.8667	0.8510

Table 18: Standard benchmark evaluation: linear probe, retrieval, zero-shot (own scripts; [15]).

Benchmark	Metric	pretrained	regr-norm	regr-brut
ImageNet-1k (Linear Probe)	Top-1 (%)	72.35	70.94	65.09
ImageNet-1k (Linear Probe)	Top-5 (%)	93.42	93.29	89.60
MVT (200382)	ZS Acc	0.8830	0.8730	0.8573
MSCOCO Captions (2014 val)	I2T R@1	0.3366	0.3748	0.3508
MSCOCO Captions (2014 val)	I2T R@5	0.7882	0.8706	0.8502
MSCOCO Captions (2014 val)	T2I R@1	0.2153	0.3264	0.3184
MSCOCO Captions (2014 val)	T2I R@5	0.5902	0.7851	0.7821
MSCOCO Captions (2014 val)	T2T R@1	0.2064	0.2423	0.2359
MSCOCO Captions (2014 val)	T2T R@5	0.5516	0.6175	0.6130
MSCOCO Captions (2014 val)	T2T_IMG R@1	0.3120	0.3506	0.3275
MSCOCO Captions (2014 val)	T2T_IMG R@5	0.7466	0.8386	0.8179
SugarCrepe (COCO 2017 val)	add_obj acc	0.7842	0.9627	0.9515
SugarCrepe (COCO 2017 val)	add_att acc	0.7168	0.9205	0.8743
SugarCrepe (COCO 2017 val)	replace_obj acc	0.9407	0.9752	0.9740
SugarCrepe (COCO 2017 val)	replace_att acc	0.7919	0.8579	0.8388
SugarCrepe (COCO 2017 val)	replace_rel acc	0.6529	0.7752	0.7696
SugarCrepe (COCO 2017 val)	swap_obj acc	0.6041	0.7224	0.6898
SugarCrepe (COCO 2017 val)	swap_att acc	0.6261	0.7282	0.7102

Table 19: Standard benchmark evaluation: SugarCrepe (COCO 2017 val): margin statistics (mean μ and std σ) corresponding to Table 18.

Perturbation	pretrained ($\mu \pm \sigma$)	regr-norm ($\mu \pm \sigma$)	regr-brut ($\mu \pm \sigma$)
add_obj	0.0139 $\pm$ 0.0187	0.0698 $\pm$ 0.0450	0.0626 $\pm$ 0.0449
add_att	0.0085 $\pm$ 0.0155	0.0408 $\pm$ 0.0344	0.0341 $\pm$ 0.0347
replace_obj	0.0506 $\pm$ 0.0349	0.1408 $\pm$ 0.0793	0.1329 $\pm$ 0.0785
replace_att	0.0183 $\pm$ 0.0235	0.0533 $\pm$ 0.0568	0.0474 $\pm$ 0.0540
replace_rel	0.0095 $\pm$ 0.0198	0.0307 $\pm$ 0.0468	0.0309 $\pm$ 0.0475
swap_obj	0.0036 $\pm$ 0.0140	0.0147 $\pm$ 0.0294	0.0134 $\pm$ 0.0285
swap_att	0.0050 $\pm$ 0.0150	0.0206 $\pm$ 0.0364	0.0169 $\pm$ 0.0330

Table 20: Standard benchmark evaluation from CLIP Benchmark [2]: zero-shot and retrieval.

Benchmark	Metric	pretrained	regr-norm	regr-brut
VOC-2007 (multilabel)	Zero-shot acc	0.7615	0.8523	0.8350
ImageNet-1k (train split)	Zero-shot acc@1	0.3270	0.4566	0.4100
ImageNet-1k (train split)	Zero-shot acc@5	0.5300	0.6817	0.6513
ImageNet-1k (train split)	Zero-shot mean per-class recall	0.3261	0.4547	0.4078
MSCOCO Captions	image retrieval R@5	0.2196	0.3510	0.3308
MSCOCO Captions	text retrieval R@5	0.3032	0.5042	0.4758
XM3600	image retrieval R@5	0.3059	0.4254	0.4138
XM3600	text retrieval R@5	0.2429	0.4091	0.3874

Table 21: Flickr8k cross-modal geometry and gap. Uniformity: more negative = better spread on the sphere. Subspace angles are principal angles between top- $k = 64$ PCA subspaces of image vs. text embeddings (in degrees).

Metric	pretrained	regr-norm	regr-brut
Modality gap (centers / distributions)			
gap_center_euclid	0.829953	0.677479	0.456744
gap_center_cosine	0.751139	1.134299	0.925754
gap_frechet_diag	0.757185	0.482206	0.256970
gap_mmd_rbf	0.262025	0.143329	0.066291
Cosine similarity structure			
IT mean (paired)	0.2754	0.3555	0.3632
IT std	0.0360	0.0588	0.0570
II mean (within-image)	0.5594	0.2034	0.1440
II std	0.0889	0.1297	0.1746
TT mean (within-text)	0.6762	0.6591	0.5837
TT std	0.1357	0.1580	0.1762
gap_mean_ii_minus_it	0.284033	-0.152096	-0.219191
gap_mean_tt_minus_it	0.400859	0.303621	0.220491
Distance between similarity distributions			
JSD(IT,TT)	0.8036	0.7176	0.6259
JSD(IT,II)	0.8136	0.5905	0.6499
JSD(TT,II)	0.3970	0.7049	0.6390
WAS(IT,TT)	0.4009	0.3039	0.2241
WAS(IT,II)	0.2840	0.1625	0.2346
WAS(TT,II)	0.1188	0.4558	0.4397
KS(IT,TT)	0.9750	0.8756	0.7385
KS(IT,II)	0.9823	0.6780	0.7411
KS(TT,II)	0.4603	0.8508	0.7712
Alignment and log-mean-exp RBF uniformity			
alignment_it	1.449250	1.288955	1.273529
uniformity_i	-1.696647	-3.022580	-3.115553
uniformity_t	-2.474447	-3.046056	-3.525195
Centers and raw norms			
$\ \mu_i\ $	0.748186	0.451401	0.379880
$\ \mu_t\ $	0.590802	0.448189	0.283353
$\cos(\mu_i, \mu_t)$	0.248861	-0.134299	0.074246
RawNorm(I) mean	18.8972	11.9725	11.0717
RawNorm(I) std	0.8472	0.7114	0.7652
RawNorm(T) mean	12.9930	11.6109	11.2062
RawNorm(T) std	1.2299	0.2277	0.2639
Within-modality geometry (summary)			
I: cone_R	0.748186	0.451401	0.379880
I: κ (approx)	1304.33	435.28	340.89
I: sphericity	0.0979	0.0339	0.0190
I: participation ratio	63.43	47.93	29.58
I: eff_rank (entropy)	178.45	120.70	79.04
T: cone_R	0.590802	0.448189	0.283353
T: κ (approx)	696.72	430.62	236.59
T: sphericity	0.0659	0.0629	0.0728
T: participation ratio	54.30	57.88	70.03
T: eff_rank (entropy)	146.79	154.00	178.68
Cross-modal subspace relationship (PCA, $k = 64$)			
principal_angles mean (deg)	60.08	42.09	51.91
principal_angles median (deg)	61.71	36.77	48.47
principal_angles min (deg)	21.78	19.79	24.48
principal_angles max (deg)	89.84	89.79	89.90

Table 22: Model geometry diagnostics (SCAM).

Metric (dataset)	pretrained	regr-norm	regr-brut
Modality gap (centers / distributions)			
gap_center_euclid (NoSCAM)	0.903562	0.653420	0.619884
gap_center_euclid (Synth)	0.883528	0.678857	0.647022
gap_center_cosine (NoSCAM)	0.722889	0.680693	0.624538
gap_center_cosine (Synth)	0.719695	0.735809	0.665648
gap_frechet_diag (NoSCAM)	0.851941	0.489420	0.435345
gap_frechet_diag (Synth)	0.815382	0.522686	0.469950
gap_mmd_rbf (NoSCAM)	0.331892	0.160035	0.144128
gap_mmd_rbf (Synth)	0.314794	0.166847	0.153658
Cosine similarity structure			
IT mean (NoSCAM)	0.2294	0.2636	0.2657
IT mean (Synth)	0.2500	0.2802	0.2767
II mean (NoSCAM)	0.5817	0.3427	0.3541
II mean (Synth)	0.5376	0.3424	0.3672
TT mean (NoSCAM)	0.6973	0.5597	0.5618
TT mean (Synth)	0.6973	0.5597	0.5618
gap_mean_ii_minus_it (NoSCAM)	0.352373	0.079104	0.088480
gap_mean_ii_minus_it (Synth)	0.287583	0.062245	0.090487
gap_mean_tt_minus_it (NoSCAM)	0.467934	0.296105	0.296115
gap_mean_tt_minus_it (Synth)	0.447257	0.279537	0.285078
Distance between similarity distributions			
JSD(IT,TT) (NoSCAM)	0.8298	0.6183	0.6698
JSD(IT,TT) (Synth)	0.8264	0.6368	0.6980
JSD(IT,II) (NoSCAM)	0.8263	0.3065	0.3314
JSD(IT,II) (Synth)	0.8136	0.2626	0.3555
JSD(TT,II) (NoSCAM)	0.4118	0.5885	0.5709
JSD(TT,II) (Synth)	0.4987	0.6148	0.5765
WAS(IT,TT) (NoSCAM)	0.4679	0.2979	0.3021
WAS(IT,TT) (Synth)	0.4473	0.2951	0.3040
WAS(IT,II) (NoSCAM)	0.3524	0.0791	0.0885
WAS(IT,II) (Synth)	0.2876	0.0622	0.0905
WAS(TT,II) (NoSCAM)	0.1156	0.2535	0.2406
WAS(TT,II) (Synth)	0.1597	0.2589	0.2346
KS(IT,TT) (NoSCAM)	0.9975	0.7055	0.7537
KS(IT,TT) (Synth)	0.9937	0.6935	0.7567
KS(IT,II) (NoSCAM)	0.9942	0.2532	0.3052
KS(IT,II) (Synth)	0.9780	0.2651	0.3605
KS(TT,II) (NoSCAM)	0.4595	0.5911	0.5632
KS(TT,II) (Synth)	0.5839	0.6161	0.5539
Alignment and uniformity			
alignment_it (NoSCAM)	1.541271	1.472806	1.468687
alignment_it (Synth)	1.499917	1.439670	1.446614
uniformity_i (NoSCAM)	-1.630123	-2.549310	-2.469731
uniformity_i (Synth)	-1.812590	-2.570200	-2.428658
uniformity_t (NoSCAM)	-1.689004	-2.485864	-2.618859
uniformity_t (Synth)	-1.691141	-2.477322	-2.617044
Centers and raw norms			
$\ \mu_i\ $ (NoSCAM)	0.762989	0.585927	0.595475
$\ \mu_i\ $ (Synth)	0.733221	0.585431	0.606607
$\ \mu_e\ $ (NoSCAM)	0.739614	0.531547	0.505807
$\ \mu_e\ $ (Synth)	0.739614	0.531547	0.505807
$\cos(\mu_i, \mu_e)$ (NoSCAM)	0.277111	0.319307	0.375462
$\cos(\mu_i, \mu_e)$ (Synth)	0.280305	0.264191	0.334352
RawNorm(I) mean (NoSCAM)	17.6180	12.1546	11.5411
RawNorm(I) mean (Synth)	17.3443	12.2670	11.4330
RawNorm(T) mean (NoSCAM)	14.6747	11.8679	11.7140
RawNorm(T) mean (Synth)	14.6747	11.8679	11.7140
Within-modality geometry (summary)			
I: cone_R (NoSCAM)	0.762989	0.585927	0.595475
I: cone_R (Synth)	0.733221	0.585431	0.606607
I: κ (NoSCAM)	1401.31	684.94	708.25
I: κ (Synth)	1216.99	683.75	736.76
I: sphericity (NoSCAM)	0.0843	0.0363	0.0209
I: sphericity (Synth)	0.0840	0.0378	0.0226
I: participation ratio (NoSCAM)	75.92	66.80	44.59
I: participation ratio (Synth)	99.51	88.24	49.11
I: eff_rank (NoSCAM)	186.75	147.13	105.19
I: eff_rank (Synth)	211.86	172.82	115.28
T: cone_R (NoSCAM)	0.739614	0.531547	0.505807
T: cone_R (Synth)	0.739614	0.531547	0.505807
T: sphericity (NoSCAM)	0.0534	0.0501	0.0537
T: sphericity (Synth)	0.0534	0.0501	0.0537
T: participation ratio (NoSCAM)	23.61	21.66	29.91
T: participation ratio (Synth)	23.61	21.66	29.91
T: eff_rank (NoSCAM)	105.00	109.33	129.84
T: eff_rank (Synth)	105.00	109.33	129.84
Cross-modal subspace relationship (PCA, $k = 64$)			
principal_angles mean (NoSCAM)	59.47	51.04	55.36
principal_angles mean (Synth)	57.41	48.76	54.38
principal_angles median (NoSCAM)	58.70	47.28	52.43
principal_angles median (Synth)	55.54	44.64	52.17
principal_angles min (NoSCAM)	26.42	21.71	26.93
principal_angles min (Synth)	26.33	21.20	26.03
principal_angles max (NoSCAM)	89.40	89.90	89.74
principal_angles max (Synth)	89.53	89.93	89.69

Table 23: Causal recoverability via subspace surgery across ViT blocks. Each cell reports recoverability using *REG-only* subspace surgery / *non-REG-only* subspace surgery.

Block	pretrained	regr-norm	regr-brut
00	0.5415 / 0.9937	0.2038 / 0.9980	0.2064 / 0.9976
01	0.5712 / 0.9989	0.2201 / 0.9990	0.2261 / 0.9985
02	0.5187 / 0.9943	0.1898 / 0.9973	0.1799 / 0.9969
03	0.5038 / 0.9982	0.2363 / 0.9981	0.2301 / 0.9976
04	0.5929 / 0.9981	0.2523 / 0.9979	0.2107 / 0.9978
05	0.5294 / 0.9973	0.2423 / 0.9970	0.2518 / 0.9965
06	0.5764 / 0.9982	0.2295 / 0.9984	0.2228 / 0.9977
07	0.5805 / 0.9981	0.2694 / 0.9976	0.2731 / 0.9970
08	0.5809 / 0.9951	0.2517 / 0.9947	0.2713 / 0.9922
09	0.6294 / 0.9792	0.3145 / 0.9472	0.2467 / 0.9435
10	0.6111 / 0.9848	0.2950 / 0.9813	0.2487 / 0.9732
11	0.6223 / 0.9377	0.3219 / 0.8904	0.2619 / 0.8444
12	0.6303 / 0.8219	0.3530 / 0.6399	0.2201 / 0.4842
13	0.5887 / 0.8410	0.2539 / 0.6651	0.2182 / 0.5034
14	0.6378 / 0.7900	0.3434 / 0.5956	0.2794 / 0.4248
15	0.6486 / 0.8054	0.3361 / 0.6797	0.2801 / 0.5752
16	0.7089 / 0.8009	0.4030 / 0.6860	0.3450 / 0.6087
17	0.7148 / 0.7801	0.4130 / 0.7121	0.3391 / 0.6520
18	0.7366 / 0.8398	0.4292 / 0.7233	0.3787 / 0.6924
19	0.7604 / 0.8584	0.4470 / 0.8001	0.4157 / 0.7246
20	0.7867 / 0.9189	0.4672 / 0.8525	0.4318 / 0.8348
21	0.8327 / 0.9365	0.5076 / 0.8622	0.4696 / 0.8626
22	0.8860 / 0.9463	0.6798 / 0.9102	0.6926 / 0.9228
23	1.0000 / 1.0000	1.0000 / 1.0000	1.0000 / 1.0000

Table 24: Cross-image consistency and late-depth register proliferation by ViT block. For each model and block we report registers-per-image statistics (mean μ with min-max), the number of retained cross-image components k (**kept_comps**), and the maximum subspace-overlap singular value (**ovMaxSV**) plus mean principal angle in degrees (**PAdegMean**).

Block	pretrained	regr-norm	regr-brut
00-09	$\mu=0.00$ [0-0], $k=0$	$\mu=0.00$ [0-0], $k=0$	$\mu=0.00$ [0-0], $k=0$
10	$\mu=2.00$ [1-3], $k=1$, $ov=0.645$, PA=71.24	$\mu=1.75$ [1-3], $k=1$, $ov=0.695$, PA=67.83	$\mu=1.75$ [1-3], $k=1$, $ov=0.699$, PA=67.86
11	$\mu=3.88$ [2-6], $k=1$, $ov=0.559$, PA=73.41	$\mu=2.88$ [1-4], $k=1$, $ov=0.556$, PA=73.02	$\mu=2.25$ [1-3], $k=1$, $ov=0.583$, PA=68.81
12	$\mu=4.00$ [2-6], $k=1$, $ov=0.808$, PA=67.73	$\mu=3.12$ [2-4], $k=1$, $ov=0.768$, PA=66.17	$\mu=2.50$ [1-4], $k=1$, $ov=0.756$, PA=64.50
13	$\mu=4.00$ [2-6], $k=1$, $ov=0.793$, PA=69.07	$\mu=5.75$ [4-9], $k=2$, $ov=0.735$, PA=73.94	$\mu=4.50$ [2-7], $k=2$, $ov=0.737$, PA=71.39
14	$\mu=6.88$ [4-11], $k=2$, $ov=0.759$, PA=73.46	$\mu=5.75$ [4-9], $k=2$, $ov=0.717$, PA=74.32	$\mu=4.50$ [2-7], $k=2$, $ov=0.721$, PA=71.39
15	$\mu=6.75$ [4-11], $k=2$, $ov=0.764$, PA=73.11	$\mu=5.25$ [3-8], $k=2$, $ov=0.726$, PA=73.19	$\mu=4.62$ [2-7], $k=2$, $ov=0.744$, PA=71.00
16	$\mu=4.00$ [2-6], $k=1$, $ov=0.743$, PA=70.16	$\mu=3.12$ [2-4], $k=1$, $ov=0.703$, PA=68.60	$\mu=3.75$ [2-6], $k=2$, $ov=0.737$, PA=70.16
17	$\mu=4.00$ [2-6], $k=1$, $ov=0.730$, PA=70.54	$\mu=3.12$ [2-4], $k=1$, $ov=0.678$, PA=68.37	$\mu=2.50$ [1-4], $k=1$, $ov=0.749$, PA=63.69
18	$\mu=4.00$ [2-6], $k=1$, $ov=0.694$, PA=71.16	$\mu=3.12$ [2-4], $k=1$, $ov=0.656$, PA=68.66	$\mu=2.50$ [1-4], $k=1$, $ov=0.707$, PA=63.99
19	$\mu=4.00$ [2-6], $k=1$, $ov=0.653$, PA=71.39	$\mu=3.12$ [2-4], $k=1$, $ov=0.640$, PA=69.18	$\mu=3.75$ [2-6], $k=2$, $ov=0.698$, PA=67.57
20	$\mu=4.00$ [2-6], $k=1$, $ov=0.698$, PA=69.21	$\mu=5.00$ [3-8], $k=2$, $ov=0.783$, PA=66.53	$\mu=3.88$ [2-6], $k=2$, $ov=0.689$, PA=67.69
21	$\mu=4.00$ [2-6], $k=1$, $ov=0.648$, PA=69.28	$\mu=5.00$ [3-8], $k=2$, $ov=0.747$, PA=67.94	$\mu=3.88$ [2-6], $k=2$, $ov=0.729$, PA=65.75
22	$\mu=4.00$ [2-6], $k=1$, $ov=0.596$, PA=71.54	$\mu=3.12$ [2-4], $k=1$, $ov=0.655$, PA=68.41	$\mu=22.62$ [2-63], $k=2$, $ov=0.890$, PA=63.85
23	$\mu=4.00$ [2-6], $k=1$, $ov=0.586$, PA=70.42	$\mu=3.00$ [2-4], $k=1$, $ov=0.571$, PA=69.27	$\mu=2.50$ [1-4], $k=1$, $ov=0.476$, PA=72.59

Table 25: Register-projection geometry in patch space. Reference register basis U_{reg} is built from **pretrained** blocks (11,12) with dimension 1024×16 (63 register vectors pooled). Per block, we report: mean projection fraction (**fracMean**), top- k share (**topkShare**), concentration (**gini**), and maximum overlap singular value with the reference basis (**ovMaxSV**).

Block	pretrained	regr-norm	regr-brut
00	f=0.1365, t=0.0795, g=0.0553, ov=0.6862	f=0.1351, t=0.0793, g=0.0548, ov=0.6876	f=0.1360, t=0.0791, g=0.0541, ov=0.6874
01	f=0.1373, t=0.0803, g=0.0623, ov=0.6553	f=0.1368, t=0.0804, g=0.0624, ov=0.6563	f=0.1390, t=0.0797, g=0.0596, ov=0.6567
02	f=0.1400, t=0.0769, g=0.0537, ov=0.6468	f=0.1399, t=0.0771, g=0.0534, ov=0.6491	f=0.1423, t=0.0765, g=0.0534, ov=0.6516
03	f=0.1319, t=0.0863, g=0.0840, ov=0.6635	f=0.1332, t=0.0865, g=0.0838, ov=0.6687	f=0.1350, t=0.0859, g=0.0825, ov=0.6694
04	f=0.1272, t=0.0878, g=0.0911, ov=0.6844	f=0.1293, t=0.0882, g=0.0916, ov=0.6925	f=0.1306, t=0.0877, g=0.0903, ov=0.6941
05	f=0.1424, t=0.0828, g=0.0821, ov=0.7086	f=0.1436, t=0.0834, g=0.0829, ov=0.7131	f=0.1453, t=0.0828, g=0.0819, ov=0.7144
06	f=0.1167, t=0.1059, g=0.1374, ov=0.7540	f=0.1178, t=0.1072, g=0.1403, ov=0.7560	f=0.1209, t=0.1045, g=0.1325, ov=0.7574
07	f=0.1029, t=0.1106, g=0.1633, ov=0.7431	f=0.1059, t=0.1114, g=0.1646, ov=0.7509	f=0.1091, t=0.1087, g=0.1553, ov=0.7519
08	f=0.0990, t=0.1414, g=0.2209, ov=0.7417	f=0.1026, t=0.1362, g=0.2141, ov=0.7522	f=0.1064, t=0.1350, g=0.2082, ov=0.7523
09	f=0.1110, t=0.2726, g=0.3682, ov=0.9034	f=0.1165, t=0.2515, g=0.3448, ov=0.9016	f=0.1203, t=0.2417, g=0.3326, ov=0.9019
10	f=0.1302, t=0.4069, g=0.4744, ov=0.9574	f=0.1366, t=0.3562, g=0.4243, ov=0.9566	f=0.1417, t=0.3196, g=0.3896, ov=0.9574
11	f=0.1492, t=0.9028, g=0.9054, ov=1.0000	f=0.1638, t=0.8691, g=0.8750, ov=1.0000	f=0.1733, t=0.8312, g=0.8417, ov=0.9999
12	f=0.1673, t=0.9853, g=0.9769, ov=1.0000	f=0.1845, t=0.9796, g=0.9734, ov=1.0000	f=0.1953, t=0.9746, g=0.9699, ov=1.0000
13	f=0.1584, t=0.9847, g=0.9765, ov=1.0000	f=0.1802, t=0.9781, g=0.9723, ov=1.0000	f=0.2032, t=0.9724, g=0.9680, ov=0.9999
14	f=0.1593, t=0.9828, g=0.9748, ov=1.0000	f=0.1850, t=0.9741, g=0.9687, ov=0.9999	f=0.2256, t=0.9633, g=0.9595, ov=0.9998
15	f=0.1604, t=0.9804, g=0.9727, ov=0.9998	f=0.1926, t=0.9688, g=0.9639, ov=0.9997	f=0.2466, t=0.9543, g=0.9514, ov=0.9996
16	f=0.1596, t=0.9744, g=0.9679, ov=0.9995	f=0.1901, t=0.9586, g=0.9558, ov=0.9994	f=0.2557, t=0.9399, g=0.9396, ov=0.9993
17	f=0.1464, t=0.9721, g=0.9660, ov=0.9994	f=0.1633, t=0.9562, g=0.9541, ov=0.9992	f=0.2236, t=0.9370, g=0.9377, ov=0.9992
18	f=0.1415, t=0.9657, g=0.9616, ov=0.9992	f=0.1474, t=0.9506, g=0.9503, ov=0.9991	f=0.1888, t=0.9352, g=0.9379, ov=0.9990
19	f=0.1354, t=0.9474, g=0.9509, ov=0.9989	f=0.1442, t=0.9216, g=0.9331, ov=0.9987	f=0.1679, t=0.9064, g=0.9215, ov=0.9986
20	f=0.1202, t=0.8955, g=0.9225, ov=0.9984	f=0.1258, t=0.8477, g=0.8872, ov=0.9983	f=0.1424, t=0.8716, g=0.8993, ov=0.9981
21	f=0.1220, t=0.8796, g=0.9086, ov=0.9970	f=0.1284, t=0.8329, g=0.8735, ov=0.9957	f=0.1048, t=0.8555, g=0.8894, ov=0.9959
22	f=0.1443, t=0.8536, g=0.8935, ov=0.9959	f=0.1278, t=0.8250, g=0.8751, ov=0.9955	f=0.1214, t=0.8886, g=0.9133, ov=0.9952
23	f=0.1742, t=0.7986, g=0.8401, ov=0.9955	f=0.1507, t=0.7582, g=0.8175, ov=0.9949	f=0.1394, t=0.8048, g=0.8368, ov=0.9947

Table 26: Registers explained variance and direction geometry (threshold $\tau=70$, $n=200$ images). Per layer l , we report: mean registers per image (m), intra-register angle in degrees (ir), inter-register angle in degrees (er), explained variance of the register direction (ev), cosine with mean register direction (cm), and probe cosine after projection (pp).

Layer	pretrained	regr-norm	regr-brut
12	m=3.620, ir=3.81, er=1.94, ev=0.797, cm=0.0140, pp=0.8397	m=3.130, ir=3.68, er=2.08, ev=0.792, cm=0.0150, pp=0.7094	m=2.695, ir=4.01, er=2.23, ev=0.808, cm=0.0184, pp=0.7464
13	m=3.620, ir=3.77, er=2.20, ev=0.748, cm=0.0145, pp=0.8401	m=3.130, ir=3.65, er=2.33, ev=0.740, cm=0.0157, pp=0.7206	m=2.695, ir=4.01, er=2.47, ev=0.764, cm=0.0196, pp=0.7636
20	m=3.625, ir=3.77, er=4.28, ev=0.432, cm=0.0227, pp=0.8616	m=3.085, ir=3.63, er=4.43, ev=0.400, cm=0.0237, pp=0.7863	m=2.695, ir=4.12, er=3.90, ev=0.541, cm=0.0251, pp=0.8417
21	m=3.640, ir=3.99, er=4.85, ev=0.401, cm=0.0280, pp=0.8631	m=3.115, ir=3.82, er=5.04, ev=0.372, cm=0.0304, pp=0.7948	m=2.695, ir=4.35, er=4.47, ev=0.500, cm=0.0298, pp=0.8525
22	m=3.645, ir=4.05, er=5.41, ev=0.349, cm=0.0340, pp=0.8683	m=3.115, ir=3.78, er=5.66, ev=0.306, cm=0.0346, pp=0.8128	m=2.710, ir=4.72, er=5.52, ev=0.409, cm=0.0486, pp=0.9101
23	m=3.695, ir=4.93, er=8.58, ev=0.242, cm=0.0778, pp=0.8644	m=3.135, ir=4.77, er=9.46, ev=0.197, cm=0.0890, pp=0.9063	m=2.745, ir=5.93, er=8.58, ev=0.330, cm=0.0866, pp=0.9285

Table 27: Registers explained variance: Aggregate diagnostics across the reported layers. We report mean projected probe cosine (pp), mean registers per image (m), mean explained variance (ev), mean inter-register angle (er), and mean cosine with mean register direction (cm).

Model	pp	m	ev	er (deg)	cm
pretrained	0.8562	3.6408	0.4949	4.5435	0.03184
regr-brut	0.8405	2.7058	0.5584	4.5282	0.03802
regr-norm	0.7884	3.1183	0.4678	4.8328	0.03474

Table 28: Register attention mass (attn) and write norm (write). Blocks 10–16.

Block	Head	pretrained		regr-norm		regr-brut	
		attn	write	attn	write	attn	write
10	0	0.0163	0.0235	0.0146	0.0213	0.0142	0.0207
	1	0.0633	0.0283	0.0648	0.0306	0.0676	0.0331
	2	0.0129	0.0212	0.0129	0.0218	0.0121	0.0206
	3	0.0179	0.0217	0.0173	0.0214	0.0163	0.0199
	4	0.0151	0.0275	0.0137	0.0249	0.0137	0.0248
	5	0.0428	0.0632	0.0408	0.0613	0.0414	0.0641
	6	0.0289	0.0493	0.0263	0.0458	0.0259	0.0456
	7	0.0123	0.0185	0.0114	0.0171	0.0103	0.0157
	8	0.0136	0.0163	0.0132	0.0157	0.0129	0.0154
	9	0.0098	0.0167	0.0096	0.0173	0.0094	0.0175
	10	0.0561	0.0315	0.0537	0.0313	0.0525	0.0318
	11	0.0097	0.0275	0.0117	0.0341	0.0118	0.0360
	12	0.0242	0.0311	0.0226	0.0300	0.0195	0.0269
	13	0.0088	0.0203	0.0081	0.0189	0.0083	0.0195
	14	0.0580	0.0272	0.0576	0.0275	0.0570	0.0265
	15	0.0042	0.0045	0.0040	0.0044	0.0038	0.0041
11	0	0.0135	0.0230	0.0173	0.0300	0.0211	0.0369
	1	0.0094	0.0128	0.0084	0.0116	0.0087	0.0119
	2	0.0620	0.0450	0.0643	0.0473	0.0763	0.0555
	3	0.0213	0.0547	0.0213	0.0553	0.0230	0.0587
	4	0.0260	0.0208	0.0326	0.0264	0.0344	0.0281
	5	0.0236	0.0407	0.0325	0.0578	0.0395	0.0699
	6	0.0244	0.0253	0.0292	0.0308	0.0344	0.0372
	7	0.0078	0.0402	0.0091	0.0468	0.0101	0.0509
	8	0.0301	0.0499	0.0362	0.0613	0.0422	0.0713
	9	0.0104	0.0245	0.0095	0.0228	0.0102	0.0244
	10	0.0142	0.0201	0.0180	0.0258	0.0218	0.0303
	11	0.0286	0.0575	0.0337	0.0678	0.0388	0.0758
	12	0.0188	0.0229	0.0196	0.0243	0.0221	0.0270
	13	0.0405	0.1033	0.0391	0.0983	0.0447	0.1079
	14	0.0627	0.0454	0.0728	0.0526	0.0833	0.0596
	15	0.0382	0.0455	0.0407	0.0502	0.0426	0.0519
12	0	0.1850	0.1378	0.1597	0.1182	0.1379	0.1010
	1	0.1085	0.2828	0.0998	0.2576	0.0853	0.2180
	2	0.1363	0.3059	0.1216	0.2705	0.1091	0.2394
	3	0.1691	0.1150	0.1302	0.0878	0.1029	0.0691
	4	0.0953	0.0482	0.0769	0.0382	0.0840	0.0417
	5	0.1572	0.1077	0.1043	0.0710	0.0768	0.0516
	6	0.0945	0.0976	0.0754	0.0765	0.0669	0.0668
	7	0.1388	0.1021	0.1209	0.0888	0.1124	0.0822
	8	0.1486	0.1122	0.1362	0.1013	0.1241	0.0918
	9	0.1129	0.2694	0.1147	0.2716	0.0971	0.2277
	10	0.1452	0.1515	0.1384	0.1420	0.1124	0.1131
	11	0.1352	0.0596	0.1239	0.0545	0.1085	0.0477
	12	0.4805	0.2052	0.3995	0.1687	0.3831	0.1611
	13	0.2261	0.1237	0.1846	0.1014	0.1552	0.0840
	14	0.5022	0.1678	0.4489	0.1517	0.4550	0.1535
	15	0.4160	0.1644	0.3876	0.1545	0.3449	0.1386
13	0	0.6869	0.1172	0.6446	0.1108	0.6621	0.1158
	1	0.4377	0.0887	0.4361	0.0890	0.4617	0.0911
	2	0.4722	0.1527	0.4897	0.1611	0.5305	0.1730
	3	0.6752	0.1029	0.6472	0.1026	0.6426	0.0982
	4	0.3157	0.1222	0.2161	0.0820	0.2296	0.0860
	5	0.7432	0.2766	0.6983	0.2459	0.7235	0.2435
	6	0.6374	0.1510	0.6166	0.1459	0.6343	0.1447
	7	0.4086	0.1628	0.3941	0.1557	0.4101	0.1587
	8	0.1474	0.1916	0.1364	0.1759	0.1271	0.1618
	9	0.6515	0.1734	0.6915	0.1797	0.7006	0.1749
	10	0.2644	0.1170	0.1774	0.0782	0.1990	0.0872
	11	0.2162	0.2519	0.1626	0.1870	0.1385	0.1567
	12	0.4220	0.1092	0.3637	0.0896	0.3188	0.0762
	13	0.6538	0.0913	0.6272	0.0920	0.6473	0.0939
	14	0.6837	0.1263	0.6478	0.1257	0.6482	0.1253
	15	0.3105	0.2238	0.2938	0.2106	0.2803	0.1982
14	0	0.5489	0.1292	0.5513	0.1254	0.6191	0.1370
	1	0.6573	0.0890	0.7112	0.1009	0.7630	0.1041
	2	0.5519	0.0865	0.4886	0.0765	0.5309	0.0820
	3	0.5883	0.1290	0.6772	0.1504	0.7316	0.1590
	4	0.4796	0.1530	0.5170	0.1679	0.5861	0.1893
	5	0.6966	0.1646	0.7109	0.1741	0.7580	0.1885
	6	0.6653	0.1327	0.6897	0.1359	0.7160	0.1401
	7	0.6063	0.1313	0.6318	0.1373	0.6520	0.1415
	8	0.5688	0.1257	0.5721	0.1259	0.5687	0.1238
	9	0.6233	0.0848	0.6452	0.0930	0.6994	0.1003
	10	0.5473	0.1130	0.5809	0.1213	0.6212	0.1271
	11	0.1764	0.1074	0.1783	0.1076	0.1783	0.1051
	12	0.1712	0.1416	0.1724	0.1411	0.1695	0.1361
	13	0.4806	0.1874	0.5235	0.2061	0.5794	0.2258
	14	0.5036	0.1366	0.5321	0.1471	0.5916	0.1620
	15	0.4874	0.1579	0.5161	0.1662	0.6423	0.2019
15	0	0.2764	0.1294	0.2905	0.1369	0.3301	0.1531
	1	0.4459	0.2263	0.3796	0.1881	0.4421	0.2129
	2	0.3681	0.1157	0.3661	0.1144	0.3923	0.1222
	3	0.5091	0.1129	0.5548	0.1228	0.6523	0.1400
	4	0.4382	0.0728	0.4716	0.0760	0.5771	0.0959
	5	0.4628	0.1328	0.4941	0.1405	0.5528	0.1586
	6	0.3369	0.3093	0.2819	0.2537	0.2657	0.2323
	7	0.6370	0.1156	0.6754	0.1234	0.7767	0.1381
	8	0.4274	0.0996	0.4894	0.1129	0.6010	0.1353
	9	0.5815	0.1164	0.5906	0.1200	0.6707	0.1374
	10	0.6042	0.1401	0.6496	0.1523	0.7351	0.1682
	11	0.3246	0.1062	0.3607	0.1172	0.4117	0.1313
	12	0.4458	0.1088	0.4969	0.1198	0.5968	0.1429
	13	0.4103	0.1289	0.4222	0.1366	0.4417	0.1419
	14	0.2402	0.1093	0.2977	0.1307	0.3570	0.1517
	15	0.4334	0.1508	0.4423	0.1535	0.4993	0.1669
16	0	0.2814	0.0642	0.3120	0.0693	0.3703	0.0822
	1	0.4293	0.2049	0.4554	0.2195	0.4738	0.2231
	2	0.5051	0.1277	0.4967	0.1272	0.5724	0.1363
	3	0.6068	0.1038	0.7153	0.1321	0.8210	0.1570
	4	0.3655	0.2309	0.4141	0.2584	0.4889	0.2907
	5	0.6630	0.1084	0.7092	0.1118	0.8142	0.1297
	6	0.2906	0.1888	0.3511	0.2246	0.4260	0.2639
	7	0.4699	0.0773	0.4935	0.0798	0.5532	0.0894
	8	0.6457	0.0947	0.6754	0.1008	0.7693	0.1126
	9	0.5045	0.0833	0.5761	0.0920	0.6047	0.0949
	10	0.7771	0.1291	0.7889	0.1387	0.8442	0.1493
	11	0.2904	0.0998	0.3013	0.1009	0.3681	0.1247
	12	0.4831	0.1744	0.6116	0.2288	0.7063	0.2507
	13	0.5674	0.1321	0.6090	0.1367	0.7137	0.1557
	14	0.6897	0.1040	0.7300	0.1132	0.8206	0.1299
	15	0.3346	0.1310	0.3667	0.1337	0.4425	0.1521

Table 29: Register attention mass (attn) and write norm (write). Blocks 17–23.

Block	Head	pretrained		regr-norm		regr-brut	
		attn	write	attn	write	attn	write
17	0	0.4791	0.0837	0.5014	0.0879	0.5745	0.0984
	1	0.5427	0.1273	0.4954	0.1177	0.5443	0.1280
	2	0.3799	0.1612	0.2907	0.1172	0.3577	0.1323
	3	0.5076	0.0870	0.5370	0.0884	0.6691	0.1077
	4	0.4528	0.1033	0.3734	0.0840	0.4616	0.1006
	5	0.4848	0.1255	0.4316	0.1009	0.5127	0.1129
	6	0.3572	0.3069	0.3073	0.2598	0.3123	0.2547
	7	0.3894	0.2941	0.3399	0.2495	0.3499	0.2459
	8	0.4664	0.0727	0.3792	0.0572	0.4337	0.0670
	9	0.4031	0.0595	0.3426	0.0519	0.4020	0.0616
	10	0.4110	0.1299	0.3776	0.1145	0.4709	0.1399
	11	0.5535	0.0967	0.5714	0.0957	0.6494	0.1050
	12	0.5579	0.1234	0.5771	0.1185	0.6266	0.1177
	13	0.3340	0.1012	0.3235	0.0920	0.3734	0.1041
	14	0.3303	0.0808	0.3097	0.0752	0.3672	0.0850
	15	0.3434	0.0550	0.3085	0.0512	0.3583	0.0575
18	0	0.3998	0.0826	0.3976	0.0853	0.5034	0.1017
	1	0.3601	0.1668	0.3089	0.1523	0.4116	0.1720
	2	0.4632	0.0861	0.4372	0.0889	0.5240	0.0925
	3	0.4225	0.0702	0.5345	0.1061	0.5817	0.1013
	4	0.4895	0.1916	0.4850	0.1981	0.5668	0.2148
	5	0.5349	0.1122	0.6070	0.1320	0.6868	0.1536
	6	0.4211	0.0804	0.3672	0.0719	0.4636	0.0905
	7	0.6346	0.1259	0.7255	0.1741	0.7642	0.1575
	8	0.2460	0.0631	0.2334	0.0576	0.2768	0.0703
	9	0.4709	0.0815	0.5910	0.1061	0.6495	0.1110
	10	0.4001	0.0823	0.3579	0.0745	0.4570	0.0872
	11	0.2138	0.0488	0.1733	0.0415	0.1683	0.0364
	12	0.3400	0.3546	0.3186	0.3395	0.4123	0.4027
	13	0.3133	0.0804	0.2809	0.0603	0.2971	0.0592
	14	0.4008	0.1095	0.3392	0.1027	0.4449	0.1246
	15	0.3261	0.0725	0.3943	0.0960	0.4135	0.0911
19	0	0.6235	0.2279	0.7024	0.2737	0.7854	0.2618
	1	0.3765	0.1079	0.4590	0.1234	0.4896	0.1292
	2	0.1939	0.0611	0.2404	0.0700	0.3028	0.0850
	3	0.2713	0.1006	0.2566	0.0857	0.3583	0.1185
	4	0.2444	0.1124	0.3231	0.1426	0.3086	0.1306
	5	0.5155	0.0979	0.6619	0.1532	0.6906	0.1505
	6	0.3292	0.1110	0.3371	0.1014	0.4007	0.1140
	7	0.3479	0.3056	0.3815	0.3410	0.3854	0.3295
	8	0.3201	0.0872	0.3195	0.0776	0.3877	0.0934
	9	0.3127	0.0853	0.2631	0.0708	0.4163	0.1079
	10	0.2747	0.0887	0.2575	0.0782	0.2879	0.0835
	11	0.7432	0.1736	0.8539	0.2054	0.8873	0.2318
	12	0.1378	0.1061	0.2106	0.1547	0.2484	0.1718
	13	0.4140	0.0874	0.5039	0.1055	0.5888	0.1184
	14	0.2828	0.0693	0.3431	0.0834	0.3886	0.0890
	15	0.4575	0.0956	0.4124	0.0888	0.5636	0.1138
20	0	0.3703	0.1528	0.5570	0.2328	0.5151	0.2195
	1	0.2478	0.0844	0.3678	0.1188	0.3476	0.1071
	2	0.2932	0.1705	0.4434	0.2631	0.4721	0.3012
	3	0.3890	0.1523	0.6096	0.2370	0.6226	0.2459
	4	0.1678	0.0805	0.1640	0.0700	0.2103	0.0835
	5	0.3287	0.3371	0.3557	0.3933	0.1893	0.1972
	6	0.3954	0.0991	0.6046	0.1463	0.5796	0.1435
	7	0.0975	0.0365	0.1975	0.0768	0.2190	0.0881
	8	0.1952	0.0613	0.2686	0.0794	0.2240	0.0617
	9	0.2774	0.1096	0.2541	0.0927	0.3125	0.1154
	10	0.4980	0.1645	0.6673	0.2254	0.6770	0.2610
	11	0.1876	0.0531	0.2575	0.0657	0.2498	0.0633
	12	0.1631	0.0608	0.1987	0.0704	0.2213	0.0779
	13	0.3416	0.1046	0.5853	0.1739	0.6115	0.1727
	14	0.4000	0.1262	0.4734	0.1549	0.4884	0.1613
	15	0.6636	0.3503	0.8183	0.4774	0.8244	0.4601
21	0	0.3229	0.1132	0.5397	0.1936	0.5888	0.2188
	1	0.1658	0.0829	0.3191	0.1495	0.3400	0.1593
	2	0.1766	0.2264	0.2597	0.3490	0.3088	0.4332
	3	0.2580	0.1287	0.3049	0.1798	0.3992	0.2689
	4	0.1649	0.3623	0.4114	0.8755	0.5133	1.0602
	5	0.2001	0.5268	0.2286	0.5955	0.2026	0.5083
	6	0.2515	0.0878	0.3465	0.1263	0.4042	0.1622
	7	0.1865	0.0793	0.3640	0.1464	0.3003	0.1179
	8	0.1993	0.2330	0.1503	0.1862	0.1709	0.1998
	9	0.2450	0.0836	0.5130	0.1916	0.4743	0.1701
	10	0.1621	0.0758	0.3403	0.1655	0.2878	0.1430
	11	0.1633	0.0704	0.3127	0.1233	0.3630	0.1440
	12	0.1144	0.0583	0.2923	0.1570	0.2113	0.0997
	13	0.1905	0.1502	0.2823	0.1978	0.3595	0.2528
	14	0.1921	0.0723	0.3955	0.1562	0.3412	0.1389
	15	0.1697	0.0821	0.3700	0.1693	0.3898	0.1909
22	0	0.1436	0.0487	0.3569	0.1884	0.3883	0.1788
	1	0.1240	0.0596	0.2430	0.1341	0.3345	0.1704
	2	0.1041	0.0420	0.3269	0.1857	0.1437	0.0761
	3	0.1319	0.0712	0.2114	0.1209	0.1539	0.0814
	4	0.1688	0.0668	0.2362	0.0991	0.3081	0.1329
	5	0.0921	0.0440	0.1122	0.0617	0.1618	0.0815
	6	0.0600	0.0264	0.0741	0.0641	0.0759	0.0526
	7	0.2051	0.1411	0.3237	0.1896	0.2704	0.1577
	8	0.1956	0.1014	0.2213	0.1198	0.2005	0.1070
	9	0.1417	0.0809	0.1131	0.0703	0.2883	0.1480
	10	0.0888	0.0919	0.3697	0.4807	0.2217	0.2479
	11	0.1358	0.0461	0.4594	0.2994	0.3612	0.1692
	12	0.1528	0.0562	0.4423	0.2385	0.3427	0.1670
	13	0.1193	0.0412	0.3697	0.2473	0.2046	0.1155
	14	0.1086	0.0325	0.2026	0.1022	0.1809	0.0733
	15	0.1135	0.0622	0.0811	0.0740	0.2289	0.1787
23	0	0.1880	0.1314	0.1586	0.1097	0.1233	0.0794
	1	0.2754	0.0960	0.0874	0.0379	0.0263	0.0108
	2	0.2263	0.1120	0.1241	0.0889	0.0235	0.0138
	3	0.2422	0.0850	0.0495	0.0185	0.0292	0.0114
	4	0.1250	0.0454	0.0066	0.0027	0.0139	0.0053
	5	0.1890	0.0977	0.0692	0.0470	0.0732	0.0411
	6	0.1449	0.0876	0.0403	0.0279	0.0628	0.0410
	7	0.1979	0.1364	0.0930	0.0730	0.0470	0.0335
	8	0.2669	0.0835	0.0540	0.0206	0.0179	0.0059
	9	0.1733	0.1428	0.0849	0.0786	0.0650	0.0563
	10	0.1470	0.1310	0.1620	0.1462	0.1065	0.0911
	11	0.2025	0.0927	0.0283	0.0144	0.0416	0.0193
	12	0.2125	0.1219	0.0247	0.0151	0.0167	0.0100
	13	0.0754	0.0714	0.0028	0.0029	0.0108	0.0112
	14	0.2662	0.1073	0.1459	0.0808	0.1001	0.0533
	15	0.0965	0.0913	0.0037	0.0039	0.0050	0.0048

Table 30: Attention routing diagnostics at Block 0 (Layer L00). Reported are (i) distribution mismatch $\text{KL}(\text{attn} \parallel \text{contrib})$, (ii) register-mass shift $\Delta \text{reg} = \text{reg}_{\text{contrib}} - \text{reg}_{\text{attn}}$, (iii) entropy shift $\Delta H = H_{\text{contrib}} - H_{\text{attn}}$ (reported via absolute-mean in the summary), and (iv) correlation between attention weights and effective contribution, $\text{corr}(x, y)$.

	pretrained	regr-norm	regr-brut
max KL (head, value)	H11, 0.09534	H11, 0.07901	H11, 0.08291
max $ \Delta \text{reg} $ (head, value)	H10, 0.03000	H10, 0.02850	H10, 0.02823
max $ \Delta H $ (head, value)	H10, -0.15790	H10, -0.12938	H10, -0.13635
mean KL	0.04530	0.03910	0.04065
mean $ \Delta \text{reg} $	0.00978	0.00920	0.00914
mean $ \Delta H $	0.05941	0.05543	0.05747

Table 31: Per-head diagnostics at Block 0 (Layer L00): entropy shift ΔH (column: d_ent), $\text{KL}(\text{attn} \parallel \text{contrib})$, register-mass shift Δreg (column: d_reg), and correlation $\text{corr}(x, y)$, shown for pretrained, regr-norm, and regr-brut.

head	pretrained				regr-norm				regr-brut			
	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr
0	-0.05754	0.06997	0.00403	0.49433	-0.05591	0.06726	0.00356	0.51742	-0.05667	0.06802	0.00325	0.51695
1	0.00162	0.01950	0.00548	0.99136	0.00203	0.01942	0.00541	0.99131	0.00403	0.01952	0.00569	0.99199
2	0.06342	0.00789	0.00635	0.99561	0.06445	0.00771	0.00621	0.99535	0.06096	0.00805	0.00658	0.99553
3	-0.00864	0.02416	0.00563	0.98598	-0.00918	0.02386	0.00565	0.98532	-0.01426	0.02374	0.00592	0.98632
4	-0.06975	0.02936	0.00780	0.82343	-0.06726	0.02773	0.00741	0.83356	-0.06495	0.02791	0.00723	0.82037
5	0.01059	0.02471	0.01461	0.85878	0.00883	0.02383	0.01445	0.86728	0.00660	0.02417	0.01470	0.88329
6	0.01158	0.06479	0.00916	0.87989	0.01938	0.04996	0.00832	0.89732	0.02043	0.04734	0.00716	0.91226
7	0.01946	0.01597	0.00481	0.99202	0.01954	0.01559	0.00495	0.99208	0.02628	0.01647	0.00465	0.99351
8	0.00433	0.07910	0.01001	0.85473	0.01153	0.05939	0.00929	0.87065	0.01881	0.06351	0.00864	0.87230
9	0.02278	0.04519	0.01297	0.87650	0.01929	0.03839	0.01182	0.89667	0.02196	0.03973	0.01227	0.90069
10	-0.15790	0.08539	0.03000	0.84858	-0.12938	0.07311	0.02850	0.83090	-0.13635	0.07642	0.02823	0.83074
11	-0.14663	0.09534	0.01121	0.87839	-0.12687	0.07901	0.01017	0.87516	-0.13064	0.08291	0.01042	0.87495
12	-0.13760	0.06971	0.01095	0.85908	-0.12179	0.05695	0.00953	0.87385	-0.13028	0.06679	0.00985	0.85639
13	-0.10151	0.04129	0.00608	0.82713	-0.09878	0.03696	0.00503	0.85315	-0.09836	0.04043	0.00536	0.83535
14	-0.09448	0.02364	0.00716	0.98755	-0.09180	0.02246	0.00659	0.98687	-0.09208	0.02286	0.00651	0.98699
15	0.04278	0.02878	0.01029	0.92287	0.04082	0.02403	0.01031	0.94120	0.03682	0.02245	0.00975	0.95413

Table 32: Attention routing diagnostics at Block 10 (Layer L10). Reported are (i) distribution mismatch $\text{KL}(\text{attn}||\text{contrib})$, (ii) register-mass shift $\Delta\text{reg} = \text{reg}_{\text{contrib}} - \text{reg}_{\text{attn}}$, (iii) entropy shift $\Delta H = H_{\text{contrib}} - H_{\text{attn}}$ (reported via absolute-mean in the summary), and (iv) correlation between attention weights and effective contribution, $\text{corr}(x, y)$.

	pretrained	regr-norm	regr-brut
max KL (head, value)	H01, 0.06462	H01, 0.06620	H01, 0.07125
max $ \Delta\text{reg} $ (head, value)	H01, 0.00998	H01, 0.00925	H01, 0.00935
max $ \Delta H $ (head, value)	H01, 0.14600	H01, 0.18605	H01, 0.23459
mean KL	0.01995	0.01984	0.02068
mean $ \Delta\text{reg} $	0.00299	0.00281	0.00276
mean $ \Delta H $	0.04923	0.05350	0.06049

Table 33: Per-head diagnostics at Block 10 (Layer L10): entropy shift ΔH (column: d_ent), $\text{KL}(\text{attn}||\text{contrib})$, register-mass shift Δreg (column: d_reg), and correlation $\text{corr}(x, y)$, shown for pretrained, regr-norm, and regr-brut.

head	pretrained				regr-norm				regr-brut			
	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr
0	-0.05180	0.01409	-0.00118	0.98859	-0.04837	0.01380	-0.00130	0.98880	-0.05092	0.01410	-0.00104	0.98790
1	0.14600	0.06462	0.00998	0.90627	0.18605	0.06620	0.00925	0.88456	0.23459	0.07125	0.00935	0.84290
2	0.01890	0.01083	0.00045	0.98251	0.01804	0.01029	0.00040	0.97935	0.02127	0.01065	0.00068	0.97942
3	0.02685	0.01180	0.00330	0.98846	0.02653	0.01156	0.00314	0.98756	0.02773	0.01150	0.00320	0.98338
4	-0.04602	0.00881	-0.00068	0.99232	-0.04386	0.00820	-0.00063	0.98988	-0.04356	0.00830	-0.00051	0.98872
5	0.05414	0.01584	0.00272	0.98485	0.05949	0.01610	0.00295	0.98513	0.06843	0.01647	0.00333	0.98766
6	0.01462	0.00950	0.00299	0.98121	0.00960	0.00912	0.00278	0.97862	0.01269	0.00888	0.00307	0.97463
7	-0.02069	0.00800	0.00177	0.98920	-0.02253	0.00801	0.00156	0.99044	-0.02465	0.00800	0.00140	0.99082
8	-0.06045	0.01395	-0.00119	0.99142	-0.06457	0.01438	-0.00085	0.98865	-0.07168	0.01531	-0.00060	0.98908
9	-0.08580	0.01042	-0.00044	0.99300	-0.08660	0.01050	-0.00054	0.98919	-0.08662	0.01050	-0.00047	0.99038
10	0.08244	0.04979	0.00586	0.98590	0.08749	0.04886	0.00557	0.98088	0.10116	0.05039	0.00553	0.97793
11	0.00353	0.00744	-0.00082	0.99231	0.00153	0.00750	-0.00060	0.99143	0.00311	0.00759	-0.00029	0.99318
12	0.01259	0.01445	0.00216	0.99413	0.01455	0.01370	0.00218	0.99360	0.00997	0.01338	0.00214	0.99319
13	-0.02696	0.00716	-0.00068	0.98686	-0.03334	0.00728	-0.00059	0.98515	-0.03297	0.00728	-0.00030	0.98497
14	0.09948	0.06062	0.00991	0.88922	0.11650	0.06069	0.00912	0.89751	0.14211	0.06607	0.00901	0.89375
15	-0.03743	0.01180	0.00376	0.99338	-0.03699	0.01130	0.00354	0.99155	-0.03634	0.01118	0.00322	0.99071

Table 34: Attention routing diagnostics at Block 12 (Layer L12). Reported are (i) distribution mismatch $\text{KL}(\text{attn}||\text{contrib})$, (ii) register-mass shift $\Delta\text{reg} = \text{reg}_{\text{contrib}} - \text{reg}_{\text{attn}}$, (iii) entropy shift $\Delta H = H_{\text{contrib}} - H_{\text{attn}}$ (reported via absolute-mean in the summary), and (iv) correlation between attention weights and effective contribution, $\text{corr}(x, y)$.

	pretrained	regr-norm	regr-brut
max KL (head, value)	H14, 0.24417	H14, 0.23918	H14, 0.23434
max $ \Delta\text{reg} $ (head, value)	H12, 0.01723	H12, 0.01692	H12, 0.01667
max $ \Delta H $ (head, value)	H14, 1.29185	H14, 1.25921	H14, 1.31162
mean KL	0.09943	0.09543	0.09241
mean $ \Delta\text{reg} $	0.00607	0.00565	0.00529
mean $ \Delta H $	0.37480	0.35369	0.33834

Table 35: Per-head diagnostics at Block 12 (Layer L12): entropy shift ΔH (column: d_ent), $\text{KL}(\text{attn}||\text{contrib})$, register-mass shift Δreg (column: d_reg), and correlation $\text{corr}(x, y)$, shown for pretrained, regr-norm, and regr-brut.

head	pretrained				regr-norm				regr-brut			
	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr
0	0.32039	0.08317	0.00362	0.97229	0.28819	0.07805	0.00244	0.97565	0.25356	0.07364	0.00196	0.97756
1	0.01048	0.01245	0.00058	0.98956	0.02218	0.01201	0.00052	0.98821	0.02051	0.01243	0.00070	0.98856
2	0.06718	0.02200	0.00291	0.98934	0.09117	0.02390	0.00298	0.98814	0.09949	0.02491	0.00313	0.98859
3	0.22768	0.09462	0.00598	0.97211	0.15915	0.08586	0.00450	0.97321	0.12961	0.07859	0.00410	0.97322
4	-0.22553	0.09628	-0.00002	0.96600	-0.28395	0.08951	-0.00139	0.96219	-0.25792	0.09663	-0.00127	0.95812
5	0.18221	0.06708	0.00393	0.98253	0.08258	0.05733	0.00255	0.97512	0.04347	0.05215	0.00172	0.98052
6	0.04697	0.03162	0.00156	0.97891	0.03491	0.02975	0.00094	0.98647	0.02341	0.02904	0.00059	0.98672
7	0.18225	0.08328	0.00364	0.96655	0.18152	0.08157	0.00318	0.97872	0.17226	0.07978	0.00265	0.98237
8	0.10121	0.07680	0.00338	0.97621	0.10154	0.07373	0.00322	0.98210	0.09356	0.07093	0.00307	0.98060
9	0.00599	0.02497	0.00306	0.99212	0.01808	0.02601	0.00282	0.99099	0.03196	0.02563	0.00316	0.99240
10	0.23339	0.06221	0.00542	0.93544	0.24499	0.06261	0.00458	0.93877	0.20076	0.06039	0.00409	0.94507
11	0.16360	0.09084	0.00661	0.97888	0.17718	0.09012	0.00679	0.98320	0.15581	0.08216	0.00607	0.98491
12	1.24448	0.23406	0.01723	0.90595	1.11447	0.22690	0.01692	0.92763	1.12194	0.22400	0.01667	0.93685
13	0.45721	0.12992	0.00682	0.96699	0.40144	0.11950	0.00546	0.96865	0.36213	0.11088	0.00500	0.96771
14	1.29185	0.24417	0.01583	0.95082	1.25921	0.23918	0.01551	0.95026	1.31162	0.23434	0.01460	0.95093
15	1.23634	0.23751	0.01659	0.94749	1.19846	0.23092	0.01664	0.95040	1.13538	0.22300	0.01595	0.95234

Table 36: Attention routing diagnostics at Block 18 (Layer L18). Reported are (i) distribution mismatch $\text{KL}(\text{attn}||\text{contrib})$, (ii) register-mass shift $\Delta\text{reg} = \text{reg}_{\text{contrib}} - \text{reg}_{\text{attn}}$, (iii) entropy shift $\Delta H = H_{\text{contrib}} - H_{\text{attn}}$ (reported via absolute-mean in the summary), and (iv) correlation between attention weights and effective contribution, $\text{corr}(x, y)$.

	pretrained	regr-norm	regr-brut
max KL (head, value)	H07, 0.68321	H09, 0.73865	H09, 0.75919
max $ \Delta\text{reg} $ (head, value)	H15, 0.06325	H15, 0.07593	H15, 0.08790
max $ \Delta H $ (head, value)	H07, 1.75077	H07, 2.03899	H07, 2.24845
mean KL	0.50833	0.53226	0.56732
mean $ \Delta\text{reg} $	0.03522	0.03808	0.04353
mean $ \Delta H $	1.08148	1.34319	1.56762

Table 37: Per-head diagnostics at Block 18 (Layer L18): entropy shift ΔH (column: d_ent), $\text{KL}(\text{attn}||\text{contrib})$, register-mass shift Δreg (column: d_reg), and correlation $\text{corr}(x, y)$, shown for pretrained, regr-norm, and regr-brut.

head	pretrained				regr-norm				regr-brut			
	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr
0	1.07352	0.58170	0.04453	0.97597	1.28180	0.59651	0.04658	0.96926	1.68784	0.66611	0.05183	0.96171
1	1.08099	0.44089	0.03425	0.96668	1.21179	0.44400	0.04042	0.97010	1.52736	0.51424	0.05111	0.94631
2	1.17200	0.60710	0.01857	0.89644	1.52600	0.61876	0.02663	0.88790	1.72606	0.68263	0.03167	0.87616
3	1.07598	0.63619	0.05918	0.97581	1.60049	0.66465	0.05803	0.95723	1.82565	0.73929	0.07221	0.95984
4	1.34604	0.38193	0.02960	0.97975	1.45233	0.37736	0.02504	0.98123	1.78415	0.40367	0.02429	0.97720
5	1.35182	0.63975	0.06189	0.98110	1.72337	0.67734	0.06153	0.97044	1.98624	0.66633	0.07054	0.96728
6	1.21036	0.58516	0.01849	0.96330	1.48116	0.63654	0.02286	0.93399	1.89777	0.67495	0.02420	0.91728
7	1.75077	0.68321	0.02215	0.95314	2.03899	0.60003	0.02381	0.96057	2.24845	0.63216	0.02440	0.95827
8	0.58104	0.38182	0.03594	0.98933	0.97414	0.48785	0.03755	0.97583	0.97137	0.46685	0.03556	0.97478
9	1.37566	0.67248	0.02604	0.92028	1.93565	0.73865	0.03297	0.91412	2.23453	0.75919	0.04211	0.91102
10	1.26766	0.62383	0.03755	0.94640	1.46985	0.63927	0.04359	0.94474	1.88297	0.71928	0.05571	0.93460
11	0.38850	0.34524	0.03927	0.99065	0.38596	0.32817	0.04215	0.99250	0.32655	0.32423	0.04838	0.98845
12	0.66959	0.12808	0.01814	0.98147	0.80736	0.13224	0.01897	0.98843	1.03560	0.14720	0.01888	0.98409
13	0.97430	0.48925	0.03776	0.96459	1.16268	0.57413	0.04013	0.96587	1.25906	0.59969	0.04643	0.96163
14	1.37267	0.46285	0.01690	0.96982	1.46303	0.46976	0.01306	0.97186	1.63233	0.50195	0.01124	0.93461
15	0.61279	0.47373	0.06325	0.97796	0.97651	0.53084	0.07593	0.97032	1.05591	0.57927	0.08790	0.96635

Table 38: Attention routing diagnostics at Block 20 (Layer L20). Reported are (i) distribution mismatch $\text{KL}(\text{attn} \parallel \text{contrib})$, (ii) register-mass shift $\Delta \text{reg} = \text{reg}_{\text{contrib}} - \text{reg}_{\text{attn}}$, (iii) entropy shift $\Delta H = H_{\text{contrib}} - H_{\text{attn}}$ (reported via absolute-mean in the summary), and (iv) correlation between attention weights and effective contribution, $\text{corr}(x, y)$.

	pretrained	regr-norm	regr-brut
max KL (head, value)	H06, 0.50552	H14, 0.56962	H06, 0.59354
max $ \Delta \text{reg} $ (head, value)	H10, 0.08199	H08, 0.07425	H06, 0.05184
max $ \Delta H $ (head, value)	H15, 1.60054	H06, 2.00739	H06, 2.03641
mean KL	0.32795	0.42378	0.41340
mean $ \Delta \text{reg} $	0.05125	0.04773	0.03352
mean $ \Delta H $	0.72693	1.40387	1.32951

Table 39: Per-head diagnostics at Block 20 (Layer L20): entropy shift ΔH (column: d_ent), $\text{KL}(\text{attn} \parallel \text{contrib})$, register-mass shift Δreg (column: d_reg), and correlation $\text{corr}(x, y)$, shown for pretrained, regr-norm, and regr-brut.

head	pretrained				regr-norm				regr-brut			
	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr
0	0.91394	0.34313	0.05182	0.97718	1.72507	0.37500	0.04194	0.98198	1.66990	0.38363	0.02811	0.98406
1	0.71048	0.35613	0.06594	0.95928	1.68808	0.49732	0.06347	0.95007	1.41256	0.48981	0.04037	0.95784
2	0.53560	0.21913	0.03786	0.96916	1.12874	0.30434	0.03557	0.95374	1.24339	0.29883	0.02311	0.94470
3	0.81141	0.37098	0.07445	0.96546	1.61520	0.43797	0.05941	0.93832	1.69323	0.43303	0.04921	0.94938
4	0.40228	0.22252	0.02858	0.97253	0.94271	0.35806	0.02857	0.96870	1.06133	0.39461	0.02057	0.97204
5	0.64254	0.12895	0.03574	0.98252	0.90810	0.11569	0.02553	0.98456	0.53976	0.09544	0.01473	0.98966
6	1.06547	0.50552	0.07145	0.91877	2.00739	0.53962	0.06260	0.89300	2.03641	0.59354	0.05184	0.90982
7	-0.03233	0.15645	0.03409	0.98774	0.67302	0.41082	0.06831	0.97685	0.61231	0.38519	0.04111	0.97693
8	0.39519	0.32531	0.06184	0.97102	1.28218	0.53622	0.07425	0.96057	0.86634	0.47462	0.04544	0.97892
9	0.64941	0.33662	0.06795	0.96177	1.26067	0.44787	0.04958	0.97009	1.18853	0.42975	0.03879	0.97051
10	1.27459	0.47128	0.08199	0.94838	1.81577	0.46462	0.05481	0.97717	1.80902	0.40324	0.03639	0.97354
11	0.42712	0.29477	0.02550	0.89571	1.20524	0.52256	0.03514	0.94551	0.99931	0.47067	0.02209	0.93767
12	0.24233	0.24347	0.03401	0.97361	0.94394	0.42391	0.04607	0.96703	0.72605	0.37337	0.03165	0.97703
13	0.82975	0.41472	0.05076	0.94238	1.76624	0.52027	0.04358	0.96008	1.90531	0.55019	0.03640	0.96937
14	1.09784	0.50179	0.04871	0.94858	1.86576	0.56962	0.04366	0.95508	1.83621	0.57686	0.03227	0.96766
15	1.60054	0.35644	0.04933	0.95101	1.63385	0.25651	0.03116	0.97433	1.67247	0.26156	0.02430	0.97285

Table 40: Attention routing diagnostics at Block 22 (Layer L22). Reported are (i) distribution mismatch $\text{KL}(\text{attn} \parallel \text{contrib})$, (ii) register-mass shift $\Delta \text{reg} = \text{reg}_{\text{contrib}} - \text{reg}_{\text{attn}}$, (iii) entropy shift $\Delta H = H_{\text{contrib}} - H_{\text{attn}}$ (reported via absolute-mean in the summary), and (iv) correlation between attention weights and effective contribution, $\text{corr}(x, y)$.

	pretrained	regr-norm	regr-brut
max KL (head, value)	H04, 0.25995	H04, 0.35578	H11, 0.36007
max $ \Delta \text{reg} $ (head, value)	H04, 0.00008	H04, 0.00104	H03, 0.04006
max $ \Delta H $ (head, value)	H08, 0.35681	H02, 1.47810	H00, 1.36347
mean KL	0.19704	0.25658	0.26955
mean $ \Delta \text{reg} $	0.00003	0.00022	0.01353
mean $ \Delta H $	0.13535	0.91679	0.88953

Table 41: Per-head diagnostics at Block 22 (Layer L22): entropy shift ΔH (column: d_ent), $\text{KL}(\text{attn} \parallel \text{contrib})$, register-mass shift Δreg (column: d_reg), and correlation $\text{corr}(x, y)$, shown for pretrained, regr-norm, and regr-brut.

head	pretrained				regr-norm				regr-brut			
	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr
0	0.09022	0.25267	-0.00002	0.99099	1.20426	0.30910	0.00022	0.98988	1.36347	0.35410	0.00859	0.94546
1	0.04169	0.20414	-0.00004	0.99477	0.69551	0.29947	0.00029	0.99825	1.14866	0.34607	0.00509	0.93773
2	0.10300	0.17914	-0.00001	0.97851	1.47810	0.26407	0.00020	0.99761	0.77807	0.23032	0.00959	0.96569
3	0.11795	0.14302	-0.00001	0.99654	0.98177	0.23495	0.00007	0.99497	0.73178	0.20613	0.04006	0.98764
4	0.24147	0.25995	0.00008	0.98997	0.74642	0.35578	0.00104	0.99782	1.02730	0.34280	0.00584	0.97275
5	0.05615	0.14740	-0.00002	0.98006	0.68660	0.26737	0.00017	0.99162	0.58921	0.27511	-0.01228	0.91947
6	-0.07678	0.12514	-0.00004	0.98824	0.54392	0.17617	-0.00002	0.97107	0.27078	0.15982	-0.02452	0.96565
7	0.28135	0.22227	-0.00000	0.99042	1.26062	0.34288	0.00029	0.99509	1.22875	0.31949	0.03324	0.97024
8	0.35681	0.19338	-0.00001	0.98656	0.90243	0.22637	0.00014	0.99558	0.84217	0.18990	0.00903	0.98005
9	0.07818	0.19274	-0.00003	0.99092	0.26472	0.18908	0.00003	0.98957	0.88782	0.31959	0.00303	0.96558
10	0.02282	0.09630	-0.00004	0.97575	0.92087	0.13706	0.00010	0.99224	0.78159	0.17582	-0.00019	0.97542
11	0.03338	0.24035	-0.00003	0.98368	1.15744	0.31084	0.00037	0.99424	1.17164	0.36007	0.00884	0.95344
12	0.29234	0.24537	0.00002	0.98741	1.33134	0.28139	0.00021	0.99714	1.18701	0.28621	0.02478	0.97659
13	0.00458	0.22569	-0.00002	0.98513	1.21073	0.26273	0.00011	0.99572	0.66114	0.25117	-0.00816	0.97012
14	0.13198	0.22859	-0.00003	0.97314	0.68234	0.31716	0.00027	0.99672	0.62530	0.29531	-0.00547	0.96380
15	0.23691	0.19649	-0.00001	0.97259	0.60164	0.13085	0.00004	0.99712	0.93784	0.20082	0.01774	0.96880

Table 42: Attention routing diagnostics at Block 23 (Layer L23). Reported are (i) distribution mismatch $\text{KL}(\text{attn}||\text{contrib})$, (ii) register-mass shift $\Delta\text{reg} = \text{reg}_{\text{contrib}} - \text{reg}_{\text{attn}}$, (iii) entropy shift $\Delta H = H_{\text{contrib}} - H_{\text{attn}}$ (reported via absolute-mean in the summary), and (iv) correlation between attention weights and effective contribution, $\text{corr}(x, y)$.

	pretrained	regr-norm	regr-brut
max KL (head, value)	H14, 0.53274	H14, 0.35649	H14, 0.27789
max $ \Delta\text{reg} $ (head, value)	H09, 0.00583	H13, 0.01241	H13, 0.01811
max $ \Delta H $ (head, value)	H09, 0.41711	H11, -0.27628	H10, 0.22410
mean KL	0.27269	0.10876	0.08748
mean $ \Delta\text{reg} $	0.00074	0.00106	0.00302
mean $ \Delta H $	0.20991	0.13520	0.12168

Table 43: Per-head diagnostics at Block 23 (Layer L23): entropy shift ΔH (column: d_ent), $\text{KL}(\text{attn}||\text{contrib})$, register-mass shift Δreg (column: d_reg), and correlation $\text{corr}(x, y)$, shown for pretrained, regr-norm, and regr-brut.

head	pretrained				regr-norm				regr-brut			
	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr	d_ent	KL	d_reg	corr
0	0.30291	0.21830	0.00071	0.94542	0.14344	0.18513	-0.00043	0.92845	0.13461	0.14398	-0.00258	0.86972
1	0.29678	0.40553	0.00038	0.89050	-0.13436	0.12546	-0.00025	0.97168	-0.10216	0.04939	-0.00171	0.96422
2	0.05086	0.44499	-0.00037	0.87533	-0.20634	0.19332	-0.00047	0.93186	-0.19530	0.07840	-0.00286	0.93417
3	0.22076	0.42067	0.00005	0.94140	-0.21923	0.10178	-0.00043	0.96694	-0.14133	0.06834	-0.00154	0.96095
4	0.04095	0.24518	-0.00051	0.96220	-0.12209	0.02400	-0.00003	0.99527	-0.14241	0.04371	-0.00048	0.98529
5	0.21852	0.26946	-0.00014	0.94609	-0.11658	0.10790	-0.00016	0.97139	-0.04793	0.12686	-0.00323	0.94781
6	0.23884	0.20039	-0.00013	0.94415	-0.12854	0.05959	-0.00021	0.96197	-0.05057	0.09272	-0.00367	0.95263
7	0.35302	0.16062	0.00091	0.95406	0.01604	0.07450	-0.00016	0.98625	-0.03368	0.04525	-0.00193	0.95994
8	0.25197	0.42710	0.00055	0.90980	-0.16325	0.09265	-0.00040	0.98191	-0.10386	0.04231	-0.00144	0.97319
9	0.41711	0.15005	0.00583	0.96780	-0.00666	0.08132	-0.00026	0.98325	-0.01605	0.07533	-0.00318	0.94751
10	0.23133	0.17122	0.00024	0.96002	0.21557	0.15262	-0.00029	0.93539	0.22410	0.12875	-0.00179	0.93886
11	0.24053	0.26905	0.00001	0.95571	-0.27628	0.06989	-0.00023	0.98390	-0.18202	0.08876	-0.00097	0.97559
12	0.22812	0.29058	0.00030	0.95442	-0.17180	0.04714	-0.00009	0.99318	-0.13988	0.04308	-0.00151	0.96608
13	-0.03526	0.05658	-0.00070	0.98549	0.05685	0.05264	0.01241	0.99264	0.13370	0.07372	0.01811	0.99058
14	0.21974	0.53274	0.00030	0.78697	0.06856	0.35649	-0.00106	0.87385	0.17846	0.27789	-0.00268	0.87721
15	0.01183	0.10061	-0.00077	0.97868	-0.11753	0.01567	-0.00004	0.99871	-0.12085	0.02112	-0.00059	0.97406